

# Lexicographic orders of the well-orderings of the reals

Ordinal spectra, coding length and representation rank,  
finite-tail geometry, characters, gaps, and self-absorption

Research report merged from three independent manuscripts  
prepared for Vladimir Reshetnikov (sources 02, 06 and 07 of batch 76)

Sources dated 2 October 2026; merged 2 October 2026

## Abstract

Let  $\kappa = 2^{\aleph_0}$ , regarded as an initial ordinal. A well-ordering of  $\mathbb{R}$  has a unique exhaustive enumeration by an ordinal of cardinality  $\kappa$ . We compare two such enumerations at their first unequal real entry. The resulting linear order  $\mathcal{W}$  has cardinality  $2^\kappa$ , contains every linear order of cardinality at most  $\kappa$ , and contains an ordinal  $\beta$  exactly when  $\beta < \kappa^+$ . Nevertheless, its least binary lexicographic coding length is  $\kappa^+$ : it contains every binary lexicographic cube of smaller length but does not contain the cube of length  $\kappa^+$ .

The restriction  $\mathcal{W}_\kappa$  to enumerations of length exactly  $\kappa$  is quite different. It is equimorphic with  $\{0, 1\}^\kappa$  in lexicographic order, as is the actual ultrafilter-enumeration index in the Kanovei–Shelah construction. We prove the exact fixed-type classification  $\mathcal{W}_{\lambda+n} \simeq_e \mathbb{B}_\lambda \times_{\text{lex}} n!$  for limit  $\lambda$  of cardinality  $\kappa$ , characterize every adjacent pair in  $\mathcal{W}$ , compute several order-topological invariants, and prove the sharp absorption theorem  $\mathcal{W}_{\text{lex}}^\gamma \hookrightarrow \mathcal{W}$  if and only if  $\gamma < \kappa^+$ . We also analyze Dedekind completion and formulate further embedding, definability, and formalization problems.

[merge] This report is built from three manuscripts written independently on the same question and delivered together in batch 76: source 02 (the base, whose abstract is the two paragraphs above), source 06 and source 07. They prove the same core three times; each shared theorem is printed once, with the other sources' arguments marked as second or third routes where the argument differs. Sources 06 and 07 add universality in every fixed-type slice; a dense set of  $2^\kappa$  isolated points with a perfect, nowhere dense remainder; the exact point-character spectrum; the exact gap spectrum of  $\mathcal{W}$ , in which every gap has a countable side, with exactly  $2^\kappa$  gaps; the gap spectrum of  $\mathcal{W}_\kappa$ , which has a  $(\text{cf } \kappa, \text{cf } \kappa)$  gap, so that  $\mathcal{W}_\kappa$  has no convex copy in  $\mathcal{W}$ ; the real representation rank  $\text{repr}_{\mathbb{R}}(\mathcal{W}) = \kappa^+$ ; the canonical lexicographic-sum decomposition of  $\mathcal{W}$  over injections; a dense finite-condensation quotient equimorphic with  $\mathcal{W}$ , with a connected completion; bounded surreal sign orders; further gaps of the Kanovei–Shelah index; and a well-ordering-based ultrafilter index. Each source answers a question of another: in ZFC,  $\mathcal{W}$  does not embed into  $\mathcal{W}_\kappa$ , which source 06 had proved only under  $2^{<\kappa} < 2^\kappa$ ; and sources 06 and 07 settle source 02's questions on cut and character spectra and on the quotient.

**Conventions and status.** All main theorems are in ZFC. All embeddings are order embeddings, not necessarily continuous maps. Equimorphism means embeddability in both directions, not isomorphism. Proofs are supplied below; they have not been independently refereed or checked by a proof assistant. The results are presented as a research synthesis with explicit derivations, not as a claim of established historical priority.

[merge] **Status of the merged report.** AI-assisted (all three sources are “prepared for Vladimir Reshetnikov”; none carries AI wording in its author line), unrefereed, and not formalized. None of the three sources claims historical priority, peer review, or machine verification; their finite checks are illustrations of finite instances, not evidence for the transfinite assertions.

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# 1 About this report

## 1.1 Three independent manuscripts

[merge] This section, the notation table, the crosswalk of [section C](#) and every paragraph tagged [merge] were added when the three manuscripts of [table 1](#) were merged. Everything untagged in [sections 2 to 14, 25 and 26](#) and [sections A and B](#) is source 02 as delivered, with its labels prefixed by `lwo:`. [Sections 15 to 24 and 27](#) print sources 06 and 07; each opens with a tagged note naming its source, and every statement in them names the source it comes from. A reference such as “06:171” means line 171 of the main `.tex` file of source 06 as delivered (archive in the arrival commit 6914ccca6; the manuscripts of sources 06 and 07 are not shipped with this report).

| Source    | Archive (batch 76)                            | Delivered title                                                                                                                                               | Size                |
|-----------|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| 02 (base) | lexicographic_well_orderings_of_reals.zip     | <i>Lexicographic Orders of Well-Orderings of the Continuum: ordinal spectra, lexicographic complexity, finite-tail geometry, and self-absorption</i>          | 1,826 lines; 24 pp. |
| 06        | Lexicographic_Well_Orderings_of_Reals (1).zip | <i>The Lexicographic Order of All Well-Orderings of the Real Line: universality, exact ordinal spectra, finite tails, local characters, and Dedekind gaps</i> | 1,522 lines; 23 pp. |
| 07        | lexicographic_well_orderings_research.zip     | <i>Lexicographic Orders of Real Well-Orderings: ordinal universality, representation rank, products, and cut geometry</i>                                     | 1,521 lines; 21 pp. |

Table 1: [merge] The three sources. The number of a source is its batch-76 manuscript number. All three are dated 2 October 2026, are “prepared for Vladimir Reshetnikov”, and pin the ProveIt revision 6ea60e3677bd847a00baf023c3c532c83884530e (source 02: [section B](#); source 06: its Section 1.1 and audit; source 07: its Section 13.2 and audit). They arrived in commit 6914ccca6 and were placed in commit 2a04b60f2.

The three texts are *independent*: they share no member file, their word 8-gram overlap is 0.65–0.75 % pairwise, and their titles, notation and proofs were written separately. None is a version of another, and none supersedes another. They nevertheless prove the same core—prefix freeness, cardinality  $2^\kappa$ , universality for orders of size at most  $\kappa$ , the exact ordinal spectrum below  $\kappa^+$ , the equimorphism  $\mathcal{W}_\kappa \simeq_e \mathbb{B}_\kappa$ , the adjacency criterion, the finite blocks of size  $n!$ , and the topology of  $\mathcal{W}$  and of  $\mathcal{W}_\kappa$ —three times. Sources 02 and 07 further share the coding rank  $\kappa^+$ , the real powers, power absorption, the fixed-length square and the Kanovei–Shelah index; sources 06 and 07 share the point-character spectrum, the gap spectrum of  $\mathcal{W}$  and the dense quotient. Source 02 is the base because it has the widest scope: every stratum  $\mathcal{W}_\alpha$ , the Kanovei–Shelah index  $\mathcal{A}$ , general alphabets in its coding-length theorem, and completion invariants.

## 1.2 How the merge is printed

[merge] Each theorem proved by more than one source is stated once. Where another source proves it by the same argument, a short note after the proof says so (“also source 06, Theorem 5.2, 06:465, same argument”). Where the argument differs, the other proof is printed as a marked *second route* (or *third route*), credited to its source. No result, proof, example, remark, question or limitation of any source is dropped: the crosswalk ([table 5](#)) lists every numbered statement of the three sources and where it stands.

Two cross-repository duplications are also recorded at their points of use. Source 02’s strict cube hierarchy ([theorem 5.3](#)) re-proves Corollary `cor:strict` of the ProveIt report *Every ordinal as a point-separating game value*; that corollary is cited first and source 02’s cylinder proof is a second

route. Source 07’s strictness of real powers ([corollary 5.6](#)) is the ordinal case of a classical theorem of Kuhlmann [6, Corollary 2.4], as source 07 itself says.

### 1.3 Questions one source answers for another

[merge] Each source left open a question that another source answers. These answers are printed as theorems, crediting the answering sources, and the questions are re-scoped in [section 26](#).

- (i) Source 06 proves that  $\mathcal{W}$  does not embed into  $\mathcal{W}_\kappa$  only under  $2^{<\kappa} < 2^\kappa$  (06:1053, its Corollary 9.5) and asks for an unconditional comparison (06:1332, its Research question 14.1). Sources 02 and 07 prove it in ZFC: [corollary 8.3](#) (source 02) and source 07’s Corollary 5.4 (07:525), which also excludes every fixed real power  $\mathbb{R}_{\text{lex}}^\gamma$  with  $\gamma < \kappa^+$ .
- (ii) Source 02 asks for the exact spectra of cuts and point characters (02:1635–1646). Sources 06 and 07 compute the point-character spectrum and the gap spectrum of  $\mathcal{W}$  ([theorems 17.3](#) and [18.5](#); 06:913, 06:802; 07:861, 07:941), and source 07 the gap spectrum of  $\mathcal{W}_\kappa$  ([theorem 20.6](#); 07:996).
- (iii) Source 02 asks whether the finite-condensation quotient  $\mathcal{W}/\sim_{\text{fin}}$  is equimorphic with  $\mathcal{W}$  (02:1604–1607). Sources 06 and 07 prove that it is ([theorem 21.1](#); 06:1133, 07:1072).

### 1.4 Notation

[merge] The three sources use different letters for the same objects and the same letters for different objects. The merged text uses the symbols of [table 2](#); every symbol taken from source 06 or 07 and renamed here is listed in its last column. No normalization changes: every renaming is a change of letter only.

Table 2: [merge] Notions and symbols. “False reading” marks a tempting confusion.

| Notion                             | Here                                                    | Sources, and readings to avoid                                                                                                                                                                                   |
|------------------------------------|---------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Continuum as an initial ordinal    | $\kappa = \mathfrak{c} = 2^{\aleph_0}$                  | 02 and 06: $\kappa$ (06 also writes $\mathfrak{c}$ ); 07: $\kappa$ (macro <code>\k</code> ). False reading: $\kappa^+$ is the successor <i>cardinal</i> , not the ordinal $\kappa + 1$ .                         |
| All well-orderings of $\mathbb{R}$ | $\mathcal{W} = \mathcal{W}(\mathbb{R})$                 | 02, 06: $\mathcal{W}$ ; 07: $\mathcal{W}$ (macro <code>\W0</code> ). The order compares enumerations at the first unequal entry in the <i>usual</i> order of $\mathbb{R}$ ; it is not an order of ordinal types. |
| Enumerations of an ordered set $X$ | $\mathcal{W}(X), \mathcal{W}_\alpha(X)$                 | 02: $ X  = \lambda$ ; 06, 07: $ X  = \nu$ . Kept as printed in each source’s text.                                                                                                                               |
| Stratum of length $\alpha$         | $\mathcal{W}_\alpha$                                    | 02: $\mathcal{W}_\alpha$ ; 06: $\mathcal{W}_\beta$ .                                                                                                                                                             |
| Minimal-type slice                 | $\mathcal{W}_\kappa$                                    | 06: $\mathcal{P}(\backslash\text{PP})$ ; 07: $\mathcal{P}(\backslash\text{Perm})$ . False reading: source 07’s $\mathcal{P}$ is <i>not</i> the power set $\mathcal{P}(\mathbb{N})$ , which source 07 also uses.  |
| Kanovei–Shelah index               | $\mathcal{A}$ , fibres $\mathcal{A}_U$                  | 02: $\mathcal{A}$ ; 07: $\mathcal{A}(\backslash\text{KS})$ ; ultrafilter: 02 $U$ , 07 $D$ , here $U$ .                                                                                                           |
| Binary cube                        | $\mathbb{B}_\theta = (\{0, 1\}^\theta, <_{\text{lex}})$ | 06: $({}^\theta 2, <_{\text{lex}})$ ; 07: $2_{\text{lex}}^\theta$ . False reading: $2^\kappa$ without a subscript is a cardinal.                                                                                 |
| Real power                         | $\mathbb{R}_{\text{lex}}^\gamma$                        | 06: $({}^\gamma \mathbb{R}, <_{\text{lex}})$ ; 07: $\mathbb{R}_{\text{lex}}^\gamma$ . All functions, no finite-support restriction.                                                                              |
| Binary coding length               | $\ell_2(L)$                                             | 02 only: $\min\{\theta : L \hookrightarrow \mathbb{B}_\theta\}$ . It equals the game value $\text{ps}(L^{\text{dd}})$ of the point-separating report (its Theorem <code>thm:embedding</code> ).                  |

| Notion                                              | Here                                         | Sources, and readings to avoid                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------------------------|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Real representation rank                            | $\text{repr}_{\mathbb{R}}(L)$                | 07 only, after Giarlotta: $\min\{\alpha : L \hookrightarrow \mathbb{R}_{\text{lex}}^{\alpha}\}$ . Both invariants are kept:<br>$\text{repr}_{\mathbb{R}}(L) \leq \ell_2(L) \leq \omega \cdot \text{repr}_{\mathbb{R}}(L)$ ; they agree on $\mathcal{W}(\kappa^+)$ and on $\mathcal{W}_{\kappa}$ and $\mathcal{A}(\kappa)$ , but not in general ( $\text{repr}_{\mathbb{R}}(\mathbb{R}) = 1$ , $\ell_2(\mathbb{R}) = \omega$ ). |
| Equimorphism                                        | $L \simeq_e M$                               | 02: $\simeq_e$ ; 07: $\equiv_{\text{emb}}$ ; 06: “mutually order-embeddable”. False reading: not isomorphism.                                                                                                                                                                                                                                                                                                                  |
| Order embedding                                     | $L \hookrightarrow M$                        | 06: $\hookrightarrow_{\text{ord}}$ . Not continuous and not convex unless said.                                                                                                                                                                                                                                                                                                                                                |
| Reverse order                                       | $L^{\text{op}}$                              | 06: $L^*$ ; 07: $L^{\text{op}}$ .                                                                                                                                                                                                                                                                                                                                                                                              |
| Prefix cylinder                                     | $C_s, C_p$                                   | 02: $C_s$ ; 06: cone $C(p)$ ; 07: $[p]$ .                                                                                                                                                                                                                                                                                                                                                                                      |
| Residual set after a prefix $p$                     | $Y_p = X \setminus \text{ran } p$            | 02: $Y$ ; 06: $A_p$ ; 07: $S \setminus \text{ran } p$ . False reading: source 06’s $A_p$ is unrelated to the index $\mathcal{A}$ .                                                                                                                                                                                                                                                                                             |
| First disagreement                                  | $\delta, \Delta(e, f)$                       | 02, 07: $\delta$ ; 06: $\Delta(e, f)$ .                                                                                                                                                                                                                                                                                                                                                                                        |
| Lower/upper branch positions                        | $D^-(e), D^+(e)$                             | 06 and 07, same meaning. False reading: unrelated to source 02’s $D_{\kappa, m} = \mathbb{B}_{\kappa} \times_{\text{lex}} \{0, \dots, m-1\}$ .                                                                                                                                                                                                                                                                                 |
| One-sided point characters                          | $\chi^-(e), \chi^+(e)$                       | 06: $\chi^{\mp}(\backslash\text{cfm}, \backslash\text{cfp})$ ; 07: $\chi^{\mp}(\backslash\text{chm}, \backslash\text{chp})$ ; 02: “left and right local cofinalities”.                                                                                                                                                                                                                                                         |
| cf $\kappa$                                         | cf $\kappa$                                  | 07: $\theta$ . Renamed because $\theta$ is an ordinal variable here ( $\mathbb{B}_{\theta}$ ).                                                                                                                                                                                                                                                                                                                                 |
| $2^{<\kappa} = \sup_{\lambda < \kappa} 2^{\lambda}$ | $2^{<\kappa}$                                | 07: $\tau$ .                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Density, cellularity, weight                        | d, cell, w                                   | 02: d, cell; 06, 07: $d, c, w$ . Cellularity is written cell, never $c$ , to avoid $\mathfrak{c}$ .                                                                                                                                                                                                                                                                                                                            |
| Infinite regular cardinals $\leq \kappa$            | $\text{Reg}(\leq \kappa)$                    | 06: “infinite regular $\rho \leq \kappa$ ”; 07: $\text{Reg}_{\infty}(\leq \kappa)$ .                                                                                                                                                                                                                                                                                                                                           |
| Gap and its character                               | $(L, U)$ ,<br>(cf $L$ , ci $U$ );<br>GapChar | 06: $(L, U)$ ; 07: $(A, B)$ ; 02 (proposition 14.1): $(L, R)$ . A gap has no endpoint on either side; a jump is not a gap.                                                                                                                                                                                                                                                                                                     |
| Finite-condensation quotient                        | $\mathcal{W}/\sim_{\text{fin}}$              | 02: $\mathcal{W}/\sim_{\text{fin}}$ ; 06: $D$ ; 07: $\mathcal{W}_{\text{fc}}$ . False reading: source 06’s $D$ is not $D^{\pm}(e)$ .                                                                                                                                                                                                                                                                                           |
| Injections $\kappa \rightarrow \mathbb{R}$          | $\mathcal{I}$                                | 06: $J$ (not necessarily onto).                                                                                                                                                                                                                                                                                                                                                                                                |
| Well-ordering-based ultrafilter index               | $\mathcal{J}$                                | 07: $J$ (bijections $\kappa \rightarrow \mathcal{P}(\mathbb{N})$ ). False reading: sources 06 and 07 both write $J$ for these two different orders.                                                                                                                                                                                                                                                                            |
| Bounded surreal sign order                          | $S_{<\theta}$                                | 06 only.                                                                                                                                                                                                                                                                                                                                                                                                                       |

## 1.5 Non-claims of the three sources

[merge] Every limitation stated by a source is kept where the source states it; together they are:

- No source claims historical priority (02: [section B](#); 06: its abstract and Section 1.1; 07: its abstract and Section 13.2). Source 07 identifies its real-power strictness as Kuhlmann’s classical theorem; source 06 attributes the characteristic-function coding to the standard literature (Moore’s Section 2); source 02 cites Knoblauch for the terminology of intrinsic length.
- The proofs are written ZFC arguments. None has been refereed or checked in Lean or Rocq. The finite checks of all three sources test finite instances only ([section A](#)).
- Repository and literature searches were targeted, not exhaustive (02: [section B](#); 06 and 07: their shipped audits).
- The research questions are proposed continuations, not an authenticated list of published open problems.

- Source 06 does not classify all gaps of  $\mathcal{W}_\kappa$  through its missing-branch construction, does not describe the automorphism group, and does not claim that full choice is necessary (its audit, Section 5).
- No source gives an intrinsic criterion for an arbitrary linear order to embed into  $\mathcal{W}$ ; absence of  $\kappa^+$  and its reverse is necessary and is not claimed sufficient (02, 06 and 07 all say so).

Placement of this report in the Cardinals collection, beside the Lean development of the Cardinals project, confers no formal status: no statement of this report has a Lean or Rocq counterpart.

## 2 Three orders that must be distinguished

### 2.1 The literal order of all well-orderings

Throughout the real case, put

$$\kappa = |\mathbb{R}| = 2^{\aleph_0},$$

and identify this cardinal with its initial ordinal. For an ordinal  $\alpha$  of cardinality  $\kappa$ , let

$$\mathcal{W}_\alpha = \{e : \alpha \longrightarrow \mathbb{R} : e \text{ is a bijection}\}, \quad \mathcal{W} = \bigcup_{\kappa \leq \alpha < \kappa^+} \mathcal{W}_\alpha.$$

A well-order relation on  $\mathbb{R}$  determines exactly one member of  $\mathcal{W}$ : its increasing enumeration in that well-order. Conversely,  $e \in \mathcal{W}_\alpha$  defines the relation

$$r \prec_e s \iff e^{-1}(r) < e^{-1}(s).$$

Thus  $\mathcal{W}$  really represents all well-orderings, not merely their ordinal isomorphism types. The order in which we compare their entries is the *usual* order of the real numbers, not one of the well-orders being compared.

For distinct  $e, f \in \mathcal{W}$ , write

$$e <_{\text{lex}} f \iff e(\delta) < f(\delta), \quad \delta = \min\{\xi < \min(\text{dom } e, \text{dom } f) : e(\xi) \neq f(\xi)\}. \quad (1)$$

The existence of  $\delta$  is proved in [lemma 4.1](#). In particular, there is no choice of an “end-of-word” convention hidden in this definition.

One could instead sort by ordinal length first. That would be a different order: choosing one enumeration of each possible length would then give an increasing sequence of type  $\kappa^+$ . This is impossible for (1). Another different construction is to compare characteristic functions of well-order relations using a chosen well-order of  $\mathbb{R}^2$ . Neither alternative is used here.

### 2.2 The fixed-length restriction

The suborder  $\mathcal{W}_\kappa$  contains only those well-orderings whose ordinal type is the initial ordinal  $\kappa$ . It is a substantial suborder, but it is not all of  $\mathcal{W}$ . For example, types  $\kappa + 1$ ,  $\kappa + 2$ , and  $\kappa \cdot 2$  also occur in  $\mathcal{W}$ .

This distinction cannot be dismissed on cardinality grounds. Both  $\mathcal{W}_\kappa$  and  $\mathcal{W}$  have cardinality  $2^\kappa$ , yet  $\mathcal{W}$  does not embed into  $\mathcal{W}_\kappa$ . Indeed, no union of strata with a bounded set of ordinal lengths can contain an embedded copy of  $\mathcal{W}$ .

### 2.3 The index in the checked Kanovei–Shelah paper

Kanovei and Shelah’s 2004 paper constructs a definable, countably saturated elementary extension of the reals. Its Section 1 uses the index

$$\mathcal{A} = \{a : \kappa \longrightarrow \mathcal{P}(\mathbb{N}) : \text{ran } a \text{ is an ultrafilter on } \mathbb{N}\}, \quad (2)$$

ordered lexicographically; subsets of  $\mathbb{N}$  are themselves compared by their binary characteristic sequences [1, Section 1]. These maps need not be injective. The definition includes principal ultrafilters. For an ultrafilter  $U$ , write  $\mathcal{A}_U = \{a \in \mathcal{A} : \text{ran } a = U\}$ .

Thus the checked source uses fixed-length ultrafilter enumerations, not literally  $\mathcal{W}$ . The two constructions share the principle of ordering *presentations* without first selecting a presentation. We study both rather than identifying them.

## 2.4 Connection to ProveIt and to existing terminology

The ProveIt report *Every ordinal as a point-separating game value* contains a binary-cylinder argument and an order-embedding interpretation of lexicographic coding [4]. We use that line of thought, but give the needed cube argument independently in section 5. None of that report's claims about compact  $T_1$  spaces is needed here.

Minimum ordinal lengths of binary lexicographic representations are also studied under the name *intrinsic length* in work of Knoblauch [2]. We fix our precise convention in definition 5.4; no classification theorem from that literature is assumed. The targeted source search did not establish a prior full classification of the specific order  $\mathcal{W}$  considered here. This is a limitation of the bibliography, not a claim that such a classification is absent from the literature.

## 3 Main conclusions at a glance

Let  $\mathbb{B}_\theta = (\{0, 1\}^\theta, <_{\text{lex}})$ , for every ordinal  $\theta$ . An expression such as  $\mathbb{R}_{\text{lex}}^\gamma$  always has ordinal-indexed coordinates, read from left to right. We use  $L \simeq_e M$  for equimorphism.

**Theorem 3.1** (Main embedding spectrum). *The following statements hold.*

- (i)  $|\mathcal{W}| = |\mathcal{W}_\kappa| = |\mathcal{A}| = 2^\kappa$ . Every linear order of cardinality at most  $\kappa$  embeds into each of these orders.
- (ii) For every ordinal  $\beta$  and each  $L \in \{\mathcal{W}, \mathcal{W}_\kappa, \mathcal{A}\}$ ,

$$\beta \hookrightarrow L \iff \beta < \kappa^+.$$

*The same equivalence holds for reverse ordinals.*

- (iii)  $\mathcal{W}_\kappa \simeq_e \mathcal{A} \simeq_e \mathcal{A}_U \simeq_e \mathbb{B}_\kappa$  for every ultrafilter  $U$  on  $\mathbb{N}$ .
- (iv)  $\mathbb{B}_\theta \hookrightarrow \mathcal{W}$  exactly when  $\theta < \kappa^+$ , whereas  $\mathbb{B}_\theta \hookrightarrow \mathcal{W}_\kappa$  exactly when  $\theta \leq \kappa$ . The latter criterion also holds with  $\mathcal{A}$  in place of  $\mathcal{W}_\kappa$ .
- (v)  $\mathbb{R}_{\text{lex}}^\gamma \hookrightarrow \mathcal{W}$  exactly when  $\gamma < \kappa^+$ . It embeds into  $\mathcal{W}_\kappa$  or  $\mathcal{A}$  exactly when  $\gamma \leq \kappa$ .
- (vi)  $\mathcal{W}_{\text{lex}}^\gamma \hookrightarrow \mathcal{W}$  exactly when  $\gamma < \kappa^+$ . For  $0 < \gamma < \kappa^+$  the two orders are equimorphic. In contrast,  $\mathcal{W}_\kappa \times_{\text{lex}} \mathcal{W}_\kappa \not\hookrightarrow \mathcal{W}_\kappa$ , and  $\mathcal{A} \times_{\text{lex}} \mathcal{A} \not\hookrightarrow \mathcal{A}$ .

These assertions are proved in sections 6 to 8 and 10. [merge] Items (i)–(iii) are also the opening theorems of sources 06 (Theorem 1.1, 06:102) and 07 (Theorem 2.1, 07:187); items (iv)–(vi) are also in source 07's Theorem 2.1, in real-power form. The exact refinement for every fixed type, proved in section 9, is

$$\mathcal{W}_{\lambda+n} \simeq_e \mathbb{B}_\lambda \times_{\text{lex}} \{0, \dots, n! - 1\}, \quad |\lambda| = \kappa, \quad \lambda \text{ limit}, \quad n < \omega.$$

Moreover, two strata compare by their limit parts first and by their final factorial block sizes second; see corollary 9.4. The structural distinction is summarized in table 3. Here  $2^{<\kappa} = \sup_{\lambda < \kappa} 2^\lambda$ , with cardinal

| Invariant                                         | $\mathcal{W}_\kappa$                     | $\mathcal{A}$                                                            | $\mathcal{W}$                |
|---------------------------------------------------|------------------------------------------|--------------------------------------------------------------------------|------------------------------|
| Cardinality                                       | $2^\kappa$                               | $2^\kappa$                                                               | $2^\kappa$                   |
| Ordinals that embed                               | $\beta < \kappa^+$                       | $\beta < \kappa^+$                                                       | $\beta < \kappa^+$           |
| Binary coding length $\ell_2$                     | $\kappa$                                 | $\kappa$                                                                 | $\kappa^+$                   |
| Dense linear order                                | yes                                      | yes                                                                      | no                           |
| Least or greatest element                         | neither                                  | neither                                                                  | neither                      |
| Cofinality                                        | $\omega$                                 | $\text{cf}(\kappa)$                                                      | $\omega$                     |
| Cointinality                                      | $\omega$                                 | $\omega$                                                                 | $\omega$                     |
| Density and cellularity                           | $2^{<\kappa}$                            | $2^{<\kappa}$                                                            | $2^\kappa$                   |
| Isolated points                                   | none                                     | none                                                                     | $2^\kappa$                   |
| Self-dual                                         | yes                                      | no                                                                       | yes                          |
| Unfilled $(\omega, \omega)$ cut                   | yes                                      | yes                                                                      | yes                          |
| Absorbs its lexicographic square                  | no                                       | no                                                                       | yes                          |
| [merge] <i>Added from sources 06 and 07</i>       |                                          |                                                                          |                              |
| Weight                                            | $2^{<\kappa}$                            | $2^{<\kappa}$                                                            | $2^\kappa$                   |
| Real representation rank $\text{repr}_\mathbb{R}$ | $\kappa$                                 | $\kappa$                                                                 | $\kappa^+$                   |
| Point characters $(\chi^-, \chi^+)$               | $(\text{cf } \kappa, \text{cf } \kappa)$ | $(\text{cf } \kappa, \text{cf } \kappa)$                                 | (15)                         |
| Gap characters                                    | <a href="#">theorem 20.6</a>             | include $(\omega, \omega)$ ,<br>$(\text{cf } \kappa, \text{cf } \kappa)$ | <a href="#">theorem 18.5</a> |
| Uncountable balanced gaps                         | yes                                      | yes                                                                      | none                         |
| Isolated points dense                             | —                                        | —                                                                        | yes                          |
| Zero-dimensional                                  | yes                                      | yes                                                                      | yes                          |
| Order-embeds into $\mathcal{W}$ with convex image | no                                       | —                                                                        | yes                          |

Table 3: Equal ordinal spectra do not imply equal lexicographic complexity. The assertions about  $\mathcal{A}$  refer to the full index (2). [merge] The rows below the rule are those of the summary tables of sources 06 (its Appendix A) and 07 (its Section 2) not already above; the gap spectrum of  $\mathcal{A}$  is not classified.

exponents; cell is the supremum of the sizes of pairwise disjoint families of nonempty open sets. All topologies in the table are order topologies.

Under CH the ordinal cutoff is  $\omega_2$ : every ordinal below  $\omega_2$  embeds, including  $\omega_1 \cdot 2$  and  $\omega_1^2$ , but  $\omega_2$  does not. Without CH, the answer is still exactly  $\kappa^+$ , not an absolute cutoff at  $\omega_2$ . For example, if  $\mathfrak{c} = \aleph_2$ , then  $\omega_2$  embeds and  $\omega_3$  does not. These are conditional substitutions into the ZFC theorem, not additional assumptions in its proof.

## 4 Exhaustive enumeration orders

Several central facts hold for any infinite linearly ordered set  $X$ , not just the real line. Write  $\lambda = |X|$  and let  $\mathcal{W}(X)$  consist of all bijections from ordinals of cardinality  $\lambda$  onto  $X$ , with the same first-difference comparison.

**Lemma 4.1** (Prefix-freeness). *No member of  $\mathcal{W}(X)$  is a proper initial segment of another. Consequently two distinct members have a first disagreement in the common part of their domains, and first-difference comparison is a linear order.*

*Proof.* Suppose  $e : \alpha \rightarrow X$  is an initial segment of the injective map  $f : \beta \rightarrow X$ , where  $\alpha < \beta$ . Since  $e$  is onto, the value  $f(\alpha)$  already occurs at some coordinate below  $\alpha$ , contrary to injectivity. If two enumerations agree throughout their common domain, one is therefore equal to the other.

For transitivity, suppose  $e < f < g$ , with first disagreements  $\delta$  and  $\varepsilon$ . Before  $\min(\delta, \varepsilon)$  all three

| Embedded or excluded object                                                    | Result                                                                             |
|--------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Every linear order $L$ with $ L  \leq \kappa$                                  | embeds in every admissible slice $\mathcal{W}_\beta$ (theorem 15.1)                |
| $\mathbb{B}_\gamma$ and $\mathbb{R}_{\text{lex}}^\gamma$ , $\gamma < \kappa^+$ | embed in $\mathcal{W}$ (theorem 8.2 and corollary 8.6)                             |
| $\alpha$ or $\alpha^{\text{op}}$                                               | embeds exactly when $\alpha < \kappa^+$ (theorem 6.4)                              |
| $S_{<\theta}$ , $\theta < \kappa^+$                                            | embeds in $\mathcal{W}$ (theorem 22.1)                                             |
| $S_{<\kappa^+}$                                                                | does not embed, although its size is $2^\kappa$                                    |
| All orders of size $2^\kappa$                                                  | not universally embedded ( $\kappa^+$ does not embed)                              |
| Full $\mathcal{W}$ into $\mathcal{W}_\kappa$                                   | excluded in ZFC (corollary 8.4); source 06 had it only if $2^{<\kappa} < 2^\kappa$ |
| $\mathcal{W}_\kappa$ into $\mathcal{W}$ with convex image                      | excluded (corollary 20.7)                                                          |

Table 4: [merge] Embedded and excluded objects; source 06’s second summary table (its Appendix A), with its row on  $\mathcal{W}$  into  $\mathcal{W}_\kappa$  updated by the in-batch answer and the last row added from source 07.

agree. If the coordinates are unequal, the earlier comparison gives  $e < g$  there. If they are equal,  $e(\delta) < f(\delta) < g(\delta)$  gives the same conclusion. Trichotomy follows from the linear order on  $X$ .  $\square$

[merge] Also source 06, Lemma 2.1 and Proposition 2.3 (06:171, 06:198), and source 07, Section 1.2 (07:132–141), by the same argument.

**Definition 4.2.** For an injective ordinal sequence  $s$  of elements of  $X$ , its cylinder is

$$C_s = \{e \in \mathcal{W}(X) : s \subseteq e\}.$$

We consider only prefixes that have an exhaustive extension. An exhaustive prefix has a singleton cylinder.

**Lemma 4.3** (Residual-set geometry). *Every cylinder is convex. If  $s$  is not exhaustive and  $Y = X \setminus \text{ran } s$ , then concatenation gives an order isomorphism*

$$\mathcal{W}(Y) \longrightarrow C_s, \quad t \longmapsto s \frown t.$$

*For finite  $Y$  of size  $m$ , the cylinder has exactly  $m!$  points.*

*Proof.* A point between two extensions of  $s$  cannot first disagree with  $s$  before its end: that disagreement would put it on the same strict side of both extensions. The remaining assertions follow because every exhaustive extension must enumerate precisely  $Y$  after  $s$ , and first differences in the tail are unchanged by concatenation.  $\square$

[merge] Sources 06 (Lemma 3.2, 06:246) and 07 (Lemma 3.1, 07:259) prove the same lemma and add its next-value decomposition: if  $Y_s = X \setminus \text{ran } s$  is nonempty, then

$$C_s \cong \sum_{a \in Y_s}^{<X} \mathcal{W}(Y_s \setminus \{a\}), \quad (3)$$

a lexicographic sum over  $Y_s$  in its given order, whose summand indexed by  $a$  is the child cylinder  $C_{s \smallfrown \langle a \rangle}$ , with  $\mathcal{W}(\emptyset)$  a one-point order. Sorting the members of  $C_s$  by their next entry proves it. Source 06 also observes that concatenation uses ordinal addition, so no ordinal subtraction convention is needed (06:242–244).

In particular, a long prefix may have already used all but two or three reals. This is why the density of the usual real order does *not* imply density of  $\mathcal{W}$ .

**Proposition 4.4** (Cardinality). *For every infinite linearly ordered set  $X$  of cardinality  $\lambda$ ,*

$$|\mathcal{W}(X)| = |\mathcal{W}_\lambda(X)| = 2^\lambda.$$

*Proof.* A well-order relation is a subset of  $X^2$ , giving the upper bound  $2^{|X^2|} = 2^\lambda$ . For the lower bound, partition  $X$  into  $\lambda$  two-element sets  $\{a_\xi, b_\xi\}$ , with  $a_\xi < b_\xi$ . For  $x \in \{0, 1\}^\lambda$ , enumerate the pair at coordinate  $\xi$  as  $(a_\xi, b_\xi)$  if  $x(\xi) = 0$ , and as  $(b_\xi, a_\xi)$  otherwise. The concatenation has length  $2 \cdot \lambda = \lambda$  and enumerates all of  $X$ . Different bit sequences give different enumerations. In fact, this is an order embedding  $\mathbb{B}_\lambda \hookrightarrow \mathcal{W}_\lambda(X)$ .  $\square$

[merge] Also source 06, Lemma 4.1 and Corollary 4.2 (06:331, 06:363), and source 07, Lemma 4.1 and Corollary 4.6 (07:329, 07:417), by the same pair-swap code. Source 06 notes that the multiplication in  $2 \cdot \nu = \nu$  is ordinal multiplication, the two positions of a block being the less significant coordinates (06:358–361). Two definitions that are *not* used are ruled out by source 06 (06:220–230): comparing characteristic functions of relations at the “least real where they differ” fails, since a set of real disagreement coordinates need not have a least member; and the order does not descend to isomorphism types, since two enumerations of any two admissible types can be made to compare either way by their first values.

## 5 Binary cubes and the cylinder obstruction

**Lemma 5.1** (Cut coding). *Every linear order  $L$  of cardinality at most an infinite cardinal  $\lambda$  embeds into  $\mathbb{B}_\lambda$ .*

*Proof.* Choose a well-ordered list  $(t_\xi)_{\xi < \mu}$  of the elements of  $L$ , where  $\mu \leq \lambda$ . For  $x \in L$ , put

$$c_x(\xi) = \begin{cases} 1 & t_\xi \leq_L x, \\ 0 & t_\xi >_L x, \end{cases} \quad (\xi < \mu),$$

and put  $c_x(\xi) = 0$  for  $\mu \leq \xi < \lambda$ . If  $x < y$ , then  $c_x \leq c_y$  coordinatewise, and the coordinate listing  $y$  distinguishes them. Their first unequal bits are consequently 0 and 1.  $\square$

[merge] Also source 06, Lemma 4.3 (06:373), and source 07, Lemma 4.2 (07:339), coding  $x$  by the characteristic function of  $\{t : t < x\}$  instead of  $\{t : t \leq x\}$ ; the argument is the same. Source 06 attributes this coding to the standard literature (Moore’s representation of Aronszajn lines, [5, Section 2]) and stresses that it uses a well-ordering of the coordinate set, not one compatible with  $L$  (06:385–387).

**Lemma 5.2** (A cut preserves a full child). *Let  $C_s$  be a full prefix cylinder in  $\mathbb{B}_\theta$ , with  $\text{dom } s < \theta$ . If  $D$  is an initial segment of  $C_s$ , then either  $C_{s \smallfrown 0} \subseteq D$  or  $C_{s \smallfrown 1} \subseteq C_s \setminus D$ . The same assertion holds for cylinders  $C_s \times F$  in  $\mathbb{B}_\theta \times_{\text{lex}} F$ , where  $F$  is any nonempty linear order.*

*Proof.* If the first inclusion fails, choose a point of the lower child outside  $D$ . Every point of the upper child lies above it, hence is also outside the initial segment. The additional terminal factor does not change the ordering of the two children.  $\square$

[merge] The next theorem is Corollary cor:strict of the ProveIt report *Every ordinal as a point-separating game value* (ordinal\_separation.tex, lines 633–641), where it is proved through the point-separating game: an embedding  $\mathbb{B}_\alpha \hookrightarrow \mathbb{B}_\beta$  with  $\beta < \alpha$  would give a winning strategy of length  $\beta$ , contrary to that report’s exact game value. Source 02’s cylinder-fusion proof below is a second route, which avoids the game.

**Theorem 5.3** (Strict hierarchy of binary cubes). *For all ordinals  $\alpha, \beta$ ,*

$$\mathbb{B}_\alpha \hookrightarrow \mathbb{B}_\beta \iff \alpha \leq \beta.$$

*Proof.* Padding with zero bits proves the forward-length direction. Suppose instead that  $f : \mathbb{B}_\alpha \hookrightarrow \mathbb{B}_\beta$  and  $\beta < \alpha$ . Construct a source prefix  $s_\xi$  of length  $\xi$ , for  $\xi \leq \beta$ , so that  $f$  is constant on its first  $\xi$  target coordinates throughout  $C_{s_\xi}$ .

At a successor step, all previous target bits are fixed. The inverse image of target bit 0 at coordinate  $\xi$ , restricted to the current cylinder, is an initial segment. By lemma 5.2, one source child lies entirely on one side of this cut. Retain that child. At a limit stage take the union of the retained prefixes. Its full cylinder is the intersection of the earlier cylinders and remains nonempty. This is a fusion argument, not an appeal to finite-intersection compactness.

After  $\beta$  steps a source cylinder of length  $\beta < \alpha$  still contains at least two points, while their images agree on all  $\beta$  coordinates. This contradicts injectivity.  $\square$

**Definition 5.4** (Binary lexicographic length). For a set-sized linear order  $L$ , define

$$\ell_2(L) = \min\{\theta : L \hookrightarrow \mathbb{B}_\theta\}.$$

This minimum exists in ZFC by lemma 5.1. A singleton and the empty order have length zero. Equimorphic orders have the same binary length.

**Lemma 5.5** (Real and binary coordinates).  $\mathbb{R} \simeq_e \mathbb{B}_\omega$ . More generally,

$$\mathbb{R}_{\text{lex}}^\gamma \simeq_e \mathbb{B}_{\omega \cdot \gamma}$$

for every ordinal  $\gamma$ .

*Proof.* List the rationals as  $(q_n)_{n < \omega}$  and send  $r$  to  $j(r)(n) = 1$  exactly when  $q_n < r$ . This is an order embedding by the same coordinatewise argument as in lemma 5.1. In the other direction, send a bit sequence  $x$  to

$$T(x) = \sum_{n < \omega} \frac{2x(n)}{3^{n+1}}.$$

At the first unequal digit the contribution dominates the sum of all later possible differences, so this map is strictly increasing. Applying these maps coordinatewise and concatenating blocks of length  $\omega$  proves the last statement. The ordinal length is  $\omega \cdot \gamma$ , not cardinal multiplication and not  $\gamma \cdot \omega$ .  $\square$

For the uncountable initial ordinal  $\kappa = 2^{\aleph_0}$ ,

$$\omega \cdot \kappa = \kappa. \tag{4}$$

Indeed, each  $\omega \cdot \xi$  for  $\xi < \kappa$  has cardinality less than  $\kappa$ , and their supremum is at least  $\kappa$  and at most  $\kappa$ . No regularity of  $\kappa$  is assumed.

[merge] The codes  $j$  and  $T$  of lemma 5.5 are also source 07's (07:436–458) and source 06's (06:433–445), and (4) is part of source 07's Lemma 3.2 (07:288), which also records  $\kappa^\kappa = 2^\kappa$ ,  $\sup_{\lambda < \kappa} \kappa^\lambda = 2^{<\kappa}$  and cf  $\kappa > \omega$ .

**Corollary 5.6** (Strictness of real powers; Kuhlmann). For all ordinals  $\alpha, \beta$ ,

$$\mathbb{R}_{\text{lex}}^\alpha \hookrightarrow \mathbb{R}_{\text{lex}}^\beta \iff \alpha \leq \beta.$$

*First proof* (from source 02's lemmas). By lemma 5.5,  $\mathbb{R}_{\text{lex}}^\alpha \simeq_e \mathbb{B}_{\omega \cdot \alpha}$  and  $\mathbb{R}_{\text{lex}}^\beta \simeq_e \mathbb{B}_{\omega \cdot \beta}$ . By theorem 5.3, an embedding exists exactly when  $\omega \cdot \alpha \leq \omega \cdot \beta$ , that is, when  $\alpha \leq \beta$ , since left multiplication by  $\omega$  is strictly increasing on ordinals.  $\square$

*Second route* (source 07, Theorem 5.2, 07:477). Extension by zero gives the forward construction when  $\alpha \leq \beta$ . For the converse, suppose  $\alpha > \beta$  and  $F : \mathbb{R}_{\text{lex}}^\alpha \hookrightarrow \mathbb{R}_{\text{lex}}^\beta$ . Recursively choose a source prefix  $s_\delta$  of length  $\delta$  and a target prefix  $t_\delta$  of the same length, for  $\delta \leq \beta$ , such that every image of a source extension of  $s_\delta$  extends  $t_\delta$ .

At step  $\delta < \beta$ , subdivide the source cylinder according to its next real value  $r$ . Images of these subcylinders are ordered by  $r$ . Their  $\delta$ -th target-coordinate sets are consequently ordered subsets of  $\mathbb{R}$ . At most countably many such sets contain two distinct values: for each such set choose a rational strictly between two of its values, and these rationals must be distinct. Choose an  $r$  outside the countable exceptional set. The coordinate is constant on this subcylinder, so it determines  $t_{\delta+1}$ .

At limits take unions. A source prefix of length at most  $\beta$  always has extensions, since  $\beta < \alpha$ . At the final step all extensions of  $s_\beta$  have exactly the same image, namely  $t_\beta$ . There are at least two such source extensions. This contradicts injectivity.  $\square$

Source 07 identifies this ordinal-exponent theorem as classical: it is the ordinal case of Kuhlmann's embedding result [6, Corollary 2.4], and source 07's coordinate fusion is its own proof of the case needed. The point-separating report has only the binary case.

## 6 The exact ordinal spectrum

The upper bound is the essential point. Cardinality alone would permit well-ordered suborders far larger than  $\kappa$ .

**Lemma 6.1** (Eventual constancy). *Let  $\lambda$  be an infinite cardinal. A nondecreasing sequence of length  $\lambda^+$  with values in a linear order of cardinality at most  $\lambda$  is eventually constant. The same holds for a nonincreasing sequence.*

*Proof.* There are at most  $\lambda$  distinct values. Take the first occurrence of each value. Their supremum, increased by one if necessary, is below  $\lambda^+$ : the supremum of at most  $\lambda$  ordinals below  $\lambda^+$  has cardinality at most  $\lambda$ . After this bound no new value can first occur. Monotonicity prevents a change to a value that already occurred before the current value. Hence the sequence is constant on a tail.  $\square$

[merge] Also source 06, Lemma 5.1 (06:454), for a regular  $\lambda$  and a target of cardinality below  $\lambda$  ("some fiber is unbounded, and a monotone unbounded fiber contains a tail"), and source 07, Lemma 4.3 (07:355).

**Theorem 6.2** (Exhaustion forbids a successor-cardinal chain). *For every infinite linearly ordered set  $X$  of cardinality  $\lambda$ ,  $\mathcal{W}(X)$  contains neither  $\lambda^+$  nor its reverse.*

*Proof.* Suppose  $(e_\eta)_{\eta < \lambda^+}$  is strictly increasing. Recursively, for each  $\xi < \lambda^+$ , we shall obtain an injective prefix  $s_\xi$  of length  $\xi$  shared by every  $e_\eta$  on a tail of the index set.

The empty prefix starts the recursion. At a limit  $\xi < \lambda^+$ , the supremum of the preceding tail bounds is still below  $\lambda^+$ , since  $|\xi| \leq \lambda$ . The union of the prefixes is injective and is shared by the resulting common tail.

Suppose the shared prefix  $s_\xi$  is exhaustive. Prefix-freeness forces all members of this tail to equal  $s_\xi$ , contrary to strict increase. Thus it is not exhaustive. Every enumeration on the tail has a value at coordinate  $\xi$ , and these values are nondecreasing because the earlier coordinates are fixed. By lemma 6.1, they are constant on a further tail. Their common value is not in  $\text{ran } s_\xi$ , by injectivity. Adjoin it to get  $s_{\xi+1}$ .

If the recursion never stops, its union is an injection  $\lambda^+ \rightarrow X$ , a contradiction. If it stops by exhaustion, we already have a contradiction. Replacing nondecreasing by nonincreasing proves the reverse-order assertion.  $\square$

[merge] Also source 06, Theorem 5.2 (06:465), and source 07, Theorem 4.4 (07:367), by the same stabilization of a common prefix. Source 06 remarks (its Remark 5.3) that the regularity used at limit stages is a ZFC fact about the successor cardinal  $\nu^+$ , not an assumption about  $\nu$ : a union of at most  $\nu$  ordinals below  $\nu^+$  has cardinality at most  $\nu \cdot \nu = \nu$ ; the continuum  $\kappa$  itself need not be regular. Source 07 notes that no separability of the alphabet is needed.

*Remark 6.3.* It would be incorrect to argue that the sequence of enumeration lengths is bounded below  $\lambda^+$ . A long family can have unbounded lengths. The proof instead stabilizes a common prefix and uses the fact that an exhaustive prefix cannot be extended. This is exactly the extra step needed for the full order, rather than a fixed-length product.

**Theorem 6.4** (Universal small orders and exact ordinals). *For every infinite linearly ordered set  $X$  of cardinality  $\lambda$ :*

- (a) *Every linear order of cardinality at most  $\lambda$  embeds into  $\mathcal{W}_\lambda(X)$  and hence into  $\mathcal{W}(X)$ .*
- (b) *For every ordinal  $\beta$ ,*

$$\beta \hookrightarrow \mathcal{W}(X) \iff \beta < \lambda^+.$$

*The same is true for reverse ordinals and for  $\mathcal{W}_\lambda(X)$ .*

*Proof.* Compose cut coding with the pair-swap embedding in [proposition 4.4](#). An ordinal has cardinality at most  $\lambda$  exactly when it is below  $\lambda^+$ . Larger ordinals contain  $\lambda^+$  as an initial segment and are excluded by [theorem 6.2](#). Reverse orders are covered by the same small-order universality and the reverse half of that theorem.  $\square$

For the real line, this proves the precise ordinal answer in [theorem 3.1](#). Notice the distinction between the external lexicographic order on well-orderings and the well-order carried by any one of its elements. The external order contains long ascending and descending chains even though every element individually codes a well-order.

[merge] Also source 06, Corollary 5.4 (06:501), which states the spectrum for every slice  $\mathcal{W}_\beta(X)$  of admissible length and notes that there is no largest embedded ordinal, and source 07, Corollary 4.5 (07:394). Source 07 adds (07:408–415) that  $\kappa + 1$ ,  $\kappa \cdot 2$  and  $\kappa \cdot \kappa$  all embed while  $\kappa^+$  does not, and that the map sending a well-order to its order type is onto  $[\kappa, \kappa^+)$  but not monotone, so choosing one well-order of each type cannot give an increasing  $\kappa^+$ -sequence. Source 06 adds (its Example 4.5) that, without CH,  $\mathcal{W}$  already contains  $\omega_1$ , its reverse, every linear order on  $\omega_1$ , every ordinal of cardinality  $\mathfrak{c}$  and every continuum-sized ordered field or linear order, as orders, without preserving field operations; and (06:530–531) that  $\mathcal{W}$  is not universal for orders of size  $2^\kappa$ , since  $\kappa^+ \leq 2^\kappa$ .

## 7 Fixed length and the actual ultrafilter index

**Theorem 7.1** (Fixed-length equivalences). *For every ultrafilter  $U$  on  $\mathbb{N}$ ,*

$$\mathcal{W}_\kappa \simeq_e \mathbb{B}_\kappa \simeq_e \mathcal{A}_U \simeq_e \mathcal{A}.$$

*Consequently, for any set-sized linear order  $L$ ,*

$$L \hookrightarrow \mathcal{W}_\kappa \iff L \hookrightarrow \mathcal{A} \iff \ell_2(L) \leq \kappa. \quad (5)$$

*Proof.* The pair-swap construction gives  $\mathbb{B}_\kappa \hookrightarrow \mathcal{W}_\kappa$ . In the other direction,  $\mathcal{W}_\kappa$  is a suborder of  $\mathbb{R}_{\text{lex}}^\kappa$ , which embeds into  $\mathbb{B}_{\omega \cdot \kappa} = \mathbb{B}_\kappa$  by [lemma 5.5](#).

Every ultrafilter  $U$  has cardinality  $\kappa$ . To see this, partition  $\mathcal{P}(\mathbb{N})$  into complementary pairs  $\{S, \mathbb{N} \setminus S\}$ . There are  $\kappa$  such pairs, and an ultrafilter contains exactly one member of each pair. Choose a surjection  $u : \kappa \rightarrow U$  and two distinct members  $u_0 < u_1$  of  $U$ , in binary lexicographic order. For  $x \in \{0, 1\}^\kappa$ , set

$$a_x(2 \cdot \xi) = u_x(\xi), \quad a_x(2 \cdot \xi + 1) = u(\xi) \quad (\xi < \kappa). \quad (6)$$

Since  $2 \cdot \kappa = \kappa$ , these are maps of the required length. Their ranges are exactly  $U$ , and the first changed coding bit gives the first changed value. Thus  $\mathbb{B}_\kappa \hookrightarrow \mathcal{A}_U$ .

Finally,  $\mathcal{A}_U \subseteq \mathcal{A} \subseteq (\mathcal{P}(\mathbb{N}))^\kappa$ . Flattening the characteristic sequences of subsets of  $\mathbb{N}$  turns the last order into  $\mathbb{B}_{\omega \cdot \kappa} = \mathbb{B}_\kappa$ . Composing these embeddings proves all the equivalences and (5).  $\square$

The fact that even one fixed fiber  $\mathcal{A}_U$  has this universality is an order-theoretic statement; it does not eliminate the role of all ultrafilters in the definable model construction. In particular, the embeddings in (6) use a chosen enumeration of  $U$ .

[merge] Also source 06, Proposition 4.6 (06:425), and source 07, (4) with its display (5.3) (07:456) and Theorem 11.2 (07:1129), where the ultrafilter coding is the same and the cardinality of an ultrafilter is its Lemma 11.1. Source 07 states the equimorphism with real coordinates as well:  $\mathcal{W}_\kappa \simeq_e \mathcal{A} \simeq_e \mathbb{B}_\kappa \simeq_e \mathbb{R}_{\text{lex}}^\kappa$ . Both stress that equimorphism is not isomorphism:  $\mathbb{B}_\kappa$  has endpoints, whereas  $\mathcal{W}_\kappa$  has none (06:447–448), and the characteristic-function order on  $\mathcal{P}(\mathbb{N})$  has endpoints and adjacent pairs, unlike the usual order of  $\mathbb{R}$ , so a bijection between these alphabets is not an order isomorphism (source 07, Remark 1.2).

**Corollary 7.2.** *The binary lengths of  $\mathcal{W}_\kappa$ ,  $\mathcal{A}$ , and  $\mathcal{A}_U$  are all  $\kappa$ . Their embedded ordinals and reverse ordinals are exactly those below  $\kappa^+$ . Moreover,*

$$\mathbb{B}_\theta \hookrightarrow \mathcal{W}_\kappa \iff \theta \leq \kappa, \quad \mathbb{R}_{\text{lex}}^\gamma \hookrightarrow \mathcal{W}_\kappa \iff \gamma \leq \kappa,$$

with the same criteria for  $\mathcal{A}$  and  $\mathcal{A}_U$ .

*Proof.* Use theorems 5.3 and 7.1. For real powers,  $\omega \cdot \gamma \leq \kappa$  is equivalent to  $\gamma \leq \kappa$ : (4) proves one direction, while  $\gamma \geq \kappa + 1$  implies  $\omega \cdot \gamma \geq \kappa + \omega > \kappa$ . The ordinal spectrum follows from equimorphism with  $\mathcal{W}_\kappa$  and theorem 6.4.  $\square$

## 7.1 An exact finite-tail refinement

The last entry of an exhaustive enumeration is forced by all preceding entries. Several final entries, however, retain a genuine finite amount of lexicographic information. This gives a more precise hierarchy than the cardinality or ordinal spectrum alone detects.

For a positive integer  $m$ , write

$$D_{\kappa, m} = \mathbb{B}_\kappa \times_{\text{lex}} \{0, \dots, m-1\}.$$

The binary sequence is the first, more significant coordinate.

**Theorem 7.3** (Finite-tail equivalence). *For every nonnegative integer  $n$ ,*

$$\mathcal{W}_{\kappa+n} \simeq_e D_{\kappa, n!}. \quad (7)$$

In particular,

$$\mathcal{W}_{\kappa+1} \simeq_e \mathcal{W}_\kappa, \quad \mathcal{W}_{\kappa+2} \simeq_e \mathbb{B}_{\kappa+1}.$$

*Proof.* Choose a fixed  $n$ -element set  $F \subseteq \mathbb{R}$ . Partition  $\mathbb{R} \setminus F$  into  $\kappa$  ordered pairs, and use the pair-swap construction for the first  $\kappa$  entries. Append the  $r$ -th permutation of  $F$  in lexicographic order, for  $0 \leq r < n!$ . This embeds  $D_{\kappa, n!}$  into  $\mathcal{W}_{\kappa+n}$ . The case  $n = 0$  uses the unique empty permutation.

Conversely, code the first  $\kappa$  entries of an enumeration by a member of  $\mathbb{B}_\kappa$ , using rational-cut codes and (4). When two such prefixes agree, the remaining set of  $n$  reals is the same. Record the rank, between 0 and  $n! - 1$ , of the final permutation of those reals. The prefix code followed by this rank is an order embedding into  $D_{\kappa, n!}$ .  $\square$

**Theorem 7.4** (Finite-fiber capacity). *For every positive finite  $m$ ,*

$$\ell_2(D_{\kappa, m}) = \kappa + \lceil \log_2 m \rceil, \quad (8)$$

$$\mathbb{B}_\theta \hookrightarrow D_{\kappa, m} \iff \theta \leq \kappa + \lfloor \log_2 m \rfloor. \quad (9)$$

Hence the same formulas hold for  $\mathcal{W}_{\kappa+n}$  with  $m = n!$ .

*Proof.* A finite increasing binary code for  $\{0, \dots, m-1\}$  proves the upper bound in (8). Suppose  $D_{\kappa, m} \hookrightarrow \mathbb{B}_{\kappa+j}$  for a finite  $j$ . Read the first  $\kappa$  target bits and apply lemma 5.2 successively to source cylinders  $C_s \times \{0, \dots, m-1\}$ . After  $\kappa$  steps, all  $m$  points over one source bit sequence have the same first  $\kappa$  target bits. They must fit into a fiber of size  $2^j$ . Thus  $m \leq 2^j$ . Targets of length below  $\kappa$  are already excluded by their failure to contain  $\mathbb{B}_\kappa$ . This proves (8).

For (9),  $2^j \leq m$  gives the direct embedding  $\mathbb{B}_{\kappa+j} = \mathbb{B}_\kappa \times_{\text{lex}} \{0, 1\}^j \hookrightarrow D_{\kappa, m}$ . Conversely, given an embedding in this direction, fuse source cylinders while reading the first  $\kappa$  bits of the target's  $\mathbb{B}_\kappa$  coordinate. The final source cylinder contains  $2^j$  points, but its target fiber has only  $m$  points. Hence  $2^j \leq m$ . Any ordinal length greater than  $\kappa + \lfloor \log_2 m \rfloor$  contains the next forbidden finite extension as an initial coordinate block, so all larger lengths are also excluded.  $\square$

**Example 7.5.** For  $n = 3$  there are six final permutations. Thus

$$\ell_2(\mathcal{W}_{\kappa+3}) = \kappa + 3, \quad \mathbb{B}_\theta \hookrightarrow \mathcal{W}_{\kappa+3} \iff \theta \leq \kappa + 2.$$

The least coding length need not itself be a cube that embeds back. This small finite-fiber example foreshadows the much larger discrepancy for  $\mathcal{W}$ : its coding length is  $\kappa^+$ , but  $\mathbb{B}_{\kappa^+}$  does not embed.

## 8 Unbounded lengths and full lexicographic complexity

**Lemma 8.1** (Arbitrarily long small-cardinal cubes). *Let  $X$  be an infinite linearly ordered set of cardinality  $\lambda$ . For every ordinal  $\theta < \lambda^+$ ,*

$$\mathbb{B}_\theta \hookrightarrow \mathcal{W}(X).$$

*Proof.* Choose pairwise disjoint pairs  $\{a_\xi, b_\xi\}$  for  $\xi < \theta$ , ordered so that  $a_\xi < b_\xi$ , leaving a residual set of cardinality  $\lambda$ . This is possible because  $|\theta| \leq \lambda$  and  $\lambda$  is infinite. For each bit sequence, list each pair in its selected orientation, in the ordinal order of  $\theta$ , and then append one fixed enumeration of the residual set. The total length is  $2 \cdot \theta + \lambda$ , an ordinal of cardinality  $\lambda$ . First changed bits give first changed entries.  $\square$

[merge] Also the second half of source 06's Lemma 4.1 (06:351) and, with real coordinates, source 07's Proposition 5.1 (07:460).

**Theorem 8.2** (The full coding length). *For every infinite linearly ordered set  $X$  of cardinality  $\lambda$ ,*

$$\ell_2(\mathcal{W}(X)) = \lambda^+, \quad \mathbb{B}_\theta \hookrightarrow \mathcal{W}(X) \iff \theta < \lambda^+. \quad (10)$$

In particular,  $\mathcal{W}$  embeds into  $\mathbb{B}_{\kappa^+}$ , but the reverse embedding does not exist.

*Proof.* Embed  $X$  into  $\mathbb{B}_\lambda$  by lemma 5.1. For an enumeration of length  $\alpha < \lambda^+$ , concatenate its coordinate codes into a binary sequence of length  $\lambda \cdot \alpha$ , and pad with zeros to length  $\lambda^+$ . Since  $|\lambda \cdot \alpha| \leq \lambda$ , this is a legitimate shorter initial block. Prefix-freeness ensures that two different exhaustive enumerations disagree before either ends, so padding introduces no ambiguity. This proves the upper bound  $\ell_2(\mathcal{W}(X)) \leq \lambda^+$ .

If  $\mathcal{W}(X)$  embedded into  $\mathbb{B}_\delta$  for  $\delta < \lambda^+$ , then lemma 8.1 would give  $\mathbb{B}_{\delta+1} \hookrightarrow \mathbb{B}_\delta$ , contrary to theorem 5.3. This proves equality.

The positive cube embeddings are lemma 8.1. To exclude  $\mathbb{B}_{\lambda^+}$ , note that this cube contains a copy of  $\lambda^+$ : map  $\eta < \lambda^+$  to the sequence with value 1 below  $\eta$  and 0 from  $\eta$  onward. This would contradict theorem 6.2. Larger cubes contain this forbidden cube.  $\square$

[merge] Source 07 proves the real-coordinate counterpart,  $\text{repr}_{\mathbb{R}}(\mathcal{W}) = \kappa^+$  and  $\text{repr}_{\mathbb{R}}(\mathcal{W}_\kappa) = \text{repr}_{\mathbb{R}}(\mathcal{A}) = \kappa$  (theorem 19.1; its Theorem 5.3, 07:505), from corollary 5.6 by the same padding argument; the two invariants are compared in table 2.

For the real case, the upper coding construction can use countable rational-cut blocks rather than blocks of length  $\kappa$ . Thus an enumeration of length  $\alpha$  is first coded at length  $\omega \cdot \alpha$ . The ultimate common padding length is still  $\kappa^+$ .

**Corollary 8.3** (No bounded-length universal suborder). *Fix  $\tau < \kappa^+$ . Then*

$$\mathcal{W} \not\hookrightarrow \bigcup_{\alpha \leq \tau} \mathcal{W}_\alpha.$$

*Consequently, the image of every order embedding  $\mathcal{W} \hookrightarrow \mathcal{W}$  has an unbounded set of enumeration lengths in  $\kappa^+$ .*

*Proof.* The bounded union embeds into  $\mathbb{B}_{\omega \cdot (\tau+1)}$  by coding and padding. Its target length is below  $\kappa^+$ , whereas  $\ell_2(\mathcal{W}) = \kappa^+$ .  $\square$

**Corollary 8.4** (No fixed power below  $\kappa^+$ ; sources 02 and 07).  *$\mathcal{W}$  embeds neither into  $\mathcal{W}_\kappa$ , nor into  $\mathcal{A}$ , nor into any real power  $\mathbb{R}_{\text{lex}}^\gamma$  or binary cube  $\mathbb{B}_\gamma$  with  $\gamma < \kappa^+$ . This holds in ZFC, without any hypothesis on  $2^{<\kappa}$ .*

*First proof* (source 02).  $\mathcal{W}_\kappa$  is the stratum  $\alpha = \kappa$ , so corollary 8.3 with  $\tau = \kappa$  excludes it;  $\mathcal{A} \simeq_e \mathcal{W}_\kappa$  by theorem 7.1. An embedding into  $\mathbb{R}_{\text{lex}}^\gamma$  would give  $\mathcal{W} \hookrightarrow \mathbb{B}_{\omega \cdot \gamma}$  by lemma 5.5, with  $\omega \cdot \gamma < \kappa^+$ , contrary to  $\ell_2(\mathcal{W}) = \kappa^+$  (theorem 8.2).  $\square$

*Second route* (source 07, Corollary 5.4, 07:525). By real representation rank:  $\text{repr}_{\mathbb{R}}(\mathcal{W}) = \kappa^+$ , whereas  $\text{repr}_{\mathbb{R}}(\mathcal{W}_\kappa) = \text{repr}_{\mathbb{R}}(\mathcal{A}) = \kappa$  and  $\text{repr}_{\mathbb{R}}(\mathbb{R}_{\text{lex}}^\gamma) = \gamma$  by corollary 5.6.  $\square$

[merge] This answers, unconditionally, source 06's Research question 14.1 (06:1332), which asked whether  $\mathcal{W}$  embeds into  $\mathcal{W}_\kappa$  or into one fixed power  $\mathbb{R}_{\text{lex}}^\gamma$  with  $\gamma < \kappa^+$  "in any model of ZFC". Source 06 had the negative answer for  $\mathcal{W}_\kappa$  only under  $2^{<\kappa} < 2^\kappa$ ; its argument is kept as a different route in the remark below. Its question about the longer fixed slices  $\mathcal{W}_\beta$  is answered as well:  $\mathcal{W}$  embeds into no bounded union of strata (corollary 8.3).

**Remark 8.5** (The cellularity route; source 06, Corollary 9.5). If  $2^{<\kappa} < 2^\kappa$ , the separation  $\mathcal{W} \not\hookrightarrow \mathcal{W}_\kappa$  also follows from cellularity. The full order has  $2^\kappa$  pairwise disjoint six-element convex blocks (proposition 12.1). An order embedding into  $\mathcal{W}_\kappa$  would send, from each block, three selected points to an ordered triple. The intervals between the first and third images would be nonempty and pairwise disjoint. This contradicts  $\text{cell}(\mathcal{W}_\kappa) = 2^{<\kappa}$  (proposition 12.2). The proof uses intervals between images, not continuity of the embedding. When  $2^{<\kappa} = 2^\kappa$  this cardinal-invariant argument says nothing; source 06 left that case open, and corollary 8.4 closes it.

**Corollary 8.6** (Real-coordinate powers). *For every ordinal  $\gamma$ ,*

$$\mathbb{R}_{\text{lex}}^\gamma \hookrightarrow \mathcal{W} \iff \gamma < \kappa^+.$$

*Proof.* For  $\gamma < \kappa^+$ , the ordinal  $\omega \cdot \gamma$  also lies below  $\kappa^+$ , so use [lemmas 5.5](#) and [8.1](#). For  $\gamma \geq \kappa^+$ , restricting all coordinates to two reals gives a copy of  $\mathbb{B}_{\kappa^+}$  and therefore of  $\kappa^+$ .  $\square$

[merge] Also source 07, Proposition 5.1 and the last assertion of Corollary 5.4 (07:460, 07:525), and, for the positive part, source 06, Proposition 4.6 (06:425).

These results give an exact classification for binary and real-coordinate lexicographic powers. They do not classify every linear order of cardinality  $2^\kappa$  that has no  $\kappa^+$ -chain in either direction. In particular, absence of the two long monotone chains is not asserted to be a sufficient embedding criterion for  $\mathcal{W}$ .

## 9 An exact classification of every fixed ordinal type

The finite-tail result extends to *every* possible ordinal length, not only to  $\kappa + n$ . The key is to exploit the shrinking alphabet after a long prefix has already enumerated most of the ground set. Coding every remaining real by a fresh block of  $\omega$  bits would miss this improvement.

**Lemma 9.1** (Countable alphabets). *If  $X$  is a countable linear order, then  $X_{\text{lex}}^\omega$  embeds into  $\mathbb{B}_\omega$ . Consequently, for every nonzero limit ordinal  $\lambda$ ,  $X_{\text{lex}}^\lambda$  embeds into  $\mathbb{B}_\lambda$ .*

*Proof.* There are countably many nonempty finite words  $t$  over  $X$ . For a word of length  $m$ , the set

$$D_t = \{x \in X^\omega : x \upharpoonright m \leq_{\text{lex}} t\}$$

is an initial segment. Enumerate these sets and code a point by bit 0 when it belongs to the corresponding  $D_t$ , and bit 1 otherwise. If  $x < y$  first differ at coordinate  $m - 1$ , then  $t = x \upharpoonright m$  gives a cut containing  $x$  but not  $y$ . Every cut code is monotone, so the resulting binary code is an order embedding.

Every limit ordinal has the form  $\lambda = \omega \cdot \eta$ . Apply the countable-alphabet code separately to its successive  $\omega$ -blocks. The concatenated binary length is  $\omega \cdot \eta = \lambda$ .  $\square$

**Theorem 9.2** (All fixed-type strata). *Let  $X \subseteq \mathbb{R}$  be infinite, with the inherited order, and let  $|\alpha| = |X|$ . Write*

$$\alpha = \lambda + n, \quad \lambda \text{ a nonzero limit ordinal, } n < \omega.$$

*Then*

$$\mathcal{W}_\alpha(X) \simeq_e \mathbb{B}_\lambda \times_{\text{lex}} \{0, \dots, n! - 1\}. \quad (11)$$

*In particular, this is an exact equimorphism classification of all strata  $\mathcal{W}_\alpha$  of the real well-ordering space.*

*Proof.* The lower embedding has a uniform construction. Fix an  $n$ -element set  $F \subseteq X$  and partition  $X \setminus F$  into pairs indexed by  $\lambda$ . Orient these pairs according to a member of  $\mathbb{B}_\lambda$  and append one of the  $n!$  permutations of  $F$ . Since  $\lambda$  is a limit ordinal,  $2 \cdot \lambda = \lambda$ , so the result has exactly the required length  $\lambda + n$ .

For the upper embedding, induct on the infinite cardinal  $\mu = |X|$ , proving the assertion simultaneously for all inherited real suborders of that size and all appropriate  $\alpha$ . If  $\mu = \aleph_0$ , code the first  $\lambda$  entries by [lemma 9.1](#), and rank the permutation of the remaining  $n$  elements. This gives the required target.

Suppose  $\mu$  is uncountable and the assertion is known for smaller infinite cardinals. Use ordinal division to write

$$\alpha = \mu \cdot \eta + \rho, \quad 0 \leq \rho < \mu, \quad \eta > 0,$$

and put  $\delta = \mu \cdot \eta$ . Since  $\omega \cdot \mu = \mu$ ,  $\omega \cdot \delta = \delta$ . Rational-cut coding therefore embeds the first  $\delta$  real entries into  $\mathbb{B}_\delta$  without increasing their ordinal length.

For a fixed such prefix  $p$ , the residual set  $Y = X \setminus \text{ran } p$  has cardinality  $|\rho|$ . If  $\rho$  is finite, its tail is just one of  $\rho!$  permutations, which finishes the construction. If  $\rho$  is infinite, write  $\rho = \sigma + n$ , with  $\sigma$  a limit ordinal. By the induction hypothesis, choose an embedding

$$\mathcal{W}_\rho(Y) \hookrightarrow \mathbb{B}_\sigma \times_{\text{lex}} \{0, \dots, n! - 1\}.$$

Choose such an embedding for each possible residual set  $Y$ ; this is a set-sized use of Choice. Append its binary part after the prefix code and retain its final finite rank. The target is

$$\mathbb{B}_{\delta+\sigma} \times_{\text{lex}} \{0, \dots, n! - 1\} = \mathbb{B}_\lambda \times_{\text{lex}} \{0, \dots, n! - 1\}.$$

Two different prefixes are compared inside the first  $\delta$  bits. Equal prefixes have the same residual set and are compared by the chosen embedding of their tails. Thus the assembled map preserves and reflects order. The induction is complete.  $\square$

**Corollary 9.3** (Exact length and cube spectrum in every stratum). *For  $\alpha = \lambda + n$  as above,*

$$\begin{aligned} \ell_2(\mathcal{W}_\alpha(X)) &= \lambda + \lceil \log_2(n!) \rceil, \\ \mathbb{B}_\theta \hookrightarrow \mathcal{W}_\alpha(X) &\iff \theta \leq \lambda + \lfloor \log_2(n!) \rfloor. \end{aligned}$$

*For  $X = \mathbb{R}$ , every stratum still has cardinality  $2^\kappa$  and ordinal spectrum precisely  $\{\beta : \beta < \kappa^+\}$ .*

*Proof.* The finite-fiber fusion proof of [theorem 7.4](#) works with any ordinal  $\lambda$  in place of  $\kappa$ ; it only uses full binary cylinders through  $\lambda$  steps. Apply it to [theorem 9.2](#). For the real case,  $|\lambda| = \kappa$  and  $\lambda \geq \kappa$ , so the stratum contains  $\mathbb{B}_\kappa$  and is a suborder of  $\mathcal{W}$ . The cardinality and ordinal conclusions follow from the earlier upper and lower bounds.  $\square$

**Corollary 9.4** (The exact comparison between strata). *Let  $\alpha = \lambda + n$  and  $\beta = \mu + m$  have cardinality  $\kappa$ , with  $\lambda, \mu$  nonzero limit ordinals and  $m, n < \omega$ . Then*

$$\mathcal{W}_\alpha \hookrightarrow \mathcal{W}_\beta \iff \lambda < \mu \text{ or } [\lambda = \mu \text{ and } n! \leq m!].$$

*Thus the fixed strata are linearly ordered by embeddability. Distinct strata are equimorphic exactly for the pairs  $\mathcal{W}_\lambda, \mathcal{W}_{\lambda+1}$  with  $\lambda$  a limit ordinal.*

*Proof.* For the same limit length, fusing source cylinders while reading the  $\lambda$  target bits forces an entire source fiber of size  $n!$  into a target fiber of size  $m!$ . Hence  $n! \leq m!$  is necessary, and it is plainly sufficient. If  $\lambda < \mu$ , then  $\lambda + \lceil \log_2(n!) \rceil < \mu$ , since  $\mu$  is a limit ordinal. Code the source into that shorter cube and then into  $\mathbb{B}_\mu$  in the target. If  $\mu < \lambda$ , the target codes into a cube of length  $\mu + \lceil \log_2(m!) \rceil < \lambda$ , whereas the source contains  $\mathbb{B}_\lambda$ . The strict cube hierarchy excludes an embedding.  $\square$

**Example 9.5.** For the following infinite-tail types, the fixed stratum is equimorphic with the binary cube of exactly the same length:

$$\mathcal{W}_{\kappa+\omega} \simeq_e \mathbb{B}_{\kappa+\omega}, \quad \mathcal{W}_{\kappa+\omega^2} \simeq_e \mathbb{B}_{\kappa+\omega^2}, \quad \mathcal{W}_{\kappa \cdot 2} \simeq_e \mathbb{B}_{\kappa \cdot 2}.$$

More generally,  $\mathcal{W}_\lambda \simeq_e \mathcal{W}_{\lambda+1} \simeq_e \mathbb{B}_\lambda$  for every limit  $\lambda$  of cardinality  $\kappa$ . The formula is about embeddability, not an order isomorphism with a cube that has endpoints.

**Proposition 9.6** (Density and cellularity across the strata). *For every  $\kappa \leq \alpha < \kappa^+$ ,*

$$d(\mathcal{W}_\alpha) = \text{cell}(\mathcal{W}_\alpha) = \begin{cases} 2^{<\kappa}, & \alpha = \kappa, \kappa + 1, \kappa + 2, \\ 2^\kappa, & \alpha \geq \kappa + 3. \end{cases}$$

*Proof.* The first three cases are proved in [proposition 12.4](#). For the others, write  $\alpha = \kappa + \tau$ , where  $\tau \geq 3$ . Choose a set  $Y \subseteq \mathbb{R}$  of cardinality  $|\tau|$  whose complement still has cardinality  $\kappa$ . There are  $2^\kappa$  distinct exhaustive prefixes of length  $\kappa$  listing  $\mathbb{R} \setminus Y$ . Each corresponding cylinder in  $\mathcal{W}_\alpha$  is isomorphic to  $\mathcal{W}_\tau(Y)$  and has at least three points. Inside each choose a nonempty open interval between three such points. These intervals are pairwise disjoint. Cardinality gives the upper bound.  $\square$

## 10 Self-absorption and closure under lexicographic constructions

The availability of arbitrary exhaustive lengths lets one concatenate entire well-orderings, rather than squeezing them into a fixed coordinate budget.

**Lemma 10.1** (Many disjoint real copies). *There are  $\kappa$  pairwise disjoint subsets of  $(0, 1)$ , each admitting an order isomorphism from  $\mathbb{R}$  onto that subset.*

*Proof.* Use the increasing injection  $j : \mathbb{R} \rightarrow \{0, 1\}^\omega$  from [lemma 5.5](#). For  $t \in \{0, 1\}^\omega$ , interleave the fixed bits of  $t$  with the variable bits of  $j(r)$ :

$$z_{t,r}(2n) = t(n), \quad z_{t,r}(2n+1) = j(r)(n).$$

For fixed  $t$ , first disagreements are those of  $j(r)$ , so  $r \mapsto T(z_{t,r})$  is increasing. Different  $t$  give disjoint images because ternary coding is injective. An affine rescaling puts all images strictly inside  $(0, 1)$ . There are  $2^{\aleph_0} = \kappa$  parameters  $t$ .  $\square$

**Theorem 10.2** (Sharp power absorption). *For every ordinal  $\gamma$ ,*

$$\mathcal{W}_{\text{lex}}^\gamma \hookrightarrow \mathcal{W} \iff \gamma < \kappa^+. \quad (12)$$

For  $0 < \gamma < \kappa^+$ ,  $\mathcal{W}_{\text{lex}}^\gamma \simeq_e \mathcal{W}$ .

*Proof.* Assume  $0 < \gamma < \kappa^+$ . Choose disjoint images  $X_\xi = h_\xi[\mathbb{R}] \subseteq (0, 1)$  for  $\xi < \gamma$ , with each  $h_\xi$  increasing. Let

$$Z = \mathbb{R} \setminus \bigcup_{\xi < \gamma} X_\xi,$$

and choose a fixed exhaustive enumeration  $z$  of  $Z$ . The set  $Z$  has cardinality  $\kappa$ , because it contains  $\mathbb{R} \setminus (0, 1)$ . For a tuple  $(e_\xi)_{\xi < \gamma} \in \mathcal{W}^\gamma$ , define

$$E((e_\xi)_{\xi < \gamma}) = \bigwedge_{\xi < \gamma} (h_\xi \circ e_\xi) \frown z. \quad (13)$$

All images being disjoint, this is an injective enumeration of all reals. Its ordinal length is the sum of at most  $\kappa$  ordinals, each of cardinality at most  $\kappa$ , followed by the length of  $z$ . The total has cardinality  $\kappa$ , so it is an allowed member of  $\mathcal{W}$ .

For two distinct tuples, let  $\xi$  be their first different component. Earlier components are equal as exhaustive enumerations, including their lengths; hence the concatenated prefixes are identical. Within component  $\xi$ , prefix-freeness supplies a disagreement before either component ends. Since  $h_\xi$  is increasing, that disagreement has exactly the required sign. Thus  $E$  is an order embedding.

The empty product is a singleton and embeds trivially. For  $\gamma \geq \kappa^+$ , choose two points of  $\mathcal{W}$  and use them as binary coordinate values. The product then contains  $\mathbb{B}_{\kappa^+}$  and hence  $\kappa^+$ , which  $\mathcal{W}$  cannot contain. Finally, for  $\gamma > 0$ , varying one coordinate and fixing all other coordinates embeds  $\mathcal{W}$  into the product.  $\square$

[merge] Also source 07, Lemma 6.1 and Theorem 6.2 (07:548, 07:563), by the same interleaved copies of  $\mathbb{R}$  and the same concatenation. Source 07 stresses that the copies must be disjoint *ordered* copies of  $\mathbb{R}$ , not disjoint intervals, of which only countably many exist (07:543–546), and that the ranges need not be convex. Absorption is mutual embeddability, not isomorphism ([remark 19.2](#)).

**Corollary 10.3** (Closure of the embedding class). *Let  $\mathcal{E}(\mathcal{W}) = \{L : L \hookrightarrow \mathcal{W}\}$ . This class is closed under reversal, under ordinal-indexed lexicographic products of at most  $\kappa$  factors, and under lexicographic sums whose index order and every summand belong to  $\mathcal{E}(\mathcal{W})$ .*

*Proof.* Negation of all real entries reverses  $\mathcal{W}$ , so the class is closed under reversal. Embed each factor into  $\mathcal{W}$  and apply [theorem 10.2](#) for products. For a sum  $\sum_{i \in I} L_i$ , choose embeddings  $f : I \rightarrow \mathcal{W}$  and  $g_i : L_i \rightarrow \mathcal{W}$ . Then

$$(i, x) \mapsto (f(i), g_i(x))$$

embeds the sum into  $\mathcal{W} \times_{\text{lex}} \mathcal{W}$ , and the latter embeds into  $\mathcal{W}$ . The family of chosen embeddings is a set-sized choice in ZFC.  $\square$

[merge] Also source 07, Corollary 6.3 (07:599).

**Corollary 10.4** (Failure of absorption at fixed length). *For every ordinal  $\gamma$ ,*

$$(\mathcal{W}_\kappa)_{\text{lex}}^\gamma \simeq_e \mathbb{B}_{\kappa \cdot \gamma}, \quad \mathcal{A}_{\text{lex}}^\gamma \simeq_e \mathbb{B}_{\kappa \cdot \gamma}.$$

*Either product embeds back into its original factor exactly when  $\gamma \leq 1$ .*

*Proof.* Equimorphisms can be applied coordinatewise, and consecutive blocks of  $\kappa$  binary coordinates have ordinal length  $\kappa \cdot \gamma$ . For  $\gamma \geq 2$ , this length exceeds  $\kappa$ ; use [theorem 5.3](#). The cases 0 and 1 are immediate.  $\square$

[merge] Source 07 states the same equimorphism with real coordinates,  $(\mathcal{W}_\kappa)_{\text{lex}}^\beta \simeq_e \mathbb{R}_{\text{lex}}^{\kappa \cdot \beta}$  and  $\mathcal{A}_{\text{lex}}^\beta \simeq_e \mathbb{R}_{\text{lex}}^{\kappa \cdot \beta}$  for  $\beta > 0$  (its Proposition 6.4 and Theorem 11.2, 07:613, 07:1129). This is the same statement:  $\mathbb{R}_{\text{lex}}^{\kappa \cdot \beta} \simeq_e \mathbb{B}_{\omega \cdot \kappa \cdot \beta}$  and  $\omega \cdot \kappa \cdot \beta = \kappa \cdot \beta$  by (4). Source 07 adds the rank  $\text{repr}_{\mathbb{R}}((\mathcal{W}_\kappa)_{\text{lex}}^\beta) = \text{repr}_{\mathbb{R}}(\mathcal{A}_{\text{lex}}^\beta) = \kappa \cdot \beta$ .

## 10.1 Many convex copies inside the full order

The preceding embeddings need not have convex images. There is also a particularly strong convex version for a large family of copies.

**Proposition 10.5.**  *$\mathcal{W}$  contains  $2^\kappa$  pairwise disjoint open convex suborders, each isomorphic to  $\mathcal{W}$ . They can be chosen so that their index order is  $\mathbb{B}_\kappa$ .*

*Proof.* Let  $I = (0, 1)$ . Partition  $\mathbb{R} \setminus I$  into  $\kappa$  ordered pairs. For  $x \in \{0, 1\}^\kappa$ , let  $s_x$  be the pair-swap enumeration of  $\mathbb{R} \setminus I$ . The cylinders  $C_{s_x}$  are pairwise disjoint and ordered according to  $x$ . By [lemma 4.3](#), each is isomorphic to  $\mathcal{W}(I)$ , and an increasing bijection  $\mathbb{R} \rightarrow I$  identifies  $\mathcal{W}(I)$  with  $\mathcal{W}$ . The order  $\mathcal{W}(I)$  has no endpoints, since  $I$  has none. Thus every point of a cylinder is bracketed by two points in that cylinder, making the convex cylinder open in  $\mathcal{W}$ .  $\square$

## 11 Adjacent pairs and finite condensation

**Lemma 11.1** (Extrema of exhaustive enumeration orders). *For any nonempty linearly ordered set  $Y$ ,  $\mathcal{W}(Y)$  has a least element exactly when the given order on  $Y$  is a well-order. That element is the increasing enumeration of  $Y$ . It has a greatest element exactly when the given order on  $Y$  is a reverse well-order, in which case it is the decreasing enumeration.*

*Proof.* In a lexicographically least exhaustive enumeration, the value at each coordinate must be the least as-yet-unused element. Otherwise a smaller unused choice, followed by any exhaustive enumeration of the residual set, would give a smaller member of  $\mathcal{W}(Y)$ . Thus a least enumeration must be increasing

and exhibit  $Y$  as a well-order. Conversely, at the first changed coordinate of any other enumeration, the increasing enumeration uses the least available value, so it is lexicographically smaller. Reversing the ground order proves the maximum assertion.  $\square$

[merge] Also source 06, Lemma 3.3 (06:263), and the first half of source 07, Lemma 8.1 (07:714).

**Theorem 11.2** (Complete adjacency criterion). *Let  $e < f$  in  $\mathcal{W}(X)$ , where  $X$  is any infinite linearly ordered set. Let  $s$  be their maximal common prefix, of length  $\delta$ , and write*

$$Y = X \setminus \text{ran } s, \quad a = e(\delta), \quad b = f(\delta).$$

*Then  $e$  and  $f$  are adjacent if and only if all of the following hold:*

- (a)  $Y$  is finite;
- (b)  $a < b$  are consecutive elements of  $Y$  in its given order;
- (c) after  $a$ , the enumeration  $e$  lists  $Y \setminus \{a\}$  in decreasing order;
- (d) after  $b$ , the enumeration  $f$  lists  $Y \setminus \{b\}$  in increasing order.

*In particular, an adjacent pair has equal enumeration lengths and differs only in a finite final segment.*

*Proof.* If  $a < c < b$  for some  $c \in Y$ , then an exhaustive extension of  $s \frown c$  lies strictly between  $e$  and  $f$ . If  $e$  is not the greatest member of its  $a$ -cylinder, a larger member of that cylinder also lies between them. Likewise,  $f$  must be the least member of its  $b$ -cylinder. By lemma 11.1,  $Y \setminus \{a\}$  must be reverse well-ordered and  $Y \setminus \{b\}$  must be well-ordered.

The intersection  $Y \setminus \{a, b\}$  is consequently both well-ordered and reverse well-ordered. It is finite: an infinite well-order contains an increasing  $\omega$ -sequence, whose range has no greatest member and cannot be reverse well-ordered. Thus  $Y$  is finite. The two extremal tails are then exactly the decreasing and increasing finite lists in the statement.

Conversely, any point between  $e$  and  $f$  would have to extend  $s$ , by convexity. Its next entry could not lie strictly between  $a$  and  $b$ . Choosing  $a$  cannot exceed the greatest tail  $e$ , and choosing  $b$  cannot precede the least tail  $f$ . No intermediate point exists.  $\square$

[merge] Also source 06, Theorem 6.2 (06:548), and source 07, Theorem 8.2 (07:747), by the same argument. Source 07 states the criterion without condition (a) and derives finiteness of  $Y$ ; its conditions (ii) and (iii) are (c) and (d) here, through lemma 11.1.

**Example 11.3** (A jump and an isolated point). Choose any exhaustive prefix  $s$  enumerating  $\mathbb{R} \setminus \{a, b\}$ , with  $a < b$ . Then

$$s \frown (a, b) < s \frown (b, a)$$

is an adjacent pair in the full order  $\mathcal{W}$ , not just in a fixed-length suborder.

If  $a < b < c$  and  $s$  enumerates  $\mathbb{R} \setminus \{a, b, c\}$ , its cylinder is the six-point chain

$$\begin{aligned} s \frown (a, b, c) &< s \frown (a, c, b) < s \frown (b, a, c) \\ &< s \frown (b, c, a) < s \frown (c, a, b) < s \frown (c, b, a). \end{aligned}$$

The four interior points are isolated in the order topology of  $\mathcal{W}$ .

[merge] This is also source 06's Example 6.1 and its Appendix B. Source 06 reads the middle jump  $s \frown (b, c, a) < s \frown (c, a, b)$  through theorem 11.2: the old suffix after  $b$  is decreasing  $(c, a)$ , the new suffix after  $c$  is increasing  $(a, b)$ , and  $b, c$  are adjacent in the unused alphabet. It warns that the example must not be extrapolated to all points: an enumeration of length  $\kappa$  has no finite final tail, lies in a singleton finite-condensation class, and has uncountable character on both sides; nor should a permutation generator be read as an algorithm that enumerates all well-orderings of  $\mathbb{R}$ .

**Definition 11.4.** The finite-condensation relation on a linear order is

$$x \sim_{\text{fin}} y \iff [\min(x, y), \max(x, y)] \text{ is finite.}$$

It is a convex equivalence relation.

**Theorem 11.5** (Exact finite blocks). *Let  $e \in \mathcal{W}$ , and write its length uniquely as  $\alpha = \lambda + n$ , where  $\lambda$  is a nonzero limit ordinal and  $n < \omega$ . If  $n = 0$ , its finite-condensation class is a singleton. If  $n > 0$ , its class consists exactly of all  $n!$  permutations of the final  $n$  reals after the fixed prefix  $e \restriction \lambda$ .*

*These classes are the maximal finite convex blocks. A point is isolated exactly when it is an interior point of one of the blocks with  $n \geq 3$ .*

*Proof.* The displayed permutations form a finite convex cylinder. If a point is adjacent to  $e$ , [theorem 11.2](#) says that the first disagreement leaves only finitely many entries. Hence the two enumerations have the same length and agree before its final finite remainder  $n$ . No adjacency can cross to a different displayed block. Every finite interval is connected by finitely many adjacency steps, so the displayed block is the full finite-condensation class.

In a linear order with no endpoints, a point is isolated exactly when it has both an immediate predecessor and an immediate successor. The first and last points of these maximal blocks have no adjacent point beyond the block; an interior point has both required neighbors. A factorial block has an interior point precisely when  $n! \geq 3$ , equivalently  $n \geq 3$ .  $\square$

[merge] Also source 06, Proposition 6.3 and Corollary 6.4 (06:590, 06:607), and source 07, Theorem 8.3 (07:770). Source 06 counts the isolated points of a block: exactly its  $n! - 2$  nonextremal permutations. Source 07 makes explicit why an adjacency chain cannot leave the block: for the length  $\lambda + n$ , a common prefix with finitely many unused values has length at least  $\lambda$ , since adding finitely many positions to an ordinal below the limit  $\lambda$  cannot reach  $\lambda$ . It observes that the finite blocks explain why extrapolating density from  $\mathcal{W}_\kappa$  to  $\mathcal{W}$  is incorrect: once a prefix has used all but finitely many reals, there is no reservoir of intermediate values left (07:813–815).

The quotient by finite condensation is a dense linear order: if two different classes were adjacent in the quotient, their union would be a finite interval and they would be the same class. We do not need to assume that this quotient is isomorphic to any familiar generalized real line. Its precise isomorphism type remains a separate question.

[merge] Sources 06 and 07 prove that  $\mathcal{W}/\sim_{\text{fin}}$  is equimorphic with  $\mathcal{W}$  ([theorem 21.1](#); source 06, Theorem 11.1, 06:1133; source 07, Proposition 10.5, 07:1072), answering the question asked in source 02's [section 26](#). Its isomorphism type is still open.

## 12 Order topology, size of intervals, and cofinalities

### 12.1 Global invariants of the full order

**Proposition 12.1.** *The order  $\mathcal{W}$  has no endpoints, is isomorphic to its reverse, and satisfies*

$$\text{cf}(\mathcal{W}) = \text{ci}(\mathcal{W}) = \omega, \quad \text{d}(\mathcal{W}) = \text{cell}(\mathcal{W}) = 2^\kappa.$$

*It has exactly  $2^\kappa$  isolated points.*

*Proof.* Every real can be the first entry of an exhaustive enumeration. Choosing first entries tending to  $+\infty$  gives a countable cofinal sequence; choosing first entries tending to  $-\infty$  gives a countable coinital sequence. Neither endpoint exists. The map  $e \mapsto -e$ , with the same ordinal domain, is an order-reversing bijection.

Fix three reals and use all pair-swap enumerations of their complement as prefixes of length  $\kappa$ . For each prefix choose one of the four interior permutations in its six-point block. This gives  $2^\kappa$  isolated points by [theorem 11.5](#). Each isolated singleton must be met by a dense set, and the singletons are pairwise disjoint nonempty open sets. The upper bounds follow from  $|\mathcal{W}| = 2^\kappa$ .  $\square$

[merge] Also source 06, Corollary 3.6 and Theorem 6.6 (06:320, 06:650), and source 07, Lemma 8.1 and Corollary 8.4 (07:714, 07:798). Both add the weight:  $w(\mathcal{W}) = 2^\kappa$ , since the interval base has at most  $|\mathcal{W}|^2 = 2^{2\kappa}$  members and  $\text{cell} \leq d \leq w$ . Source 06 also shows that the isolated points are dense ([theorem 16.4](#)); sources 06 and 07 obtain  $\text{cf } \mathcal{W} = \text{ci } \mathcal{W} = \omega$  from the stronger fact that every residual order  $\mathcal{W}(Y)$ ,  $Y \subseteq \mathbb{R}$ , has cofinality and coinitality in  $\{1, \omega\}$  ([lemma 16.2](#)).

There is a second, independent reason for the large cellularity: [proposition 10.5](#) gives  $2^\kappa$  disjoint open copies of  $\mathcal{W}$ . Thus the phenomenon is not solely a by-product of isolated finite tails. For example, selecting one point from each finite-condensation class still leaves  $2^\kappa$  disjoint nontrivial interval regions coming from those large cylinders.

It would nevertheless be false to say that *every* interval in  $\mathcal{W}$  is large. Some intervals are empty, some are finite, and a cylinder leaving countably infinitely many unused reals has cardinality only  $2^{\aleph_0}$ . The residual-set lemma, not the ambient cardinality, controls such examples.

## 12.2 The dense fixed-length order

**Proposition 12.2.** *The order  $\mathcal{W}_\kappa$  is dense, has no endpoints, and is self-dual. Its global cofinality and coinitality are  $\omega$ , and*

$$d(\mathcal{W}_\kappa) = \text{cell}(\mathcal{W}_\kappa) = 2^{<\kappa}. \quad (14)$$

*At every point  $e$ , the cofinality of the left side and the coinitality of the right side are both  $\text{cf}(\kappa)$ .*

*Proof.* Let  $e < f$  first disagree at  $\delta < \kappa$ . The common prefix has used fewer than  $\kappa$  reals, whereas the interval  $(e(\delta), f(\delta))$  has cardinality  $\kappa$ . Choose an unused real strictly between the two entries. Extending the resulting prefix to a bijection of length  $\kappa$  gives an intermediate point. This proves density. Endpoints, global cofinalities, and self-duality are treated as for  $\mathcal{W}$ .

There are at most

$$\sum_{\xi < \kappa} \kappa^{|\xi|} = 2^{<\kappa}$$

prefixes of length below  $\kappa$ . Here finite nonzero exponents contribute  $\kappa = 2^{\aleph_0}$ , and for infinite  $\mu < \kappa$ ,  $\kappa^\mu = (2^{\aleph_0})^\mu = 2^\mu$ . Choose one exhaustive extension for each such prefix. The resulting set is dense, since the intermediate prefix in the previous paragraph determines a whole cylinder inside the given interval.

For every infinite cardinal  $\mu < \kappa$ , orient  $\mu$  disjoint pairs to obtain  $2^\mu$  incompatible prefixes of length  $2 \cdot \mu < \kappa$ . Their cylinders are nonempty, convex, and have no endpoints: one can change a later coordinate upward or downward while fewer than  $\kappa$  values have been used. The cylinders are therefore open. This proves the lower cellularity bound after taking the supremum over  $\mu < \kappa$ . Cellularity never exceeds density, proving (14).

Finally, for every prefix length  $\xi < \kappa$ , there are points on both sides of  $e$  agreeing with it below  $\xi$ : change a still later coordinate to an unused smaller or larger real and complete the enumeration. Choose such points at a cofinal set of  $\text{cf}(\kappa)$  prefix lengths. They are cofinal on the left and coinital on the right. Conversely, fewer than  $\text{cf}(\kappa)$  points on one side have a bounded set of first-disagreement coordinates with  $e$ . A perturbation agreeing longer than that bound lies closer to  $e$  than all of them. This proves both local lower bounds.  $\square$

[merge] Also source 06, Theorems 9.2 and 9.3 (06:973, 06:1001), and source 07, Theorem 7.1 (07:637), by the same short-prefix arguments. Both add  $w(\mathcal{W}_\kappa) = 2^{<\kappa}$ , since the short-prefix cylinders, at most  $2^{<\kappa}$  of them, are clopen and form a base (theorem 20.2 and proposition 20.3).

*Remark 12.3* (Regularity is not assumed). The formula uses  $\text{cf}(\kappa)$ , not automatically  $\kappa$ . However,  $\text{cf}(2^{\aleph_0}) > \omega$ . Here is a direct diagonal proof. If  $\{0, 1\}^\omega = \bigcup_{n < \omega} X_n$  with each  $|X_n| < 2^{\aleph_0}$ , partition  $\omega$  into infinite sets  $I_n$ . Choose a pattern on  $I_n$  that is not the restriction of any member of  $X_n$ . The union of these patterns belongs to none of the  $X_n$ , a contradiction. A countable cofinal sequence of smaller cardinals would produce just such a cover. Consequently  $\mathcal{W}_\kappa$  is not first countable at any point.

[merge] Sources 06 (06:1029–1038) and 07 (Lemma 3.2, 07:305–310) prove  $\text{cf } \kappa > \omega$  through König's inequality: if cardinals  $\kappa_n < \kappa$  were cofinal in  $\kappa$ , then  $\kappa = \sum_n \kappa_n < \prod_n \kappa_n^+ \leq \kappa^{\aleph_0} = \kappa$ , a contradiction; the strict inequality is the same diagonal argument.

### 12.3 A topology refinement for finite tails

**Proposition 12.4.** *For finite  $n \geq 0$ , the following hold.*

- (a) *For  $n = 0, 1$ ,  $\mathcal{W}_{\kappa+n}$  is dense and has no isolated points.*
- (b) *For  $n = 2$ , it has  $2^\kappa$  adjacent pairs and no isolated points. Its density and cellularity are still  $2^{<\kappa}$ .*
- (c) *For  $n \geq 3$ , it has  $2^\kappa$  isolated points and hence density and cellularity  $2^\kappa$ .*

*For  $n = 0, 1$ , density and cellularity are  $2^{<\kappa}$  as well. The weight of  $\mathcal{W}_{\kappa+2}$  is  $2^\kappa$ , even when its density is smaller.*

*Proof.* If two points first disagree below  $\kappa$ , the unused-real argument in proposition 12.2 gives an intermediate point of the same length. If their first  $\kappa$  entries agree, they lie in one finite permutation block. The block sizes are 1, 1, 2, 6, . . . , proving the assertions about adjacency and isolation. Varying a pair-swap prefix while fixing the final finite set gives  $2^\kappa$  blocks.

For  $n \leq 2$ , a nonempty open interval cannot be finite: a finite open set in a  $T_1$  space would contain an isolated point. Such an interval therefore contains two points with different prefixes below  $\kappa$ , since each full  $\kappa$ -prefix has at most two extensions. As before, a shorter cylinder fits between those points. Choosing one extension of every prefix of length below  $\kappa$  gives the upper density bound, and incompatible short prefixes give the lower cellularity bound.

For the weight assertion, every adjacent pair  $a < b$  requires a basic open set containing  $b$  and contained in  $(a, \infty)$ . Different adjacent pairs require different such basic sets: if  $a < b \leq c < d$ , one set cannot both contain  $b$  and be contained in  $(c, \infty)$ . Thus any base has size at least  $2^\kappa$ . The set of all endpoint intervals supplies the reverse bound.  $\square$

## 13 Finer properties of the Kanovei–Shelah index

Equimorphism with  $\mathbb{B}_\kappa$  gives the exact embeddable orders, but does not determine global cofinality or topology. Those require separate arguments.

**Lemma 13.1.** *Every nonempty cylinder in  $\mathcal{A}$  determined by fewer than  $\kappa$  coordinates has no endpoints. In particular,  $\mathcal{A}$  is a dense linear order without endpoints.*

*Proof.* Fix  $a \in \mathcal{A}_U$  and a prefix of length  $\xi < \kappa$ . Every ultrafilter contains  $\mathbb{N}$  and has  $\kappa$  members. Since the prefix lists fewer than  $\kappa$  values, arbitrarily far out in the tail there is a value other than  $\mathbb{N}$ . Replacing such a value by  $\mathbb{N}$ , and then enumerating all of  $U$  on the remaining coordinates, gives a larger point with the prescribed prefix.

For a smaller point, if  $U$  has a least member, the tail cannot consist only of this member; choose a later larger value and replace it by the least one. If  $U$  has no least member, replace any chosen later value by a smaller member of  $U$ . In either case complete the remaining coordinates by an enumeration of  $U$ . Thus there are both larger and smaller points in the cylinder.

If  $a < b$  first differ at  $\delta$ , apply the preceding construction to  $a$  with its prefix through  $\delta$  fixed. A larger extension in that cylinder still lies below  $b$ .  $\square$

**Proposition 13.2.** *The full index  $\mathcal{A}$  satisfies*

$$|\mathcal{A}| = 2^\kappa, \quad d(\mathcal{A}) = \text{cell}(\mathcal{A}) = 2^{<\kappa}, \quad \text{cf}(\mathcal{A}) = \text{cf}(\kappa), \quad \text{ci}(\mathcal{A}) = \omega.$$

*At every point, the left and right local cofinalities are both  $\text{cf}(\kappa)$ . In particular,  $\mathcal{A}$  is not isomorphic to its reverse and is not isomorphic to  $\mathcal{W}_\kappa$ .*

*Proof.* Cardinality follows from the embeddings in [theorem 7.1](#). For density, there are at most  $2^{<\kappa}$  prefixes of length below  $\kappa$ . By [lemma 13.1](#), every nonempty interval contains a cylinder with such a prefix; choose one extension for each extendible prefix. Conversely, fix  $U$  and two of its members. All binary choices for a prefix of an infinite length  $\mu < \kappa$  extend to members of  $\mathcal{A}_U$ . Their full  $\mathcal{A}$ -cylinders are disjoint nonempty open sets. This proves the cellularity and density formulas.

Let  $\nu(a)$  be the first coordinate at which  $a$  is not  $\mathbb{N}$ . It exists because  $\text{ran } a$  is an ultrafilter with  $\kappa$  members. For a cofinal set of ordinals  $\delta < \kappa$ , construct a map beginning with  $\delta$  copies of  $\mathbb{N}$ , then a fixed non- $\mathbb{N}$  member of a fixed ultrafilter, then an exhaustive tail. These maps are cofinal in  $\mathcal{A}$ . A family of fewer than  $\text{cf}(\kappa)$  maps has its values of  $\nu(a)$  bounded below  $\kappa$ ; a sufficiently long initial string of  $\mathbb{N}$  lies above the whole family. Hence  $\text{cf}(\mathcal{A}) = \text{cf}(\kappa)$ .

For each natural number  $n$ , take a member of  $\mathcal{A}$  whose first value is  $\{n\}$  and whose range is the principal ultrafilter at  $n$ . The singletons  $\{n\}$  descend coininitially among the nonempty subsets of  $\mathbb{N}$  in binary lexicographic order. Therefore these maps are coininitial in  $\mathcal{A}$ , which has no least element. This proves  $\text{ci}(\mathcal{A}) = \omega$ .

Local cofinalities follow from the same bounded-disagreement argument used for  $\mathcal{W}_\kappa$ , now using [lemma 13.1](#) for perturbations after arbitrary prescribed prefixes. Finally,  $\text{cf}(\kappa) > \omega$ , so the two global cofinalities are unequal. They rule out self-duality and also rule out an isomorphism with  $\mathcal{W}_\kappa$ .  $\square$

[merge] Also source 07, Theorem 11.3 (07:1163), with the same cofinality and coininitiality witnesses; source 07 adds the weight, a clopen base of short prefixes, and coininitiality witnesses for the free-ultrafilter variant of  $\mathcal{A}$  ([proposition 23.1](#)).

For comparison, a fixed principal-ultrafilter fiber  $\mathcal{A}_U$  has coininitiality  $\text{cf}(\kappa)$ : it has a least available coordinate value, which can be repeated for arbitrarily long initial segments. A nonprincipal fiber has coininitiality  $\omega$ , because the cofinite sets form a decreasing coininitial sequence in its coordinate order. All fibers nevertheless have the same embeddability class by [theorem 7.1](#).

## 14 Explicit countable gaps and Dedekind completion

### 14.1 A gap caused by a repeated real

**Proposition 14.1.** *Both  $\mathcal{W}$  and  $\mathcal{W}_\kappa$  have an unfilled cut of cofinality type  $(\omega, \omega)$ .*

*Proof.* Divide either order into the following initial and final segments:

$$\begin{aligned} L &= \{e : e(0) < 0 \text{ or } [e(0) = 0 \text{ and } e(1) < 0]\}, \\ R &= \{e : e(0) > 0 \text{ or } [e(0) = 0 \text{ and } e(1) > 0]\}. \end{aligned}$$

Every exhaustive enumeration has at least two entries, and injectivity excludes  $e(0) = e(1) = 0$ . Thus this is a partition and  $L < R$ . For  $n \geq 1$ , complete  $(0, -1/n)$  and  $(0, 1/n)$  to exhaustive enumerations of length  $\kappa$ . The first family is cofinal in  $L$ ; the second is coinital in  $R$ . Neither side has an endpoint, and a point between all the bounds would have to start  $(0, 0)$ , which is forbidden.  $\square$

[merge] Also source 07, Example 7.3 (07:703), with the same bounds; source 07 concludes that neither order has the countable interpolation property of an  $\eta_1$ -order.

This also illustrates why large local character at every point does not imply countable saturation as a dense order. The obstruction is an omitted cut, rather than a point with small local character.

## 14.2 A gap caused by incompatible ultrafilter values

**Proposition 14.2.** *The order  $\mathcal{A}$  also has an unfilled  $(\omega, \omega)$  cut.*

*Proof.* Let  $u$  be the even natural numbers and  $v$  the odd natural numbers. Put

$$v_n^- = (v \setminus \{2n + 1\}) \cup \{2n + 2\}, \quad v_n^+ = v \cup \{2n\}.$$

In binary lexicographic order,  $v_n^-$  increases toward  $v$  from below, and  $v_n^+$  decreases toward  $v$  from above. Each pair  $u, v_n^-$  belongs to the principal ultrafilter at  $2n + 2$ ; each pair  $u, v_n^+$  belongs to the principal ultrafilter at  $2n$ . Therefore both two-term prefixes  $(u, v_n^-)$  and  $(u, v_n^+)$  extend to members of  $\mathcal{A}$ .

A map between all these bounds must have first value  $u$  and second value  $v$ . Indeed, if a binary sequence differs from  $v$ , its first differing bit is eventually shared by neither bounding side in the required direction. But no ultrafilter contains the disjoint sets  $u$  and  $v$ . Thus the cut is not realized. The initial segment of maps with first value below  $u$ , or first value  $u$  and second value below  $v$ , has the lower sequence cofinal; the complementary final segment has the upper sequence coinital.  $\square$

[merge] Source 07 proves the same statement with different bounds, and adds a gap of character  $(\text{cf } \kappa, \text{cf } \kappa)$  (proposition 23.2; its Proposition 11.4, 07:1218).

These statements concern the index orders. They do not contradict the countable saturation of the elementary extension constructed from the index in the Kanovei–Shelah theorem.

## 14.3 Completion does not create longer ordinals

We use the Dedekind completion that fills unrealized two-sided cuts and keeps existing points and jumps. Adding endpoints is optional and does not change the following cardinal or ordinal conclusions.

**Lemma 14.3** (Cuts have small cofinal witnesses). *Every nonempty initial segment of  $\mathcal{W}$  has a cofinal subset of cardinality at most  $\kappa$ . Every nonempty final segment has a coinital subset of cardinality at most  $\kappa$ .*

*Proof.* If an initial segment had no such cofinal subset, at every stage  $\xi < \kappa^+$  one could choose a point larger than all previously chosen points: the preceding set has cardinality at most  $\kappa$  and is not cofinal. This constructs the forbidden increasing sequence of length  $\kappa^+$ . Reverse the order for the other assertion.  $\square$

[merge] Also source 06, Corollary 5.5 (06:520), for every nonempty suborder of  $\mathcal{W}(X)$ , and source 07, Corollary 4.6 (07:417).

**Theorem 14.4** (Completion invariants). *Let  $\widehat{\mathcal{W}}$  be a Dedekind completion of  $\mathcal{W}$ . Then*

$$|\widehat{\mathcal{W}}| = 2^\kappa, \quad \beta \hookrightarrow \widehat{\mathcal{W}} \iff \beta < \kappa^+, \quad \ell_2(\widehat{\mathcal{W}}) = \kappa^+.$$

*The ordinal equivalence also holds for reverse ordinals. Every two-sided cut of  $\mathcal{W}$  has each of its two cofinalities at most  $\kappa$ .*

*Proof.* By lemma 14.3, every lower half of a cut is determined by a cofinal subset of size at most  $\kappa$ . Thus the number of cuts is at most

$$|\mathcal{W}|^\kappa = (2^\kappa)^\kappa = 2^\kappa.$$

The lower cardinal bound follows from the copy of  $\mathcal{W}$  already present.

Suppose the completion contained an increasing sequence of length  $\kappa^+$ . Between the first and third members of each successive three-point block, choose a point of  $\mathcal{W}$ . Such a point always exists: if the middle point is old, use it; if it is an added cut, the cut property supplies an old point between the outer endpoints. The resulting points form a strictly increasing sequence of length  $\kappa^+$  in  $\mathcal{W}$ , a contradiction. The same argument applies in reverse. The positive ordinal embeddings were already present in  $\mathcal{W}$ .

For binary length,  $\mathcal{W} \hookrightarrow \widehat{\mathcal{W}}$  gives the lower bound. The cube  $\mathbb{B}_{\kappa^+}$  is Dedekind complete, as verified below. An embedding  $f : \mathcal{W} \rightarrow \mathbb{B}_{\kappa^+}$  extends over each unrealized cut  $(L, R)$  by assigning it  $\sup f[L]$ . This value is strictly above every  $f(l)$  because  $L$  has no greatest element, and strictly below every  $f(r)$  because  $R$  has no least element. Distinct cuts remain distinct, by an old point between them. Thus the extension is an order embedding. If endpoints are to be added, first put  $f[\mathcal{W}]$  inside a proper binary cylinder by a fixed finite prefix and then use the ambient endpoints. Its length remains  $\kappa^+$  because finite  $+\kappa^+ = \kappa^+$ .  $\square$

[merge] The cardinality also follows from source 07's count of initial segments (Corollary 4.6, 07:417) and from source 06's count of proper gaps (theorem 18.6; Theorem 7.6, 06:824), which shows that there are exactly  $2^\kappa$  of them. Source 06 also proves the ordinal assertion by the same interval argument (06:851–856).

**Lemma 14.5** (Completeness of binary lexicographic cubes). *Every  $\mathbb{B}_\theta$  is Dedekind complete and has both endpoints.*

*Proof.* The constant sequences give endpoints. For a nonempty subset  $S$ , construct its supremum by retaining a nonempty set of candidates with a common prefix. At a successor step choose bit 1 if some current candidate has bit 1, and retain those candidates; otherwise choose bit 0 and retain all current candidates. At a limit, intersect the candidate sets.

If the first empty intersection occurs at a limit  $\lambda \leq \theta$ , complete the accumulated prefix by zeros. Every member of  $S$  was discarded at an earlier first disagreement where it had 0 and the chosen prefix had 1, so the completed sequence is an upper bound. Any strictly smaller sequence first disagrees with this upper bound before  $\lambda$ ; at that stage there was a candidate with the chosen 1, which is larger than the proposed smaller bound. Thus the completed sequence is the least upper bound. If no empty intersection occurs through  $\theta$ , the final sequence belongs to  $S$  and is its maximum. This covers all cases.  $\square$

*Remark 14.6.* The completion theorem does not prove  $\widehat{\mathcal{W}} \hookrightarrow \mathcal{W}$ . Both orders have the same cardinality, ordinal spectrum, and binary length, but these invariants do not in general decide embeddability. Whether the completion can be absorbed is one of the natural next questions.

## 15 Universality in every slice

[merge] This section and the next five are source 06's, with source 07's proofs of the same statements recorded as routes. The cardinality, universality and ordinal spectrum of sections 4 and 6 hold for every infinite linear alphabet  $X$ ; source 06 sharpens universality to every single slice  $\mathcal{W}_\beta(X)$ .

**Theorem 15.1** (Universality in every fixed-type slice; source 06, Theorem 4.4). *Let  $X$  have infinite cardinality  $\nu$ , and let  $\beta$  be any ordinal of cardinality  $\nu$ . Every linear order  $L$  of cardinality at most  $\nu$  embeds in  $\mathcal{W}_\beta(X)$ . Consequently  $|\mathcal{W}_\beta(X)| = 2^\nu$ .*

*Proof.* Write  $\beta = \nu + \tau$ , where  $\tau$  is the order type of the interval of positions after the initial  $\nu$  positions. Partition  $X = S \sqcup T$  so that  $|S| = \nu$  and  $|T| = |\tau|$ , taking  $T = \emptyset$  if  $\tau = 0$ . Pair-code all bits of  $\mathbb{B}_\nu$  in  $S$  over the first  $\nu$  positions (proposition 4.4), then append a fixed bijection  $\tau \rightarrow T$ . The resulting embeddings all have domain  $\nu + \tau = \beta$ . Compose with lemma 5.1. The same construction embeds all  $2^\nu$  bit strings, proving the cardinality claim together with the relation-code upper bound.  $\square$

For  $X = \mathbb{R}$  this contains corollary 9.3's statement that every real stratum has cardinality  $2^\kappa$  and ordinal spectrum  $\{\beta : \beta < \kappa^+\}$ , by a direct construction instead of the all-strata classification.

*Remark 15.2* (Larger embedded orders; source 06, 06:415–423). Cardinality at most  $\nu$  is a sufficient condition, not a necessary one:  $\mathbb{B}_\nu$  itself has size  $2^\nu$  and embeds. Another useful sufficient condition is a separating family  $(I_\xi)_{\xi < \nu}$  of initial segments: assume that whenever  $x < y$ , some  $I_\xi$  contains  $x$  but not  $y$ . Then the nonmembership vectors  $(\mathbf{1}_{x \notin I_\xi})_{\xi < \nu}$  are coordinatewise increasing and strictly distinguish points, giving an embedding into  $\mathbb{B}_\nu$ . Separately, the collection of all initial segments of a linear order of size  $\nu$ , ordered by inclusion, embeds by the characteristic functions of those initial segments.

## 16 Countable ends and dense isolation

**Lemma 16.1** (Real subsets have countable monotone length; source 06, Lemma 3.4). *Every strictly increasing or strictly decreasing ordinal-indexed sequence of real numbers has countable domain. A subset of  $\mathbb{R}$  without a greatest element has countable cofinality; the dual statement holds for coinitality.*

*Proof.* For a strictly increasing sequence choose a rational strictly between each successive pair of terms. These rational intervals are disjoint, so there are only countably many successor pairs. An uncountable ordinal has uncountably many successor positions, a contradiction. The decreasing case is identical. For the last assertion, a subset unbounded above has a countable cofinal sequence chosen past the integers. A nonempty bounded subset with no maximum has supremum  $r$ , and points chosen above  $r - 1/n$  form a countable cofinal set. A finite cofinal set would imply a maximum.  $\square$

**Lemma 16.2** (Countable ends of every residual permutation order; source 06, Lemma 3.5). *For every nonempty  $Y \subseteq \mathbb{R}$ , each of  $\text{cf}(\mathcal{W}(Y))$  and  $\text{ci}(\mathcal{W}(Y))$  is either 1 or  $\omega$ . More precisely, the former is 1 exactly when  $Y$  is reverse well-ordered, and the latter is 1 exactly when  $Y$  is well-ordered.*

*If  $\mathcal{W}(Y)$  has no greatest element, it has a cofinal family of cylinders with infinite residual alphabet. The dual assertion holds at the lower end.*

*Proof.* Repeatedly remove the greatest still remaining element of  $Y$ , taking unions at limits. This greedy decreasing sequence cannot continue for  $\omega_1$  steps, by lemma 16.1. It therefore stops at a countable ordinal  $\tau$ , either because  $Y$  has been exhausted or because the remaining nonempty set  $B$  has no maximum. In the first case the decreasing enumeration is greatest. In the second, let  $s$  be the removed prefix. Every enumeration not extending  $s$  is smaller than every member of  $C_s$ : at its first disagreement it failed to choose the current maximum. Thus  $C_s$  is a final segment. Choose a strictly increasing cofinal sequence  $b_n$  in  $B$ . The cylinders  $C_{s \smallfrown \langle b_n \rangle}$  are cofinal in  $\mathcal{W}(Y)$ , and each has infinite residual alphabet  $B \setminus \{b_n\}$ . Choosing one completion in each cylinder gives countable cofinality. There is no greatest element by lemma 11.1. Reverse all comparisons for the lower end.  $\square$

[merge] Source 07 proves the same statement by the same greedy stripping (second half of its Lemma 8.1, 07:731–742), including the empty set. Applied to  $Y = \mathbb{R}$ , it gives a second route to the global invariants  $\text{cf } \mathcal{W} = \text{ci } \mathcal{W} = \omega$  of proposition 12.1 (source 06, Corollary 3.6). Source 06 adds that every increasing bijection  $h : \mathbb{R} \rightarrow \mathbb{R}$  induces an automorphism  $e \mapsto h \circ e$  of  $\mathcal{W}$ , and that these natural symmetries and negation need not exhaust the automorphism group (06:325–327).

**Lemma 16.3** (Infinite-disagreement intervals contain large cylinders; source 06, Lemma 6.5). *Let  $e < f$  in  $\mathcal{W}$  have maximal common prefix  $s$ , residual set  $Y_s$ , and first entries  $a = e(\delta)$ ,  $b = f(\delta)$  as in theorem 11.2. If  $Y_s$  is infinite then  $(e, f)$  contains a cylinder with infinite residual alphabet. In particular, such an interval contains a six-element finite completion cylinder, with isolated internal points.*

*Proof.* If some  $c \in Y_s$  lies strictly between  $a$  and  $b$ , the cylinder with next value  $c$  is contained in  $(e, f)$  and still has infinite residual alphabet. Otherwise consider the two end child cylinders.

The orders  $\mathcal{W}(Y_s \setminus \{a\})$  and  $\mathcal{W}(Y_s \setminus \{b\})$  cannot have, respectively, a greatest and a least element: by the proof of theorem 11.2, that would force  $Y_s$  finite. Hence either the  $a$ -child has no maximum or the  $b$ -child has no minimum. In the first case, lemma 16.2 supplies a cylinder with infinite residual alphabet wholly above  $e$  in the  $a$ -child. In the second it supplies such a cylinder wholly below  $f$  in the  $b$ -child. These cylinders are within  $(e, f)$ .

In any cylinder with infinite residual set  $B$ , extend its prefix by listing all but three elements of  $B$ . The remaining completion cylinder has  $3! = 6$  elements and is still contained in  $(e, f)$ . Its four internal points have immediate neighbors and are isolated globally.  $\square$

**Theorem 16.4** (Dense isolation and cardinal invariants; source 06, Theorem 6.6). *The set  $I$  of isolated points of  $\mathcal{W}$  is dense and has cardinality  $2^\kappa$ . Moreover,*

$$d(\mathcal{W}) = \text{cell}(\mathcal{W}) = w(\mathcal{W}) = 2^\kappa.$$

*Proof.* Consider any nonempty open interval  $(e, f)$ . If its first-disagreement residual  $Y_s$  is infinite, use lemma 16.3. If  $Y_s$  is finite,  $[e, f]$  is a finite chain, and every point strictly between its endpoints has immediate neighbors. Every nonempty open set contains a nonempty open interval because  $\mathcal{W}$  has no endpoints. Thus  $I$  is dense.

Fix three reals and take all enumerations of their complement of length  $\kappa$ . There are  $2^\kappa$  such prefixes, by proposition 4.4. Their six-element completion cylinders are disjoint and each contributes four isolated points. Hence  $|I| = 2^\kappa$ . Every dense set must contain every open singleton, so  $d(\mathcal{W}) \geq 2^\kappa$ ; these singletons also witness the same lower bound for cellularity. The usual interval base has size at most  $|\mathcal{W}|^2 = 2^\kappa$ , proving the weight upper bound and all equalities.  $\square$

[merge] The counts are those of proposition 12.1 (source 02) and of source 07's Corollary 8.4 (07:798), proved there by the same reserved three reals; the density of  $I$  is source 06's alone.

**Proposition 16.5** (The perfect remainder and zero-dimensionality; source 06, Proposition 6.7). *The first derived set  $\mathcal{W}' = \mathcal{W} \setminus I$  is closed, nowhere dense, has cardinality  $2^\kappa$ , and has no isolated points in its subspace topology. Thus  $\mathcal{W}'' = \mathcal{W}'$ . The whole space  $\mathcal{W}$  is zero-dimensional.*

*Proof.* The set  $I$  is open and dense, so its complement is closed and nowhere dense. All enumerations of limit length  $\kappa$  are nonisolated, giving  $|\mathcal{W}'| = 2^\kappa$ .

There is a nonisolated point between any two different finite-condensation classes. Indeed, the closed interval between representatives of distinct classes is infinite, so its first-disagreement residual is infinite by theorem 11.2 and convexity of finite cylinders. Apply lemma 16.3, and complete the resulting prefix by an enumeration of its infinite residual set of initial-cardinal length. The resulting total domain is a limit ordinal, so the point has singleton finite block and is nonisolated. Inside a nonsingleton finite block,  $\mathcal{W}'$  retains exactly the first and last points. Neither can become isolated in  $\mathcal{W}'$ , since its outward side has no adjacent different class. Singleton classes have the same property on both sides. This proves  $\mathcal{W}'' = \mathcal{W}'$ .

For zero-dimensionality, the cut at a jump yields clopen initial and final rays. Near a nonisolated point, any side lacking an immediate neighbor has isolated points arbitrarily close, by density of  $I$ ; their jumps supply such clopen cuts. A side with an immediate neighbor already has a jump cut. Combining a

lower and an upper clopen cut gives a clopen convex neighborhood inside any prescribed neighborhood. Isolated points themselves have clopen singleton neighborhoods.  $\square$

The derived-set statement is topological, not an assertion that the order is scattered:  $\mathcal{W}$  contains a copy of  $\mathbb{Q}$ , and much larger dense orders.

## 17 The point-character spectrum of the full order

[merge] This answers, for  $\mathcal{W}$ , the point-character half of source 02's research topic "Exact spectra of cuts and point characters" (section 26). Sources 06 and 07 prove it by the same disagreement-position argument; the statement and proof below are source 06's, and source 07's differing realizations are recorded after the proof.

For  $e : \alpha \rightarrow \mathbb{R}$  in  $\mathcal{W}$ , define

$$\begin{aligned} D^-(e) &= \{\xi < \alpha : \exists a \notin \text{ran}(e \upharpoonright \xi), a < e(\xi)\}, \\ D^+(e) &= \{\xi < \alpha : \exists a \notin \text{ran}(e \upharpoonright \xi), a > e(\xi)\}. \end{aligned}$$

These sets record positions at which an enumeration can branch below or above  $e$ . Write

$$\chi^-(e) = \text{cf}\{f : f < e\}, \quad \chi^+(e) = \text{ci}\{f : e < f\}.$$

Both sides are nonempty. The value 1 means an immediate neighbor.

**Theorem 17.1** (Character from available disagreement positions; source 06, Theorem 8.1). *If  $D^-(e)$  has no largest element and  $\lambda^- = \sup\{\xi + 1 : \xi \in D^-(e)\}$ , then*

$$\chi^-(e) = \text{cf } \lambda^-.$$

*If  $D^-(e)$  has a largest element, then  $\chi^-(e) \in \{1, \omega\}$ . The identical statements with  $+$  and reversed order determine  $\chi^+(e)$ .*

*Proof.* Every  $f < e$  has first disagreement at a position in  $D^-(e)$ , and every such position permits a lower completion. Later first disagreements lie strictly closer to  $e$ . If the position set has no largest element, choosing one lower completion at each position of a cofinal subset gives a cofinal family of size  $\text{cf } \lambda^-$ , and fewer than  $\text{cf } \lambda^-$  points have bounded first-disagreement positions and miss a closer lower branch. This gives the formula.

If the last lower-branch position is  $\delta$ , the sector of all points first disagreeing at  $\delta$  is cofinal below  $e$ . It is a lexicographic sum over the real set of available next values below  $e(\delta)$ , with residual permutation orders as summands. A next-value set with no maximum has cofinality  $\omega$ ; with a maximum, use the last residual summand and lemma 16.2. Hence the sector has cofinality 1 or  $\omega$ . The upper-side proof is dual.  $\square$

[merge] Source 07's Proposition 9.1 (07:832) is the same statement, written  $\chi^-(e) = \text{cf } D^-(e)$ , with the same proof. It adds when an immediate predecessor occurs: the set  $B$  of available next values below  $e(\delta)$  at the last position  $\delta$  must have a greatest value, and the final fiber over it must have a greatest enumeration.

**Proposition 17.2** (No unequal uncountable point characters; source 06, Proposition 8.2). *If both  $\chi^-(e)$  and  $\chi^+(e)$  are uncountable, then they are equal.*

*Proof.* Both disagreement-position sets have no largest element. Let their suprema as in theorem 17.1 be  $\lambda^-$  and  $\lambda^+$ . Suppose  $\lambda^- < \lambda^+$ . At every position at least  $\lambda^-$ ,  $e$  chooses the least unused real. Thus this

whole final part of  $e$  is strictly increasing and has countable length, by [lemma 16.1](#). Every strictly later limit position in that tail, in particular  $\lambda^+$ , has countable cofinality. This contradicts uncountability of  $\chi^+(e)$ . The reverse inequality is handled by a decreasing tail. Therefore the suprema agree, and so do their cofinalities.  $\square$

**Theorem 17.3** (Exact point-character spectrum; sources 06 and 07). *Let  $K = \{1\} \cup \text{Reg}(\leq \kappa)$ , where  $\text{Reg}(\leq \kappa)$  is the set of infinite regular cardinals at most  $\kappa$ . The realized pairs  $(\chi^-(e), \chi^+(e))$ ,  $e \in \mathcal{W}$ , are exactly*

$$\{(u, v) \in K^2 : u = v \text{ or } \min(u, v) \leq \omega\}. \quad (15)$$

*In particular, every infinite regular cardinal at most  $\kappa$  occurs as a one-sided point character, and a point can have pair  $(\rho, \rho)$  for every uncountable regular  $\rho \leq \kappa$ , whereas a proper gap cannot have two uncountable cofinalities ([theorem 18.3](#)).*

*Proof.* The restriction to  $K$  follows from [lemma 14.3](#) (source 06’s Corollary 5.5): every nontrivial infinite cofinality of a linear order is regular, and the sides of a point are bounded by  $\kappa$ . [Proposition 17.2](#) gives the remaining exclusion. We now realize every allowed pair.

For infinite regular  $\rho \leq \kappa$  set  $\theta = \kappa \cdot \rho$ , an ordinal of cardinality  $\kappa$  and cofinality  $\rho$ . To realize  $(\rho, \rho)$ , partition  $\mathbb{R}$  into four-element sets indexed by  $\theta$ . Within each set label the values  $a < b < c < d$  and use the block  $\langle b, a, c, d \rangle$ . The lower branches at  $b$  and upper branches at  $c$  are cofinal in the resulting limit domain  $4 \cdot \theta = \theta$ . The character formula gives  $(\rho, \rho)$ .

To realize  $(\rho, 1)$ , reserve  $\{0, 1\}$ , enumerate the complement using cofinally many decreasing two-element blocks over length  $\theta$ , and append  $\langle 0, 1 \rangle$ . The lower branch positions are cofinal in  $\theta$  and stop there. The last two-element block gives an immediate successor. Reversing real values gives  $(1, \rho)$ .

To realize  $(\rho, \omega)$ , reserve  $\mathbb{N}$ , enumerate  $\mathbb{R} \setminus \mathbb{N}$  in decreasing pairs over length  $\theta$ , and append  $0, 1, 2, \dots$  in its usual order. Lower branches have supremum  $\theta$ ; upper branches are cofinal in the final  $\omega$ -tail. Negation gives  $(\omega, \rho)$ . These constructions include the cases  $\rho = \omega$ . Finally an internal point of a six-element final block realizes  $(1, 1)$ .  $\square$

*Second route* (source 07, Theorem 9.2, 07:861). The exclusions are argued as above. For the realizations fix an infinite regular  $\mu \leq \kappa$  and put  $\lambda = \kappa \cdot \mu$ . Any  $\kappa$ -sized subset of  $\mathbb{R}$  can be partitioned into triples indexed by  $\lambda$ . List each sorted triple as *middle, least, greatest*. The flattened enumeration has length  $3 \cdot \lambda = \lambda$ , and at the first position of each triple both a smaller and a larger value are still unused, so both divergence sets are cofinal in  $\lambda$ , whose cofinality is  $\mu$ . Apply this to all of  $\mathbb{R}$  for  $(\mu, \mu)$ ; to  $\mathbb{R} \setminus \mathbb{N}$ , followed by  $\mathbb{N}$  in increasing order, for  $(\mu, \omega)$ , and negate for  $(\omega, \mu)$ . For  $(\mu, 1)$  reserve two reals  $a < b$ , use the triple construction on their complement, and append  $(a, b)$ ; appending  $(b, a)$  instead gives  $(1, \mu)$ . An interior point of a six-element block gives  $(1, 1)$ .  $\square$

Source 07 notes that if  $\kappa$  is singular,  $\kappa$  itself is not a character. A point’s neighborhood-base cardinality is the maximum of its two one-sided values; when both are 1, it is an isolated point with a one-element local base (source 06). The two-sided spectrum is substantially more informative than this single-cardinal invariant: it detects one-sided endpoints of finite blocks and the asymmetry of countable tails (source 07).

## 18 Gaps of the full order and the size of its completion

[merge] This answers, for  $\mathcal{W}$ , the cut half of source 02’s research topic “Exact spectra of cuts and point characters”. The proof is source 06’s cut spine; source 07’s prefix tracking (its Lemma 10.1 and Theorem 10.2, 07:921–992) follows the same straddling cylinders, and its different realization of the gaps is printed as a second route.

**Definition 18.1** (Source 06, Definition 7.1). A *proper Dedekind gap* is a partition  $\mathcal{W} = L \sqcup U$  into nonempty sets with  $L < U$ , such that  $L$  has no greatest element and  $U$  no least element. Its cofinality pair, or *character*, is  $(\text{cf } L, \text{ci } U)$ ;  $\text{GapChar}(M)$  is the set of characters of the gaps of a linear order  $M$ . A jump is not a proper gap. The Dedekind completion here adjoins one point for each proper gap and keeps all original points and jumps; endpoints may separately be added if desired.

**Lemma 18.2** (Cut-spine alternative; source 06, Lemma 7.2). *Every proper gap is described in one of the following two ways:*

- (i) *At an injective prefix  $p$ , a nonempty proper initial segment  $B$  of  $Y_p$  divides the next-value child cylinders: those with next value in  $B$  lie on the lower side, and the other child cylinders lie on the upper side.*
- (ii) *At a limit-length injective prefix  $p$ , the whole cylinder  $C_p$  lies on one side and is cofinal or cointial on that side. The gap is the lower or upper boundary of  $C_p$ .*

*The length of the describing prefix is below  $\kappa^+$ .*

*Proof.* Say that a cylinder *straddles* the gap if it meets both  $L$  and  $U$ . Start with the empty prefix, whose cylinder is all of  $\mathcal{W}$ . In any straddling cylinder, at most one child cylinder straddles: the children are linearly arranged by their next real value. If none straddles, the children partition into a nonempty initial family contained in  $L$  and a nonempty final family contained in  $U$ . This is (i). If one straddles, extend the prefix by its next value.

At a limit stage take the union of the prefixes. It is an injection and has a completion. If its cylinder still straddles, continue. If the cylinder lies entirely in  $L$ , it is cofinal in  $L$ . To see this, suppose a lower-side point lay above every member of the cylinder. Its first disagreement with the union prefix would occur at an earlier stage. The straddling cylinder immediately after that coordinate would be wholly below this lower-side point, but would also contain an upper-side point, a contradiction. The upper-side case is dual. This gives (ii).

The process cannot continue for  $\kappa^+$  stages, since it would then have chosen  $\kappa^+$  distinct reals. Nor can a singleton exhausted cylinder straddle. Thus it stops at one of the stated configurations.  $\square$

[merge] Source 07's Lemma 10.1 (07:921) is the straddling step and the limit step of this proof; it adds that the first difference used at a limit exists because a complete enumeration cannot properly end inside an injective prefix that still extends further.

**Theorem 18.3** (Gap restriction; source 06, Theorem 7.3). *Every proper gap of  $\mathcal{W}$  has countable cofinality on at least one side. Both sides have infinite regular cofinality at most  $\kappa$ .*

*Proof.* In case (i) of lemma 18.2, the lower-side part inside  $C_p$  is cofinal in  $L$  because  $C_p$  straddles the gap. It is a lexicographic sum over  $B$  of residual permutation orders, by (3). If  $B$  has no maximum, countably many next values are cofinal by lemma 16.1. If  $B$  has a maximum, the last child cylinder has cofinality at most  $\omega$  by lemma 16.2. Therefore  $\text{cf } L \leq \omega$ . The dual argument gives  $\text{ci } U \leq \omega$ . Neither side has an endpoint, so both equal  $\omega$ .

In case (ii), the side on which the whole cylinder is cofinal or cointial has cofinality at most  $\omega$  by lemma 16.2. If that value were 1, the side itself would have an endpoint, contrary to properness. Thus it is  $\omega$ .

The other side is bounded by  $\kappa$  using lemma 14.3. The cofinality of an endpoint-free linear order is an infinite regular cardinal: a cofinal set can be replaced by an increasing cofinal sequence of minimal cardinality, and a singular indexing cardinal would admit a smaller cofinal subsequence. Apply the same fact to the reverse upper side.  $\square$

Thus there is no proper  $(\omega_1, \omega_1)$ -gap, or any proper gap with both cofinalities uncountable. This does not assert countable saturation:  $(\omega, \omega)$ -gaps exist (proposition 14.1), and jumps exist.

**Lemma 18.4** (Boundary cofinality of a partial enumeration; source 06, Lemma 7.4). *Let  $p : \theta \rightarrow \mathbb{R}$  be injective, with  $\theta$  limit, and let*

$$D^-(p) = \{\xi < \theta : \text{some real unused before } \xi \text{ is smaller than } p(\xi)\}.$$

*If  $D^-(p)$  is cofinal in  $\theta$ , then the set of all full enumerations strictly below the entire cylinder  $C_p$  has cofinality  $\text{cf } \theta$ . The reversed statement holds for the upper boundary and  $D^+(p)$ .*

*Proof.* Every enumeration below the cylinder first disagrees with  $p$  at a coordinate in  $D^-(p)$ . Points that first disagree at a later coordinate are closer to the cylinder and greater than those disagreeing earlier. Choosing one completion at each coordinate in a cofinal subset of  $D^-(p)$  gives a cofinal family of size  $\text{cf } \theta$ . Conversely, fewer than  $\text{cf } \theta$  first-disagreement coordinates are bounded in  $\theta$  and miss a closer available lower branch.  $\square$

**Theorem 18.5** (Exact gap spectrum of  $\mathcal{W}$ ; sources 06 and 07).

$$\text{GapChar}(\mathcal{W}) = \{(\omega, \rho), (\rho, \omega) : \rho \in \text{Reg}(\leq \kappa)\}.$$

*In particular a gap cannot have two uncountable sides.*

*Proof.* Theorem 18.3 gives the inclusion  $\subseteq$ . For each infinite regular  $\rho \leq \kappa$  put  $\theta = \kappa \cdot \rho$ , an ordinal of cardinality  $\kappa$  and cofinality  $\rho$ . Partition  $S = (-\infty, 0]$  into pairs indexed by  $\theta$ , which is possible since  $|\theta| = \kappa$ . List each pair in decreasing order, and concatenate the blocks. The resulting prefix  $p$  enumerates  $S$  and has length  $2 \cdot \theta = \theta$ . Its lower-branch set  $D^-(p)$  is cofinal, since the first element of each decreasing pair leaves a smaller partner unused. The residual alphabet is  $T = (0, \infty)$ .

At the lower boundary of  $C_p$ , the lower side has cofinality  $\rho$  by lemma 18.4; it has no greatest element. The upper side has  $C_p \cong \mathcal{W}(T)$  as a coinital suborder, and this order has no minimum and has coinitality  $\omega$  by lemma 16.2. Hence this is a proper  $(\rho, \omega)$ -gap. Negation reverses the pair.  $\square$

*Second route for the realization* (source 07, Theorem 10.2, 07:976–991). Fix an infinite regular  $\mu \leq \kappa$  and put  $\lambda = \kappa \cdot \mu$ . Reserve  $S = (-1, 1)$ , and enumerate  $\mathbb{R} \setminus S$  in  $\lambda$  positions by the middle-least-greatest triple construction of the second route of theorem 17.3. Denote this prefix by  $p$  and its cylinder by  $C$ . It is isomorphic to  $\mathcal{W}(S)$  and has no greatest point, with cofinality  $\omega$ . Form the cut immediately above  $C$ :

$$L = \{x : \exists c \in C (x \leq c)\}, \quad U = \mathcal{W} \setminus L.$$

A point in  $U$  first differs positively from  $p$  at a coordinate below  $\lambda$ . Such positive differences can be made at cofinally many triple starts. Perturbations at a cofinal set of these starts form a coinital family in  $U$  of size  $\mu$ . Fewer than  $\mu$  differences are bounded and can all be improved by a later one, so  $\text{ci } U = \mu$  and  $U$  has no least point. Since  $C$  is cofinal in  $L$ ,  $\text{cf } L = \omega$ . Negation gives  $(\mu, \omega)$ .  $\square$

**Theorem 18.6** (Completion cardinality; source 06, Theorem 7.6). *There are exactly  $2^\kappa$  proper gaps of  $\mathcal{W}$ . Its Dedekind completion, with or without two added endpoints, has cardinality  $2^\kappa$ .*

*Proof.* There are at most  $2^\kappa$  injective prefixes. For each length  $\theta < \kappa^+$  there are at most  $\kappa^{|\theta|} \leq \kappa^\kappa = 2^\kappa$  functions, and there are  $\kappa^+ \leq 2^\kappa$  lengths. A subset of  $\mathbb{R}$  has at most  $\kappa$  initial segments: an initial segment is determined by its supremum in the extended real line, together with whether a possible boundary point is included. The descriptions in lemma 18.2 therefore give at most  $2^\kappa$  proper gaps.

For the reverse inequality, let  $p$  range over all length- $\kappa$  enumerations of  $S = (-\infty, 0]$ . There are  $2^\kappa$  choices. The lower-branch set  $D^-(p)$  is cofinal in  $\kappa$ : otherwise a tail of  $p$  would always choose its least unused value, hence would be an increasing sequence of  $\kappa$  reals, contrary to lemma 16.1. By lemma 18.4,

the lower boundary of  $C_p$  has no greatest lower point. Its upper side has the minimum-free cylinder  $\mathcal{W}((0, \infty))$  coinital. Thus every such boundary is a proper gap. Distinct prefixes of the same length have disjoint ordered cylinders and distinct lower boundaries. This produces  $2^\kappa$  gaps.

Adding one point for each proper gap to the original  $2^\kappa$  points leaves the cardinality unchanged. Two endpoints make no difference.  $\square$

The cardinality is also part of [theorem 14.4](#) (source 02) and of source 07's Corollary 4.6. Source 06 notes, as source 02 does, that the completion still has no increasing or decreasing sequence of length  $\kappa^+$ : such a sequence would give  $\kappa^+$  separated nonempty intervals, from which original points could be chosen in order, since every nonempty open interval of the completion meets the original order.

Under CH the gap spectrum is  $\text{GapChar}(\mathcal{W}) = \{(\omega, \omega), (\omega, \omega_1), (\omega_1, \omega)\}$  (source 07, 07:1057–1063). These are gap characters, not point characters; conflating the two would reverse some of the conclusions.

## 19 Real powers and representation rank

[merge] This section and [sections 23](#) and [24](#) are source 07's. Source 07 measures lexicographic complexity by real coordinates, following the representability terminology of Giarlotta [7]:

$$\text{repr}_{\mathbb{R}}(L) = \min\{\alpha : L \hookrightarrow \mathbb{R}_{\text{lex}}^\alpha\}.$$

For an ordinal exponent every real-valued function is allowed; there is no finite-support restriction. The binary invariant  $\ell_2$  of [definition 5.4](#) satisfies  $\text{repr}_{\mathbb{R}}(L) \leq \ell_2(L) \leq \omega \cdot \text{repr}_{\mathbb{R}}(L)$  ([table 2](#)).

**Theorem 19.1** (Representation ranks; source 07, Theorem 5.3).

$$\text{repr}_{\mathbb{R}}(\mathcal{W}_\kappa) = \text{repr}_{\mathbb{R}}(\mathcal{A}) = \kappa, \quad \text{repr}_{\mathbb{R}}(\mathcal{W}) = \kappa^+.$$

*In fact  $\mathcal{W}$  embeds in  $\mathbb{B}_{\kappa^+}$  as well as in  $\mathbb{R}_{\text{lex}}^{\kappa^+}$ .*

*Proof.*  $\mathcal{W}_\kappa \simeq_e \mathcal{A} \simeq_e \mathbb{R}_{\text{lex}}^\kappa$  ([theorem 7.1](#) and the note after it) and [corollary 5.6](#) give the first identity. If  $\mathcal{W}$  embedded in  $\mathbb{R}_{\text{lex}}^\gamma$  for some  $\gamma < \kappa^+$ , [corollary 8.6](#) would imply  $\mathbb{R}_{\text{lex}}^{\gamma+1} \hookrightarrow \mathbb{R}_{\text{lex}}^\gamma$ , a contradiction.

For the upper bound, extend each complete enumeration  $e : \alpha \rightarrow \mathbb{R}$  by zero to a function on  $\kappa^+$ . Distinct complete enumerations already differ before either ends, so padding changes no comparison. For a binary target, first replace every real coordinate by its  $j$ -code ([lemma 5.5](#)) and then pad by zeros to length  $\kappa^+$ . Each code length is  $\omega \cdot \alpha < \kappa^+$ , and the same prefix-free argument applies.  $\square$

[merge] The binary half of the upper bound, and  $\ell_2(\mathcal{W}) = \kappa^+$ , are source 02's [theorem 8.2](#).

Thus  $\mathcal{W}$  contains every real power strictly below the critical exponent ([corollary 8.6](#)), but not the real power at that exponent. Its representation rank is the critical exponent nonetheless. Ordinal spectrum and representation rank measure genuinely different aspects of an order (source 07, 07:536–539). For fixed-length powers source 07 adds  $\text{repr}_{\mathbb{R}}((\mathcal{W}_\kappa)_{\text{lex}}^\beta) = \text{repr}_{\mathbb{R}}(\mathcal{A}_{\text{lex}}^\beta) = \kappa \cdot \beta$  for every nonzero ordinal  $\beta$  (its Proposition 6.4 and Theorem 11.2): by the note after [corollary 10.4](#) both powers are equimorphic with  $\mathbb{R}_{\text{lex}}^{\kappa \cdot \beta}$ , and [corollary 5.6](#) applies.

*Remark 19.2* (Absorption is not isomorphism; source 07, 07:628–633). Mutual embeddability must not be upgraded to isomorphism.  $\mathcal{W}_{\text{lex}}^\omega$  is dense: between two tuples, increase a coordinate of the smaller tuple strictly after their first difference, keeping all earlier coordinates fixed. This is possible because  $\mathcal{W}$  has no greatest element. The full order  $\mathcal{W}$  is not dense, by its finite blocks ([theorem 11.5](#)). Thus  $\mathcal{W}_{\text{lex}}^\omega \simeq_e \mathcal{W}$  ([theorem 10.2](#)) but  $\mathcal{W}_{\text{lex}}^\omega \not\cong \mathcal{W}$ .

## 20 More on the fixed-length order

[merge] Sources 06 and 07 sharpen [proposition 12.2](#). The full order allows prefixes that have already used  $\kappa$  reals but have not exhausted  $\mathbb{R}$ . In  $\mathcal{W}_\kappa$ , every proper prefix has length less than  $\kappa$ , hence uses fewer than  $\kappa$  values. This elementary difference controls its topology (source 06, 06:956–959).

**Lemma 20.1** (Short residuals remain locally large; source 06, Lemma 9.1). *If  $p : \theta \rightarrow \mathbb{R}$  is injective and  $\theta < \kappa$ , then every nonempty open real interval contains  $\kappa$  elements of  $Y_p$ . Every such prefix has extensions in  $\mathcal{W}_\kappa$ .*

*Proof.* An open real interval has cardinality  $\kappa$ . Removing fewer than  $\kappa$  points leaves cardinality  $\kappa$ , without any regularity assumption on  $\kappa$ . Enumerate  $Y_p$  with domain  $\kappa$  and concatenate; since  $\theta + \kappa = \kappa$ , the resulting full domain is  $\kappa$ .  $\square$

Source 07 states the same fact (07:281–286) and stresses that it is what makes short-prefix cylinders of  $\mathcal{W}_\kappa$  behave differently from long-prefix cylinders of  $\mathcal{W}$ .

**Theorem 20.2** (Dense local universality and a clopen base; source 06, Theorem 9.2).  *$\mathcal{W}_\kappa$  is dense without endpoints. Every nonempty open interval in  $\mathcal{W}_\kappa$  contains a copy of  $\mathbb{B}_\kappa$ , and hence contains every linear order of size at most  $\kappa$ ; in particular every such interval has cardinality  $2^\kappa$ . The short-prefix cylinders*

$$C_p \cap \mathcal{W}_\kappa, \quad \text{dom } p < \kappa,$$

*are clopen and form a base for its order topology.*

*Proof.* Given  $e < f$ , their first disagreement occurs at  $\delta < \kappa$ . By [lemma 20.1](#), an unused real lies strictly between their values there. The corresponding next-value cylinder is inside  $(e, f)$ . Its residual set has cardinality  $\kappa$ , so pair coding followed by the fixed prefix embeds the binary power, still with total length  $\kappa$ . Varying the first real also shows that there are no endpoints.

A short-prefix cylinder has no endpoints: after its prescribed prefix, one can choose unused next values both below and above any given next value. The cylinder is therefore open. If a point lies outside it, its first disagreement with the prefix occurs at a coordinate below the prefix length; a longer short-prefix neighborhood of that outside point remains outside. Thus the complement is open. Finally, inside a prescribed interval around  $e$ , fix its prefix beyond its first disagreements with both bounds. The resulting cylinder lies inside the interval and proves the base assertion.  $\square$

[merge] Source 07 proves the clopen base in its Theorem 7.1 and the copy of  $\mathbb{B}_\kappa$  in every interval in its Corollary 7.2 (07:637, 07:680), by the same between-values construction.

**Proposition 20.3** (Weight and convergence; sources 06 and 07).  *$w(\mathcal{W}_\kappa) = 2^{<\kappa}$ , so that  $d(\mathcal{W}_\kappa) = \text{cell}(\mathcal{W}_\kappa) = w(\mathcal{W}_\kappa) = 2^{<\kappa}$ . No sequence of distinct points of  $\mathcal{W}_\kappa$  converges in its order topology.*

*Proof.* The number of short prefixes is at most  $\kappa^{<\kappa} = 2^{<\kappa}$ : for infinite  $\lambda < \kappa$ ,  $(2^{\aleph_0})^\lambda = 2^\lambda$ , and the finite lengths do not exceed the same supremum. The clopen base of [theorem 20.2](#) gives the weight upper bound; the lower bound is  $\text{cell}(\mathcal{W}_\kappa) = 2^{<\kappa}$  ([proposition 12.2](#)) and  $\text{cell} \leq d \leq w$ .

Given a point  $e$  and countably many other points, their first disagreement coordinates with  $e$  are bounded below  $\kappa$ , since  $\text{cf } \kappa > \omega$ . A longer short-prefix neighborhood of  $e$  excludes them all. Hence a sequence of distinct points cannot converge to  $e$  (source 06, Theorem 9.3).  $\square$

**Corollary 20.4** (A  $P$ -space; source 07, Corollary 7.2). *Intersections of fewer than  $\text{cf } \kappa$  open subsets of  $\mathcal{W}_\kappa$  are open. In particular  $\mathcal{W}_\kappa$  is a zero-dimensional  $P$ -space, is not first countable, and every convergent sequence is eventually constant.*

*Proof.* At a point of an intersection of fewer than  $\text{cf } \kappa$  open sets, choose a prefix cylinder inside each set. Their lengths have supremum below  $\kappa$ , so their union is still a short prefix and its cylinder lies in the intersection. Recall  $\text{cf } \kappa > \omega$ . A  $T_1$   $P$ -space has no non-eventually-constant convergent sequence: otherwise remove a countable set of distinct nonlimit sequence values by intersecting their open complements. This gives a neighborhood of the alleged limit that misses a subsequence. The character computation of [proposition 12.2](#) rules out first countability, and a clopen base gives zero-dimensionality.  $\square$

*Remark 20.5* (Do not confuse global and local cofinality; source 06, Remark 9.4). Although every point of  $\mathcal{W}_\kappa$  has uncountable two-sided character, the whole order has countable cofinality and coinitality: choose enumerations whose first values tend to  $+\infty$  or  $-\infty$ . These are different invariants.

## 20.1 The gaps of the fixed-length order

[merge] This answers, for  $\mathcal{W}_\kappa$ , the cut half of source 02's research topic on spectra.

**Theorem 20.6** (All gaps of the minimal-type stratum; source 07, Theorem 10.3).

$$\text{GapChar}(\mathcal{W}_\kappa) = \{(\omega, \mu), (\mu, \omega) : \mu \in \text{Reg}(< \kappa)\} \cup \{(\text{cf } \kappa, \text{cf } \kappa)\},$$

where  $\text{Reg}(< \kappa)$  is the set of infinite regular cardinals below  $\kappa$ .

*Proof.* Track a gap  $(L, U)$  by short prefixes exactly as in [lemma 18.2](#), but only up to length  $\kappa$ . Every prefix of length below  $\kappa$  has a completion in  $\mathcal{W}_\kappa$ . A split between next-value fibers again gives  $(\omega, \omega)$ : short-prefix cylinders, as well as their child cylinders, have countable cofinality and coinitality by varying the next unused real through a cofinal or coinital sequence.

Suppose instead that at a limit length  $\delta < \kappa$  the union-prefix cylinder  $C$  first lies wholly in  $L$ . It is cofinal in  $L$ , so  $\text{cf } L = \omega$ . The first differences of points in  $U$  from this union prefix are cofinal in  $\delta$ , because every earlier cylinder was straddling. Later such differences are closer to  $C$  than earlier ones. Choosing a cofinal set of indices gives  $\text{ci } U \leq \text{cf } \delta$ , and a family of fewer than  $\text{cf } \delta$  points has bounded differences and is not coinital. Thus the character is  $(\omega, \text{cf } \delta)$ . This second cardinal is regular and below  $\kappa$ . The case  $C \subseteq U$  is dual.

It remains possible that the process reaches length  $\kappa$ , producing an injective sequence  $p : \kappa \rightarrow \mathbb{R}$ . If it is onto, it is a point of  $\mathcal{W}_\kappa$ . The straddling property at every shorter stage then makes that point an endpoint of one side, contrary to the gap assumption. If it is not onto, there is no member of  $\mathcal{W}_\kappa$  extending it. Every member differs from  $p$  somewhere, and comparison with this virtual branch divides  $\mathcal{W}_\kappa$  into the two gap sides. Downward and upward perturbations are possible at every coordinate after fixing any shorter prefix. The bounded-difference argument of [proposition 12.2](#) gives character  $(\text{cf } \kappa, \text{cf } \kappa)$ .

For realization of  $(\omega, \mu)$  with  $\mu < \kappa$  regular infinite, take any injective prefix of length  $\mu$  and cut immediately above its  $\mathcal{W}_\kappa$ -cylinder. This cylinder has cofinality  $\omega$ ; outside positive perturbations approach it with coinitality  $\mu$ . Negation gives the reversed pair. Finally choose a bijection  $p : \kappa \rightarrow \mathbb{R} \setminus \{0\}$ . Its virtual branch gives the  $(\text{cf } \kappa, \text{cf } \kappa)$  gap just described.  $\square$

The last construction illustrates what the full order adds:  $p \frown (0)$  is a genuine point of  $\mathcal{W}$  lying in that previously unfilled cut. This fills the gap by allowing a longer *injective* enumeration, not by introducing repetitions. Longer prefixes also create the mixed gaps of [theorem 18.5](#) whose uncountable side can be any regular cardinal at most  $\kappa$  (source 07, 07:1040–1044). Under CH,  $\text{GapChar}(\mathcal{W}_\kappa) = \{(\omega, \omega), (\omega_1, \omega_1)\}$ .

**Corollary 20.7** (No convex copy; source 07, Corollary 10.4). *Although  $\mathcal{W}_\kappa \subseteq \mathcal{W}$ , there is no order embedding of  $\mathcal{W}_\kappa$  into  $\mathcal{W}$  with convex image.*

*Proof.* cf  $\kappa > \omega$ . The (cf  $\kappa$ , cf  $\kappa$ ) gap of  $\mathcal{W}_\kappa$  would induce a gap with the same character in  $\mathcal{W}$  if its image were convex: any point between its two sides would have to be in the image. [Theorem 18.5](#) forbids such a gap.  $\square$

## 20.2 How longer enumerations fill gaps of the minimal slice

[merge] Source 06's Section 10 relates the full order and its minimal-type slice beyond shared embedding spectra. It uses the order  $\mathcal{I}$  of all injections  $\kappa \rightarrow \mathbb{R}$ ; source 06 calls it  $J$ , and the letter is changed here because source 07's different  $J$  appears in [section 24](#).

**Definition 20.8** (Source 06, Definition 10.1). Let  $\mathcal{I}$  be the lexicographically ordered set of all injections  $p : \kappa \rightarrow \mathbb{R}$ , without requiring surjectivity. Thus  $\mathcal{W}_\kappa \subseteq \mathcal{I}$ .

**Theorem 20.9** (Canonical lexicographic-sum decomposition; source 06, Theorem 10.2). *The first- $\kappa$ -coordinates map gives an order isomorphism*

$$\mathcal{W} \cong \sum_{p \in \mathcal{I}} \mathcal{W}(\mathbb{R} \setminus \text{ran } p), \quad (16)$$

with  $\mathcal{W}(\emptyset)$  interpreted as a singleton. Further,  $\mathcal{W}_\kappa$  is order-dense in  $\mathcal{I}$ , and  $\mathcal{I}$  and  $\mathcal{W}_\kappa$  are mutually order-embeddable.

*Proof.* Every full enumeration has length at least  $\kappa$ . Its restriction to the first  $\kappa$  positions is in  $\mathcal{I}$ , and deleting those positions identifies the corresponding fiber with the residual permutation order. Distinct prefixes compare at a coordinate below  $\kappa$ , so the fibers are arranged exactly as the lexicographic sum in (16).

Given  $p < q$  in  $\mathcal{I}$ , choose an unused real strictly between their values at the first disagreement and complete the resulting short prefix within  $\mathcal{W}_\kappa$ . This gives a point of  $\mathcal{W}_\kappa$  between them. Finally,  $\mathcal{W}_\kappa \subseteq \mathcal{I} \subseteq \mathbb{R}_{\text{lex}}^\kappa$ , and the last order embeds into  $\mathcal{W}_\kappa$  ([theorem 7.1](#) and [lemma 5.5](#)).  $\square$

**Proposition 20.10** (The missing exhaustive branch; source 06, Proposition 10.3). *Every  $p \in \mathcal{I} \setminus \mathcal{W}_\kappa$  determines a proper gap of  $\mathcal{W}_\kappa$  with cofinality pair (cf  $\kappa$ , cf  $\kappa$ ). Inside  $\mathcal{W}$ , the set of all points strictly between the two sides of this gap is exactly  $C_p$ .*

*Proof.* No member of  $\mathcal{W}_\kappa$  agrees with  $p$  on all  $\kappa$  positions, since  $p$  is not onto. Thus comparison with  $p$ , as functions of the same length, partitions  $\mathcal{W}_\kappa$  into a lower and upper part. The available lower and upper branch positions of  $p$  are cofinal in  $\kappa$ : a bound on either would make a  $\kappa$ -long tail of  $p$  monotone in  $\mathbb{R}$ . Every short branch can be completed in  $\mathcal{W}_\kappa$ . The disagreement-coordinate argument therefore gives cofinality cf  $\kappa$  on both sides, with no endpoints.

Every member of  $C_p$  lies above all lower-side members of  $\mathcal{W}_\kappa$  and below all upper-side members. Conversely, a full enumeration not extending  $p$  first disagrees with it at some  $\xi < \kappa$ . A permitted disagreement on the same side at a later coordinate gives a member of  $\mathcal{W}_\kappa$  strictly between that enumeration and  $C_p$ . Hence the enumeration does not lie between *all* lower-side and *all* upper-side members.  $\square$

This identifies an actual mechanism behind the absence of uncountable balanced gaps in the full order: some such gaps in the minimal slice are replaced by entire residual permutation orders, not merely by one point. The decomposition is exact; source 06 does not claim that every gap of  $\mathcal{W}_\kappa$  comes from an injective missing branch, and horizontal countable-cofinality gaps remain a separate phenomenon (06:1122–1127). [merge] [Theorem 20.6](#) (source 07) shows that the missing branches give the only uncountable balanced gaps of  $\mathcal{W}_\kappa$ ; its other gaps have one countable side.

## 21 The dense quotient and its connected completion

[merge] This answers source 02's question whether  $\mathcal{W}/\sim_{\text{fin}}$  is equimorphic with  $\mathcal{W}$  (section 26). Part (a) is source 06's Theorem 11.1; part (b) is source 07's Proposition 10.5 beyond (a).

**Theorem 21.1** (Removing finite permutation blocks; sources 06 and 07).

(a) *The order  $\mathcal{W}/\sim_{\text{fin}}$  is dense without endpoints, has cardinality  $2^\kappa$ , and has exactly the same ordinal embedding spectrum as  $\mathcal{W}$ . In fact  $\mathcal{W}/\sim_{\text{fin}}$  and  $\mathcal{W}$  are mutually order-embeddable. Its Dedekind completion is a linear continuum of cardinality  $2^\kappa$ , with no endpoints and with no copy of  $\kappa^+$  or  $(\kappa^+)^{\text{op}}$ .*

(b)  $d(\mathcal{W}/\sim_{\text{fin}}) = \text{cell}(\mathcal{W}/\sim_{\text{fin}}) = w(\mathcal{W}/\sim_{\text{fin}}) = 2^\kappa$ , and  $\text{GapChar}(\mathcal{W}/\sim_{\text{fin}}) = \text{GapChar}(\mathcal{W})$ .

*Proof of (a)* (source 06). The classes are nonempty and finite, so their number is  $2^\kappa$ . There are no first or last classes, since an endpoint class would have an endpoint in  $\mathcal{W}$ . If two distinct classes had no class strictly between them, their finite union would be convex, contradicting maximality of the classes. Thus  $\mathcal{W}/\sim_{\text{fin}}$  is dense. Choosing one representative from each class embeds  $\mathcal{W}/\sim_{\text{fin}}$  in  $\mathcal{W}$ .

For the reverse embedding choose an increasing bijection  $h : \mathbb{R} \rightarrow (0, 1)$  and a fixed enumeration  $t : \kappa \rightarrow \mathbb{R} \setminus (0, 1)$ . For  $e : \alpha \rightarrow \mathbb{R}$ , let  $E(e) = (h \circ e) \smallfrown t$ . This is a full enumeration, and its domain  $\alpha + \kappa$  is a limit ordinal of cardinality  $\kappa$ . Distinct  $e, f$  disagree before either finishes, by lemma 4.1; hence applying  $h$  preserves their first disagreement, and appending  $t$  does not alter it. Thus  $E$  is an order embedding into  $\mathcal{W}$ . Every  $E(e)$  has a singleton finite-condensation class because its domain is a limit ordinal. Composing with the quotient map is consequently an order embedding  $\mathcal{W} \hookrightarrow \mathcal{W}/\sim_{\text{fin}}$ .

A proper gap of  $\mathcal{W}/\sim_{\text{fin}}$  lifts to a proper gap of  $\mathcal{W}$  by taking unions of its classes. Neither lifted side can gain an endpoint, since that endpoint's class would be an endpoint on the corresponding side of  $\mathcal{W}/\sim_{\text{fin}}$ . Therefore  $\mathcal{W}/\sim_{\text{fin}}$  has at most  $2^\kappa$  proper gaps. Its completion has cardinality  $2^\kappa$ , since  $\mathcal{W}/\sim_{\text{fin}}$  already does. Being a complete dense order, it is a linear continuum; the usual order-topological connectedness proof applies. For clarity, if a separation of a nontrivial interval existed, the supremum of the part on one side would violate openness of one of the separating sets. Finally, a  $\kappa^+$ -long monotone sequence in this completion would yield one in its dense suborder  $\mathcal{W}/\sim_{\text{fin}}$  by choosing intervening points, and hence one in  $\mathcal{W}$ , which is impossible.  $\square$

*Proof of (b), and a second route to the equimorphism* (source 07). Source 07 embeds  $\mathcal{W}/\sim_{\text{fin}}$  into  $\mathcal{W}$  by the least point of each block, and  $\mathcal{W}$  into  $\mathcal{W}/\sim_{\text{fin}}$  by  $e \mapsto (h \circ e) \smallfrown q \smallfrown n$ , with  $h : \mathbb{R} \rightarrow (-2, -1)$  increasing,  $q$  an enumeration of  $\mathbb{R} \setminus ((-2, -1) \cup \mathbb{N})$  and  $n$  the increasing enumeration of  $\mathbb{N}$ ; every image has a limit domain, so lies in a singleton block.

For the density lower bound, fix  $S = [0, 1]$ . There are  $2^\kappa$  distinct prefixes of type  $\kappa$  enumerating  $\mathbb{R} \setminus S$ . Their disjoint convex cylinders each contain at least three distinct singleton finite blocks, obtained from limit-length completions. After projection each contains a nonempty open interval. These intervals witness cellularity  $2^\kappa$ ; cardinality gives the weight upper bound as before.

A gap cannot cut through a finite block, because a nontrivial split of a finite block would supply endpoints. Conversely a gap of the quotient lifts to a gap of  $\mathcal{W}$ . Infinite cofinalities on either side are unchanged by replacing points by finite blocks. This proves the spectrum assertion.  $\square$

Adding two endpoints to this completion gives a compact connected ordered space of the same cardinality. This connected object is different from the completion of  $\mathcal{W}$  itself: the latter retains jumps. Finite condensation is an essential step in the connected construction (source 06).

The quotient and  $\mathcal{W}_{\text{lex}}^\omega$  (remark 19.2) are both dense and mutually embeddable with  $\mathcal{W}$ , but they are not isomorphic to each other. Every point of  $\mathcal{W}_{\text{lex}}^\omega$  has character  $(\omega, \omega)$ , by perturbing later and later factors. The quotient retains, for example, the  $(\omega_1, \omega_1)$  character of a singleton limit-domain block constructed in theorem 17.3 (source 07, 07:1103–1108).

## 22 Bounded surreal sign orders

[merge] This section is source 06’s. The repository’s sign-sequence foundations suggest a useful comparison, but the termination convention differs. A surreal sign sequence is compared using

$$- < \text{termination} < +$$

at the first place where the sequences cease to agree. In contrast, exhaustive enumerations are prefix-free, so termination never participates in their comparison. For an ordinal  $\theta$ , define

$$S_{<\theta} = \bigcup_{\alpha < \theta} \{-, +\}^\alpha,$$

with this sign order. One may view it as the order of surreal numbers whose sign-sequence lengths, or birthdays, are below  $\theta$ ; no field structure is used here.

**Theorem 22.1** (Every bounded sign order embeds, but their full union does not; source 06, Theorem 12.1). *For every  $\theta < \kappa^+$ ,  $S_{<\theta} \hookrightarrow \mathcal{W}$ . However,*

$$S_{<\kappa^+} \not\hookrightarrow \mathcal{W}, \quad |S_{<\kappa^+}| = |\mathcal{W}| = 2^\kappa.$$

*Proof.* Pad a sign sequence  $s$  of length  $\alpha < \theta$  to length  $\theta$  by using the alphabet  $\{0, 1, 2\}$ : write 0 for  $-$ , 2 for  $+$ , and 1 at all coordinates from  $\alpha$  onward. At the first disagreement this implements exactly the sign/termination convention, so it is an order embedding into  $(\{0, 1, 2\}^\theta, <_{\text{lex}})$ . Replace the symbols 0, 1, 2 by the binary blocks 00, 01, 10. This embeds into  $\mathbb{B}_{2, \theta}$ , which embeds into  $\mathcal{W}$  because  $|2 \cdot \theta| \leq \kappa$  (lemma 8.1).

For each  $\alpha < \kappa^+$ , the all-plus sequence of length  $\alpha$  is strictly below the all-plus sequence of any greater length. Thus the full union contains  $\kappa^+$ , excluded by theorem 6.2. Its cardinality is  $2^\kappa$ : the length- $\kappa$  level gives the lower bound, and there are  $\kappa^+$  levels, each of size at most  $2^\kappa$ .  $\square$

This is a concrete warning against a tempting union argument. Every bounded stage embeds, yet the set-sized union of all stages below  $\kappa^+$  does not, even though its cardinality is no larger than the target’s. The issue is the order structure of the union, not cardinality or a proper-class paradox. The construction above supplies no coherent family of embeddings whose union would be an embedding of the full sign order, and the theorem shows that such coherence is impossible.

The sign-simplicity module inspected by source 06, `Algebra/SurrealNumbers/Surreal/Foundations/SignSequenceSimplicity.lean`, proves uniqueness of the minimum-birthday element of a nonempty convex set of sign sequences and separates that statement from existence of cut separators [13]. That distinction is relevant here: prefix and simplicity arguments alone do not establish cut filling. The cut-filling and non-filling statements of this report instead come from exhaustive completion and the cut spine. [merge] The module is context only: none of its declarations states a result of this report.

[Added 2 October 2026, batch 80: for an infinite cardinal  $\theta$  the surreal-well-orders report takes  $S_{<\theta}$  itself as the alphabet and computes the binary length of its minimal slice; see the dated note after the question “Other ground orders” in section 26.]

## 23 The Kanovei–Shelah index: further topology, gaps and fibres

[merge] Source 07’s Section 11 studies the index  $\mathcal{A}$  of (2) with the binary characteristic-function order on  $\mathcal{P}(\mathbb{N})$ . Its embedding type (source 07’s Lemma 11.1 and Theorem 11.2) is theorem 7.1 and theorem 19.1, and its cofinality, coinitality, density, cellularity and point characters are proposition 13.2. The formal definition includes principal ultrafilters; the embedding and topological results remain true when the family is restricted to free ultrafilters, with the changes indicated below. Thus a difference in raw enumeration rules does not change the embedding class at length  $\kappa$ , but it does change the order-isomorphism type.

**Proposition 23.1** (Base, weight and the free variant; source 07, Theorem 11.3). *The order topology of  $\mathcal{A}$  has a clopen base of admissible prefixes of length below  $\kappa$ , and  $w(\mathcal{A}) = 2^{<\kappa}$ . The free-ultrafilter variant of  $\mathcal{A}$  also has coinitality  $\omega$ . Intersections of fewer than  $\text{cf } \kappa$  open subsets of  $\mathcal{A}$  are open, and every convergent sequence in  $\mathcal{A}$  is eventually constant.*

*Proof.* Fix  $a \in \mathcal{A}$  with range  $U$ . Arbitrarily late coordinates have value strictly below  $\mathbb{N}$ : otherwise the range would have cardinality below  $\kappa$ . At any such coordinate replace the value by  $\mathbb{N}$ , and complete the remaining coordinates so that the range is still  $U$ . This gives arbitrarily late upward perturbations. There are also arbitrarily late downward perturbations. If  $U$  has no least member in its binary order, any coordinate value has a smaller member of  $U$ . If  $U$  has a least member, the tail cannot be constantly equal to it, again because  $|U| = \kappa$ . Choose a late nonleast value and replace it by a smaller member of  $U$ . The unused coordinates have order type  $\kappa$ , so a full surjection onto  $U$  can always be appended after a short altered prefix. The same argument as in [theorem 20.2](#) gives clopen prefix cylinders and a base; there are at most  $2^{<\kappa}$  short prefixes, and  $w \geq \text{cell} = 2^{<\kappa}$  by [proposition 13.2](#).

For coinitality of the free variant, replace the singletons  $\{n\}$  of the proof of [proposition 13.2](#) by the cofinite sets  $[n, \infty) \cap \mathbb{N}$  and complete to any fixed free ultrafilter. The  $P$ -space assertions follow as in [corollary 20.4](#). These topological facts do not supply order-theoretic saturation.  $\square$

**Proposition 23.2** (Two explicit gaps in the ultrafilter index; source 07, Proposition 11.4). *The order  $\mathcal{A}$  has gaps of characters  $(\omega, \omega)$  and  $(\text{cf } \kappa, \text{cf } \kappa)$ . In particular it is not an  $\eta_1$ -order.*

*Proof* (the  $(\omega, \omega)$  gap is a second route to [proposition 14.2](#)). Fix a free ultrafilter  $U$ . Let  $a_m$  begin with  $\mathbb{N} \setminus \{m\}$ , and let  $b_n$  begin with  $\mathbb{N}, [n, \infty) \cap \mathbb{N}$ , completing both to surjections onto  $U$ . The  $a_m$  increase, the  $b_n$  decrease, and  $a_m < b_n$  for every  $m, n$ . A point between them all would have to have first value  $\mathbb{N}$  and second value  $\emptyset$ . The latter is in no ultrafilter. More explicitly, a first value below  $\mathbb{N}$  lies below some  $\mathbb{N} \setminus \{m\}$ ; if the first value is  $\mathbb{N}$ , a nonempty second value lies above some  $[n, \infty) \cap \mathbb{N}$ . Thus the two countable families really determine complementary gap sides, both with countable character.

For the other gap, put  $v = \mathbb{N} \setminus \{0\}$  and consider the constant virtual branch  $p(\xi) = v$  for  $\xi < \kappa$ . Its singleton range is not an ultrafilter, so every member of  $\mathcal{A}$  differs from it somewhere. Every short constant prefix has extensions on both sides: at the next coordinate use  $\mathbb{N} \setminus \{0, 1\} < v$  or  $\mathbb{N} > v$  and then enumerate  $U$ . Arbitrarily late deviations give cofinal and coinital families of size  $\text{cf } \kappa$ ; fewer deviations have bounded indices and are not sufficient. Comparing with  $p$  therefore gives a  $(\text{cf } \kappa, \text{cf } \kappa)$  gap.  $\square$

*Remark 23.3* (The labelling fibres are not convex; source 07, 07:1243–1247). The fibers of the labelling map  $a \mapsto \text{ran}(a)$  are not convex. For example, let  $U_0, U_1$  be principal at 0, 1 respectively. Two enumerations of  $U_0$  can start with  $\{0\}$  and  $\mathbb{N}$ , while an enumeration of  $U_1$  starts with  $\{0, 1\}$  and lies between them. Thus identifying all repeated copies of each ultrafilter is not an ordered quotient of  $\mathcal{A}$  by convex classes.

## 24 A well-ordering-based ultrafilter index

[merge] This section is source 07's Section 12. It connects the literal well-ordering interpretation to ultrafilters directly, as a separate construction, not as a claim about the notation or exact index of [1]. Source 07 writes  $J$  for the order below; it is  $\mathcal{J}$  here, because source 06's  $J$  is  $\mathcal{I}$  ([definition 20.8](#)).

Let

$$\mathcal{J} = \{e : \kappa \rightarrow \mathcal{P}(\mathbb{N}) : e \text{ is bijective}\},$$

with the characteristic-function alphabet and the first-difference order. Every  $e \in \mathcal{J}$  canonically determines a free ultrafilter  $U_e$ . Begin with the cofinite filter. At stage  $\xi < \kappa$ , adjoin  $e(\xi)$  and close under the filter operations if this remains a proper filter; otherwise leave the filter unchanged. At limit stages take unions.

**Theorem 24.1** (Source 07, Theorem 12.1). *The resulting  $U_e$  is a free ultrafilter, is definable from  $e$ , and every free ultrafilter on  $\mathbb{N}$  equals  $U_e$  for some  $e \in \mathcal{J}$ . Moreover  $\mathcal{J} \simeq_e \mathbb{R}_{\text{lex}}^\kappa$ .*

*Proof.* If adjoining a set  $X$  to a filter  $F$  makes it improper, some  $Y \in F$  has  $Y \cap X = \emptyset$ , so  $\mathbb{N} \setminus X \in F$  already. Thus every set or its complement is present after that set has been considered. Increasing unions at limits are proper filters because every finite intersection already occurs at some earlier stage. At the end the filter decides every set and is an ultrafilter. It contains every cofinite set, so cannot be principal. All choices during this recursion are determined by  $e$ , giving definability from that parameter.

Fix a free ultrafilter  $U$ . Order the complementary pairs in type  $\kappa$ , and in each pair list the  $U$ -member first and its complement second. The resulting enumeration has domain  $2 \cdot \kappa = \kappa$ . Inductively the current filter stays inside  $U$ . Its next  $U$ -member can always be adjoined properly, and that member prevents adjoining its complement at the following step. Consequently the final filter is  $U$ .

Pair switches on the alphabet  $\mathcal{P}(\mathbb{N})$  give  $\mathbb{B}_\kappa \hookrightarrow \mathcal{J}$ , just as in [proposition 4.4](#); the ternary coordinate map  $T$  of [lemma 5.5](#) gives  $\mathcal{J} \hookrightarrow \mathbb{R}_{\text{lex}}^\kappa$ , and  $\mathbb{R}_{\text{lex}}^\kappa \simeq_e \mathbb{B}_\kappa$  proves the last assertion.  $\square$

This supplies a definable linearly ordered family of labels covering every free ultrafilter, without choosing a preferred global well-order. It is a literal well-order-enumeration counterpart of the repeated-index idea.

In the published finite-support construction, a finite set of labels is sorted before applying its ultrafilter quantifiers; no well-order of the entire label set is required. A further  $\omega_1$ -stage iteration supplies countable saturation [[1](#), §§2–4]. The gaps proved here belong to index orders, not to that final hyperreal field. An embedding or even an isomorphism of unlabelled index orders need not preserve their ultrafilter labels or the resulting ultrapower construction.

## 25 Foundational scope and interpretation

### 25.1 A definable order is not a selected well-order

The set of all well-order relations on  $\mathbb{R}$  is definable from  $\mathbb{R}$ , and first-difference comparison can be expressed without choosing one member of that set as privileged. This observation does not produce a definable well-ordering of  $\mathbb{R}$ . It defines a linear order *on the collection* of all its well-orderings. Likewise, forming all the presentations in (2) does not select one ultrafilter or one presentation.

Our proofs nevertheless use Choice. It supplies well-orderings of the real line, partitions into pairs, selected enumerations of residual sets, and simultaneous choices of embeddings. In ZF alone,  $\mathbb{R}$  need not be well-orderable, so the literal collection  $\mathcal{W}$  may be empty. Even when that collection is nonempty, the arguments using families of choices should not automatically be read as proofs in a weaker theory.

The set-sized nature of the construction is also essential. Every possible length is below  $\kappa^+$ , and every well-ordering is a subset of  $\mathbb{R}^2$ . There is no proper class of elements hidden in  $\mathcal{W}$ , despite its unbounded range of lengths below  $\kappa^+$ . In contrast, the phrase “all ordinals” denotes a proper class and does not describe a suborder of this set.

### 25.2 Which aspects are genuinely independent?

The proofs separate four kinds of information. The ordinal spectrum records how far a single monotone well-ordered chain can run. Binary length measures the depth needed to distinguish all points through ordered binary cuts. Adjacency records what happens when only finitely many ground elements remain. Finally, topology and cut cofinalities describe local approach and missing limits. These quantities interact, but none should be silently substituted for another.

For example,  $\mathcal{W}_\kappa$  and  $\mathcal{A}$  have identical embedding classes yet unequal global cofinalities. The full order and its completion have the same ordinal spectrum and binary length, but differ in completeness. The strata  $\mathcal{W}_{\kappa+2}$  and  $\mathcal{W}_{\kappa+3}$  have different binary lengths and different finite blocks, though their cardinalities and ordinal spectra coincide. Under some cardinal-arithmetic assumptions even density fails to distinguish those last two orders; their isolated points still do.

All embeddings in this article are purely order-theoretic. An order embedding need not be continuous for the two order topologies. In particular, the fact that a cube or a complete order embeds into a dense order does not transfer its order-topological compactness to the ambient space. No topological or model-theoretic consequence is inferred from an embedding without a separate argument.

### 25.3 Choice and cardinal arithmetic in sources 06 and 07

[merge] Source 06 (its Section 13.1) and source 07 (its Section 13.1) make the same points independently, and add the following. The order comparison is defined from canonical ordinal enumerations and the given real order; it does not choose a preferred well-ordering of  $\mathbb{R}$ . ZFC is used to ensure that these enumerations exist, to choose pair partitions and background completions, and to use the regularity of the successor cardinal in the stabilization proof. Once a background well-order of an alphabet is fixed, every one of its subsets has a specified completion by restriction, which avoids a fresh choice at every prefix (source 06). The no-long-chain argument applies to every infinite well-orderable alphabet; it does not rely on the completeness or density of  $\mathbb{R}$  (source 07).

Neither source proves that full AC is necessary for every theorem, and replacing ZFC by a particular weaker choice hypothesis would require a separate audit of the cardinal and cofinality arguments. In ZF the defining formula for the family of well-order relations on  $\mathbb{R}$  still makes sense, but its nonemptiness says that  $\mathbb{R}$  is well-orderable. A definable *family* is not a definable *chosen member*; the absence of a least point in the lexicographic order explains why taking “the lexicographically least well-order” cannot perform such a selection. No assertion here gives a Borel real code for a well-order of all reals, and no least choice principle sufficient for every theorem has been identified (source 07).

No continuum hypothesis, generalized continuum hypothesis, forcing axiom, large cardinal, or regularity assumption on  $\mathfrak{c}$  is used. In particular  $|\mathcal{W}| = 2^\mathfrak{c}$  must not be replaced by  $\mathfrak{c}^+$  without an additional cardinal-arithmetic hypothesis. Under CH the ordinal barrier is  $\omega_2$  and  $d(\mathcal{W}_\kappa) = \omega_1$ , while  $|\mathcal{W}| = 2^{\omega_1}$  can still require a separate cardinal-arithmetic specification (source 06).

## 26 Further research questions and topics

The following are questions proposed by this investigation. We have not verified that every one is open in the published literature. The solved fixed-stratum *equimorphism* problem is deliberately excluded from the list; the next issues concern sharper structure and broader universality.

[merge] Source 06’s seven and source 07’s seven research questions are merged into source 02’s ten topics; each is printed once, under the topic it shares, citing every source that asks it. Neither source 06 nor source 07 advertises its questions as catalogued open problems: source 06 ties each to a proved limitation or construction of its own text, and source 07 states them so that a solution does not repeat a result already proved. Source 06’s Research question 14.1 (“Unconditional comparison with one fixed lexicographic power”) is answered in ZFC by [corollaries 8.3 and 8.4](#) and is not repeated. Source 06 had suggested “an ordinal-valued lexicographic-dimension invariant rather than cardinality alone” as the approach, warning that any argument must respect the known limitations of converses based solely on absence of long monotone sequences [8]; the two answers use exactly such invariants,  $\ell_2$  and  $\text{repr}_{\mathbb{R}}$ . The topic on spectra of cuts and point characters is re-scoped below, and the quotient part of the second topic is answered. Source 07 names the square-isomorphism question ([research question 26.5](#)) as the most direct next mathematical target and the ordinal-spectrum theorem as the most direct verification target.

## 26.1 An intrinsic embedding criterion for the full order

Find a necessary and sufficient condition on a set-sized linear order  $L$  for  $L \hookrightarrow \mathcal{W}$ . We have an exact answer when  $\ell_2(L) \leq \kappa$  and exact answers for the binary and real-coordinate powers, but not for every order of cardinality  $2^\kappa$  and binary length  $\kappa^+$ . Absence of  $\kappa^+$  and its reverse is necessary; this article does not prove it sufficient.

A concrete route is to study trees of convex partitions whose branches consume distinct symbols from an alphabet of size  $\kappa$ . The exhaustion argument shows what a forbidden branch would do. The remaining problem is to characterize when a compatible coding can be organized across all branches, not merely along each branch separately.

[merge] Sources 06 and 07 ask the same question.

**Research question 26.1** (An intrinsic embedding criterion for large orders; source 06, Research question 14.2). Which linear orders of cardinality between  $\kappa$  and  $2^\kappa$  embed into  $\mathcal{W}$ ?

Source 06 has several sufficient classes: all orders of size at most  $\kappa$ , all binary and real powers of fixed length below  $\kappa^+$ , and orders with small separating families (remark 15.2). Necessary conditions include the absence of  $\kappa^+$  and its reverse. A useful characterization should explain the additional role of an exhaustion-compatible tree representation. Neither necessity nor sufficiency beyond what is proved should be assumed.

**Research question 26.2** (An exact criterion for embedding in  $\mathcal{W}$ ; source 07, Research question 14.1). Characterize the linear orders  $L$  that embed in  $\mathcal{W}$ . The necessary conditions

$$|L| \leq 2^\kappa, \quad \kappa^+ \not\hookrightarrow L, \quad (\kappa^+)^{\text{op}} \not\hookrightarrow L, \quad \text{repr}_{\mathbb{R}}(L) \leq \kappa^+$$

are not shown to be sufficient. Every order with representation rank *strictly below*  $\kappa^+$  does embed (corollary 8.6), so the unresolved part of this criterion is the boundary rank  $\kappa^+$ . Seek either a sufficient intrinsic prefix-tree condition or a counterexample satisfying these necessary bounds.

## 26.2 Can the full order absorb its completion?

Decide whether

$$\widehat{\mathcal{W}} \hookrightarrow \mathcal{W}.$$

The affirmative direction would make the two orders equimorphic, since the other embedding is automatic. The invariants in theorem 14.4 do not obstruct it, and the power-absorption theorem suggests considerable room for coding cuts. A proof must nevertheless assign codes to different cuts coherently. Independently choosing a cofinal sequence for each cut need not preserve their lexicographic comparison.

A related problem asks whether the dense finite-condensation quotient  $\mathcal{W}/\sim_{\text{fin}}$  is equimorphic with  $\mathcal{W}$ , and whether its completion has a more recognizable presentation.

[merge] *Answered in part.* The quotient is equimorphic with  $\mathcal{W}$  (theorem 21.1, sources 06 and 07), and its completion is a linear continuum of cardinality  $2^\kappa$ . A recognizable presentation of the completion, and whether  $\widehat{\mathcal{W}} \hookrightarrow \mathcal{W}$ , remain open; sources 06 and 07 ask the following.

**Research question 26.3** (Internal structure of the dense quotient and its completion; source 06, Research question 14.5). Can the dense finite-condensation quotient  $\mathcal{W}/\sim_{\text{fin}}$ , or its connected completion, be characterized by a natural uniqueness theorem?

Cardinality, the ordinal spectrum, and the gap spectrum alone need not characterize a linear order. The canonical injection-indexed decomposition (16) suggests stronger local invariants. One can also ask which compact connected ordered spaces arise from replacing the residual fibers by their own dense quotients before completion.

**Research question 26.4** (A canonical description of the completions; source 07, Research question 14.4). Give a normal form for every cut using the prefix-tracking proof, including its first stopping stage and remaining alphabet. The cardinalities and gap characters of the completions are known, but an explicit isomorphism-level description is not. Compare the completion of  $\mathcal{W}_\kappa$  with expansions of its non-surjective terminal branches into full residual-enumeration cylinders. The latter process must also account for the new mixed gaps of  $\mathcal{W}$ .

### 26.3 Isomorphism types beyond equimorphism

Determine the order-isomorphism types of individual strata and their natural quotients. In particular, equimorphism of  $\mathcal{W}_\lambda$  and  $\mathcal{W}_{\lambda+1}$  does not settle whether they are isomorphic. Their presentations impose different exhaustion conditions, and an isomorphism need not preserve the represented ordinal length.

The comparison theorem [corollary 9.4](#) supplies a complete baseline: the task is no longer to decide which fixed strata embed, but to identify additional invariants capable of distinguishing equimorphic ones. Cut spectra, automorphism orbits, and local characters are plausible candidates.

[merge] The cut and character spectra of  $\mathcal{W}$  and  $\mathcal{W}_\kappa$  are now known ([theorems 17.3, 18.5 and 20.6](#)). Source 07 asks the analogous question for the powers of  $\mathcal{W}$ .

**Research question 26.5** (Isomorphism rather than mutual embeddability; source 07, Research question 14.2). Determine whether  $\mathcal{W}_{\text{lex}}^2 \cong \mathcal{W}$ . More generally classify the nonzero successor exponents  $\beta < \kappa^+$  for which  $\mathcal{W}_{\text{lex}}^\beta \cong \mathcal{W}$ . The countable limit power is already excluded by density ([remark 19.2](#)), and all these powers are mutually embeddable with  $\mathcal{W}$ . Finite-block size, global cofinality, and representation rank alone do not settle the square.

### 26.4 Automorphisms and recovery of the prefix structure

Every increasing bijection  $h : \mathbb{R} \rightarrow \mathbb{R}$  induces an automorphism  $e \mapsto h \circ e$  of  $\mathcal{W}$ . Determine the full automorphism group and whether some canonical fragment of the prefix tree can be reconstructed from the external linear order alone.

Finite-condensation classes and their sizes are intrinsic and therefore preserved. It is less clear how much of the non-finite ordinal remainder of an enumeration can be recovered. One useful intermediate target is a classification of the orbits of endpoints and interior points of the finite blocks, together with the singleton-block points.

[merge] Sources 06 and 07 ask the same question.

**Research question 26.6** (Automorphisms and recoverable structure; source 06, Research question 14.3). Which features of an enumeration are recoverable from the abstract order  $\mathcal{W}$ , and what is its full automorphism group?

Finite-condensation class size and one-sided cofinalities are order invariants. Ordinary increasing bijections of  $\mathbb{R}$  give automorphisms, but that construction alone does not show that the ordinal length of an enumeration is invariant under every automorphism. A first concrete target is the orbit structure of singleton finite blocks with equal point characters.

**Research question 26.7** (Automorphisms and finite condensation; source 07, Research question 14.3). Describe the order-automorphism group of  $\mathcal{W}$  and of its dense finite-block quotient. Increasing bijections of  $\mathbb{R}$  act on enumerations, and negation reverses the order, but these need not exhaust the possibilities. Are block size, position inside the block, and the two external point characters enough to determine an orbit? Determine which information about an enumeration's ordinal domain can be recovered from the unlabelled order structure.

## 26.5 Exact spectra of cuts and point characters

[merge] *Re-scoped.* For  $\mathcal{W}$  this topic is answered by sources 06 and 07: the point-character spectrum is (15) (theorem 17.3) and the gap spectrum is  $\{(\omega, \rho), (\rho, \omega) : \rho \in \text{Reg}(\leq \kappa)\}$  (theorem 18.5). For  $\mathcal{W}_\kappa$  the point characters are constant (proposition 12.2) and source 07 computes the gap spectrum (theorem 20.6). For  $\mathcal{A}$  the point characters are constant (proposition 13.2), but only two of its gap characters are known (propositions 14.2 and 23.2). The topic therefore stays open for the longer strata  $\mathcal{W}_\beta$  with  $\kappa < \beta < \kappa^+$ , for the gap spectrum of  $\mathcal{A}$ , and for the convex-suborder criteria of the second paragraph below. Source 02's original text follows.

Classify all pairs of left and right cofinalities occurring at points and unrealized cuts of  $\mathcal{W}$  and its fixed strata. We have computed the local pair (cf  $\kappa$ , cf  $\kappa$ ) throughout  $\mathcal{W}_\kappa$ , exhibited missing  $(\omega, \omega)$  cuts, and bounded every side of a full-order cut by  $\kappa$ . Those facts leave many cases undecided.

A productive decomposition may separate three sources of cuts: a missing next coordinate in an ordered residual set, a limit of increasingly long injective prefixes, and exhaustion at the end of a finite block. This could also lead to exact criteria for convex suborders to be complete, separable, or first countable.

[merge] The cut-spine proof of lemma 18.2 realizes source 02's proposed decomposition: case (i) is the missing next coordinate in an ordered residual set, case (ii) the limit of increasingly long injective prefixes.

## 26.6 Other ground orders

The ordinal cutoff and full binary length were proved for every infinite linearly ordered ground set  $X$ . The all-strata classification used  $X \subseteq \mathbb{R}$ , particularly countable separating cuts and rational coordinate codes. Determine the weakest hypotheses on  $X$  under which

$$\mathcal{W}_{\lambda+n}(X) \simeq_e \mathbb{B}_\lambda \times_{\text{lex}} n!$$

still holds. For orders with large intrinsic binary length, develop a replacement formula that records both coordinate complexity and the shrinking size of the residual alphabet.

This question naturally includes nonseparable real-closed fields and set-sized suborders of surreal numbers. The distinction between the underlying order and any algebraic structure must remain explicit.

[merge] Sources 06 and 07 ask the same question.

[Added 2 October 2026, batch 80: answered in part for one family of set-sized suborders of surreal numbers, at the minimal stratum  $n = 0$ . The surreal-well-orders report, `Algebra/SurrealNumbers/docs/foundations-and-computation/surreal-well-orders/`, merged from batch-80 manuscripts 08, 09, 11 and 12, takes  $X = S_{<\theta}$  for an infinite cardinal  $\theta$ , so that  $\lambda = |S_{<\theta}| = 2^{<\theta}$  (it writes  $\text{No}_{<\theta}$ ; the order is the sign order of section 22). Its Theorem 8.3 (label `swo:st:thm:transition`, source 12's) proves  $\mathcal{W}_\lambda(S_{<\theta}) \simeq_e \mathbb{B}_\lambda$ , which is the case  $n = 0$  of the display above, except when  $\theta$  is a singular strong-limit cardinal. Then  $\lambda = \theta$  and  $\mathcal{W}_\theta(S_{<\theta}) \simeq_e \mathbb{B}_{\theta \cdot \theta}$ , with the ordinal product, so  $\ell_2(\mathcal{W}_\theta(S_{<\theta})) = \theta \cdot \theta > \theta$  and the display fails for  $n = 0$ . Its Example 8.5 (`swo:st:n10.5`) takes  $\theta = \beth_\omega$ , a singular strong limit in ZFC alone. The question stays open for  $n > 0$ , for the longer strata, for cutoffs  $\theta$  that are not cardinals, for other alphabets, and for the weakest hypotheses on  $X$ . The same report reprints, with pointers to the labels here and no novelty claim, its sources' re-proofs of results that this report proves for every infinite linear alphabet: lemmas 4.1, 5.1, 6.1, 8.1 and 11.1, proposition 4.4, and theorems 5.3, 6.2, 6.4, 8.2, 11.2, 11.5 and 15.1. No result here uses that report.]

**Research question 26.8** (Other alphabets and exact gap spectra; source 06, Research question 14.4). For a general infinite linear alphabet  $X$ , how is the gap spectrum of  $\mathcal{W}(X)$  determined by the cuts and monotone sequences of  $X$ ?

The cardinality, universality, and ordinal spectrum are already settled for all such alphabets ([theorems 6.4](#) and [15.1](#)). The real-specific gap result uses that extremal stripping has countable length and residual permutation orders have countable ends. Replacing those properties by explicit cofinality bounds should yield a general cut-spine transfer theorem. Dense nonseparable alphabets, well-ordered alphabets, and ordered fields with large cofinality provide informative tests.

**Research question 26.9** (Other alphabets and other lengths; source 07, Research question 14.5). Extend the detailed classification from  $\mathbb{R}$  to subsets of  $\mathbb{R}$  and then to arbitrary linear alphabets. The no- $\nu^+$  theorem already works for any infinite alphabet of cardinality  $\nu$ . The endpoint, density, and rank arguments require additional structure. Countable alphabets deserve a separate treatment: for example  $\omega \cdot \omega \neq \omega$ , so the real-to-binary block flattening used at the uncountable initial ordinal  $\kappa$  cannot simply be copied at length  $\omega$ .

[merge] For subsets  $X \subseteq \mathbb{R}$ , source 02's [theorem 9.2](#) already classifies every stratum up to equimorphism, including countable  $X$  through [lemma 9.1](#).

## 26.7 Ultrafilter fibers and the hyperreal presentation

Among nonprincipal ultrafilters  $U$ , determine whether the order-isomorphism type of  $\mathcal{A}_U$  depends on  $U$ , even though its embedding class does not. Principal and nonprincipal fibers are already distinguished by their coinitalities, so an undifferentiated claim that all fibers are isomorphic would be false.

One can also ask which transformations of the index preserve a specified finite-support ultrapower presentation, rather than merely providing an order embedding of indices. The index order by itself does not encode all ultrafilter data used by the construction; equimorphism of index orders is not a proof of isomorphism of the associated nonstandard models.

[merge] Source 07 asks the same question.

**Research question 26.10** (Ultrafilter labels and changes of index; source 07, Research question 14.6). Find useful conditions under which an embedding between labelled linear indices induces an elementary embedding, or an isomorphism, between the associated finite-support ultrapowers. The maps must respect the ultrafilter labels and the order of the finite quantifier iterations, not just the underlying order type. The greedy labelling of  $\mathcal{J}$  ([theorem 24.1](#)) supplies a concrete test case against the actual repeated-range index  $\mathcal{A}$ .

## 26.8 Weak choice and uniform definability

Determine the choice principles needed for the separate results: existence of a nonempty exhaustive order, the successor-cardinal chain obstruction, universality for small orders, optimal fixed-stratum coding, and power absorption. Distinguish existence of arbitrary embeddings from embeddings definable uniformly from their input orders.

Herzberg, Kanovei, Katz, and Lyubetsky develop a modification of the hyperreal construction with transfer under countable choice and a further well-ordering hypothesis for properness [3]. Their work provides motivation for separating the choice strength of the index-order analysis from that of the nonstandard model. No equivalence with their precise axiomatic hypotheses is asserted here.

[merge] Sources 06 and 07 ask the same question.

**Research question 26.11** (Choice strength; source 06, Research question 14.6). Which conclusions remain valid in ZF plus the assertion that  $\mathbb{R}$  is well-orderable, and which require stronger choice or regularity principles?

The existence of a background well-order already supplies all subset completions of  $\mathbb{R}$ . The more delicate issues are the successor-cardinal stabilization argument, sizes of collections of prefixes, and

extraction of cofinal sequences. A dependency-by-dependency analysis is preferable to a blanket claim that choice is either indispensable or irrelevant.

**Research question 26.12** (Choice strength and a verified core; source 07, Research question 14.7). Determine the exact weak-choice assumptions needed for the main positive codings and for prefix completion over  $\mathbb{R}$ . In parallel, formalize the no-successor-sized-chain theorem and the pair-switch embedding first; these would give a small, auditable formal core with a complete answer to the ordinal-embedding part of the problem. Treat the gap spectra as a later, separate verification milestone.

## 26.9 Cardinal arithmetic and stability of the classification

The embeddability theorems are stated in ZFC, but their numerical interpretation depends on the continuum. Investigate which finer isomorphism invariants depend on  $2^{<\kappa}$ ,  $2^\kappa$ , or  $\text{cf}(\kappa)$ . For instance, the density distinction between the first three real strata and later ones is strict exactly when  $2^{<\kappa} < 2^\kappa$ ; the adjacency and isolation distinctions do not require that inequality.

A model-comparison study should specify whether the real set, the collection of its well-orderings, or only selected cardinal parameters are preserved. Keeping the reals fixed does not by itself keep every subset of  $\mathbb{R}^2$ , and hence every member of  $\mathcal{W}$ , fixed.

## 26.10 A proof-assistant development for ProveIt

A formalization can be divided into reusable modules. First formalize prefix-free ordinal words and their first-difference order. Then establish cut coding, the full-cylinder obstruction, and the successor-cardinal stabilization argument. On that foundation, formalize ordinal block concatenation and finite-fiber capacity, followed by the cardinal induction for all fixed strata. Adjacency and finite condensation form a largely independent final module.

The most delicate obligations are not numerical. They are the existence of common tail bounds at limit stages, correct ordinal rather than cardinal multiplication, and preservation of a full cylinder under transfinite fusion. The finite verification program accompanying this article checks none of these infinitary obligations and should not be presented as a substitute for them.

[merge] Sources 06 and 07 propose formalization plans of their own. Source 06 (its Section 13.4) suggests a relation-based carrier, a structure consisting of a well-order relation on  $\mathbb{R}$  from which the ordinal enumeration is derived, or alternatively an ordinal-indexed bijection carrier with an explicit bound on its domain. It orders the work as follows: the prefix-free lemma, first-disagreement transitivity, cylinder convexity and the pair encoder; then initial-segment coding and the regular-cardinal stabilization lemma; the exhaustion barrier as the main transfinite-recursion milestone; for the real-specific theory, countability of monotone real sequences, greedy extremal stripping, the jump criterion and disagreement-position cofinalities; and the cut-spine recursion last, followed by its counting and cofinality corollaries. The existing sign-sequence modules are useful architectural examples, but copying a sign-order definition without changing its termination semantics would be incorrect, and the repository's distinction between a mathematical report and a checked theorem must be preserved [11, 13]. Source 07 (its Section 13.3) first isolates four reusable components—prefix-free ordinal sequences, order embeddings by the first difference, regular-cardinal tail stabilization, and completion of injective prefixes—which suffice for the exact ordinal spectrum; a second stage adds the coordinate-fusion proof and pair and block codings for representation ranks and product absorption; the topology comes later: the endpoint lemma, adjacency, and finally cut tracking with explicit successor and limit stopping cases. No unspecified theorem about “lexicographic universality” should substitute for the actual coding, and no assumed continuity should enter proofs about arbitrary order embeddings. Both are proof plans, not claims that those steps have been executed.

**Research question 26.13** (Formalizing the cut spine; source 06, Research question 14.7). Can one produce a reusable, kernel-checked theorem for lexicographic leaf orders of prefix trees with no branch longer than their resource alphabet?

The proof should separate three inputs: linear ordering of child sets, closure of compatible prefixes under unions, and eventual exhaustion. Specializing it to injective real sequences would yield the main gap theorem; specializing the stabilization lemma would yield the ordinal barrier. This is a more reusable formalization target than encoding only the particular real-number instance. Source 07's [research question 26.12](#) proposes the verified ordinal core.

## 27 Conclusions of sources 06 and 07

[merge] Source 02 has no concluding section; sources 06 and 07 close as follows.

*Source 06.* The lexicographic order of all well-orderings of  $\mathbb{R}$  is simultaneously large and sharply bounded. It has  $2^c$  elements and embeds every continuum-sized linear order, yet it cannot contain the successor-continuum ordinal. Its exact ordinal spectrum is therefore determined in ZFC without any continuum hypothesis. The exhaustion requirement also creates structure invisible in a bare lexicographic power. Finite residual alphabets create factorial-sized jump blocks and a dense set of isolated points. Continuing beyond the initial cardinal length fills certain uncountable balanced gaps of the minimal slice by whole residual orders. Nevertheless countable-sided gaps remain, and the cut-spine analysis classifies all possible gap cofinalities. The contrast between points, gaps, the minimal slice, and the dense quotient shows why none of these objects should be substituted for another without an explicit argument.

*Source 07.* The lexicographic order of all well-orderings of  $\mathbb{R}$  is not remotely a well-order, but its well-ordered suborders have a sharp ceiling: it contains every ordinal of cardinality at most the continuum and none larger. Within that ceiling it has unexpectedly large embedding capacity, reaching every real lexicographic exponent below  $\kappa^+$  and absorbing all corresponding powers of itself. At the same time its local structure records exhausted finite tails, long divergence patterns, and a rigid restriction on unfilled cuts: one side of every gap is countable. The fixed-length stratum and the actual ultrafilter index share the ordinal ceiling but have smaller representation rank and different topologies. Keeping these orders separate makes the connection with definable hyperreals more precise and prevents cardinality, saturation, and isomorphism from being conflated with mere mutual embeddability.

## A Proof dependencies and reproducible finite checks

The central arguments have a short dependency structure:

$$\begin{aligned} \text{prefix-freeness} + \text{eventual constancy} &\implies \text{exact ordinal cutoff,} \\ \text{full-child lemma} + \text{fusion} &\implies \text{strict cube hierarchy,} \\ \text{pair swaps} + \text{cut coding} &\implies \text{small-order universality,} \\ \text{countable codes} + \text{cardinal induction} &\implies \text{all fixed-stratum equivalences,} \\ \text{disjoint real copies} + \text{exhaustive concatenation} &\implies \text{full power absorption.} \end{aligned}$$

These arguments are self-contained apart from elementary ZFC facts about well-ordering, cardinal arithmetic, and transfinite recursion, which are explained at their points of use.

[merge] Source 06's dependency map (its Section 13.2) adds the chains

$$\begin{aligned} \text{extremal enumerations} &\implies \text{finite-tail jump classification,} \\ \text{greedy extremal stripping} + \text{cut spine} &\implies \text{gap restriction and completion size,} \\ \text{disagreement positions} &\implies \text{point characters and gap realizations,} \end{aligned}$$

and notes that the comparison with surreal sign orders depends only on binary coding and the ordinal barrier. Source 07's audit gives the analogous dependencies of its representation ranks (real-to-binary coding, real-power strictness, fixed-length padding of prefix-free codes), of its fixed-length gaps (prefix tracking up to  $\kappa$ , with special care for non-surjective branches of length exactly  $\kappa$ ) and of its index results (surjectivity onto an ultrafilter of size  $\kappa$ , arbitrarily late perturbations, explicit missing branches); no hyperreal saturation claim is inferred from index topology.

The archive contains `finite_checks.py`, a Python program using only the standard library, together with its actual output in `finite_checks.json`. Run it with

```
python finite_checks.py -output finite_checks.json
```

It checks pair-swap order preservation, cut coding of finite labeled orders, every candidate adjacent pair of permutations through size six, consecutive permutation pairs through size eight, finite concatenation, and the integer capacities used in the terminal-factor formulas. The supplied run passes **552,450** exact checks. Counts and parameter bounds are recorded by the program, rather than inferred from an informal estimate.

[merge] In this report the program is shipped as `code/02-strata-finite_checks.py` and its recorded output as `data/02-strata-finite_checks.json`; the text of the program still names its delivered output `finite_checks.json`. Its default output path is `finite_checks.json` in the current directory, so a rerun should pass an explicit scratch path, for example `pycode/02-strata-finite_checks.py --output<scratch>/finite_checks.json`, and never the shipped `data/` file. A rerun on a copy during placement reproduced the recorded JSON, identical after removing the carriage returns that Python writes in text mode on Windows. The finite checks of sources 06 and 07 are shipped as `code/06-spectra-finite_checks.py` and `code/07-rank-finite_checks.py`, with their recorded outputs in `data/`; they too test finite instances only.

[merge] *Source 06's checks* (its Section 13.3) use only the Python standard library. They check the pair encoder on all binary inputs for small block counts, verify convexity and factorial cardinalities of finite prefix cylinders, compare the jump criterion with consecutive lexicographic permutations, and exhaustively check the converse jump criterion for all pairs of permutations up to a stated finite size. The recorded run passed 511 binary encodings (with 43,435 first-disagreement comparisons), 16,072 finite prefix cylinders, 46,225 consecutive-pair checks through alphabet size eight, and 266,272 all-pair comparisons through alphabet size six, and checked initial-segment coding for 5,914 finite linear orders through size seven. These checks can catch indexing, orientation, and finite-suffix mistakes. They do not verify the transfinite stabilization proof, the cut-spine argument, cardinal arithmetic, or any claim about an uncountable order.

*Source 07's checks* test finite permutation adjacency (46,225 adjacent pairs, 7,432 exhaustive pairs and 30,000 pairs sampled with a recorded fixed seed), pair-switch coding (2,047 words), initial-segment cut coding (5,913 coordinate orders), concatenation of interleaved ordered copies (258 product elements), and a nonconvex ultrafilter fibre on a three-point universe ([remark 23.3](#)). They do not check transfinite induction, choice, cardinal arithmetic, topology, or the gap classification.

**What the checks establish.** They are exact finite tests of the stated constructions and permutation formulas. They do not verify a transfinite theorem, establish a literature-priority claim, or constitute a Lean or Rocq formalization. The mathematical evidence for the infinite claims is the written proof, subject to ordinary review.

## B Source audit and manuscript status

The Kanovei–Shelah definition was checked directly in the paper, including the displayed index and its comparison rule on page 2 of the arXiv PDF. [merge] Source 07 locates the same definition in [1, §2,

p. 160], on the printed page of the published version (its audit, item 1), whereas source 02 cites Section 1 of the paper; both quote the same index and the same comparison rule. The section numbering was not rechecked at the merge. The paper’s bibliographic information was cross-checked against Shelah’s official paper archive. The weak-choice follow-up was checked through its authors’ arXiv record. Knoblauch’s work is cited for the related research terminology and bibliographic context, not as an uninspected source for any specific theorem proved here.

The ProveIt source was inspected at the immutable commit

6ea60e3677bd847a00baf023c3c532c83884530e.

Its relevant report is the binary-cube and point-separation manuscript cited below. Only its elementary cylinder perspective is reused; the required lemma is reproved. No existing formal verification of the new article’s theorems is claimed, and this work does not modify the repository.

The literature search was targeted rather than exhaustive. It identified related primary sources, but it does not settle historical priority for the classification and absorption statements. The questions in [section 26](#) are proposed next steps, not an authenticated list of published open problems. All websites and repository material were consulted for this manuscript on 2 October 2026.

[merge] Sources 06 and 07 record their own audits in the shipped files `06-spectra-research_audit.md` and `07-rank-source_audit.md`, both pinned to the same ProveIt revision and both dated 2 October 2026. Neither claims priority, peer review or machine verification, and both describe their repository search as targeted rather than exhaustive. In summary:

- Source 06 inspected the Cardinals README, the report index [11] and the module [13] of the repository [10]; it read but did not compile the module, and uses it as context only. It cites Moore [5, Section 2] for the characteristic-function representation, Ramakrishnan [8] for the literature on lexicographic representation (including Fleischer’s paper and its correction, whose original source 06 did not obtain), and Tao [9] for background on ordinals; consequently it does *not* infer an arbitrary-order embedding theorem from the absence of long monotone sequences. No nontrivial unverified assertion from a repository report is used as a premise.
- Source 07 inspected the Kanovei–Shelah definition on the printed page of the published version, Kuhlmann’s Corollary 2.4 on a rendered preprint page (the parsed text being garbled; the preprint’s support convention permits all functions for an ordinal exponent), and Giarlotta’s first page for the terminology of the representability number [7]. In the repository it read the Cardinals README [12], which separates unformalized reports from the Lean library, and a recorded result `lexicographic_tails_result.json` of the cofinal-strata report, which concerns a different lexicographic sum and is not used. It separates published input (the index definition, Kuhlmann’s theorem, the terminology) from its own deductions, which are not attributed to the cited papers merely because those papers motivated the definitions.

## C Theorem crosswalk of the three sources

[merge] Every numbered statement of the three sources, by topic. Source numbers are those of the delivered PDFs (T theorem, L lemma, P proposition, C corollary, D definition, E example, R remark, Q research question), followed by the line of the delivered `.tex`. “Here” is where the statement is printed in this report. Unnumbered statements of a source are cited by line.

Table 5: [merge] Crosswalk.

| Result                                                                 | Source 02                                     | Source 06                       | Source 07                              | Here                                                            |
|------------------------------------------------------------------------|-----------------------------------------------|---------------------------------|----------------------------------------|-----------------------------------------------------------------|
| Main results at a glance                                               | T2.1 (191)                                    | T1.1 (102)                      | T2.1 (187)                             | <a href="#">3.1</a>                                             |
| Definition; prefix freeness;<br>linear order                           | L3.1 (269)                                    | L2.1 (171), D2.2,<br>P2.3 (198) | D1.1 (118), §1.2                       | <a href="#">4.1</a>                                             |
| Cylinders and residual sets;<br>next-value sum                         | D3.2, L3.3 (297)                              | D3.1, L3.2 (246)                | L3.1 (259)                             | <a href="#">4.3</a> , <a href="#">(3)</a>                       |
| Cardinality $2^\lambda$ by pair<br>swaps                               | P3.4 (318)                                    | L4.1 (331), C4.2<br>(363)       | L4.1 (329), C4.6<br>(417)              | <a href="#">4.4</a>                                             |
| Cut (initial-segment)<br>coding                                        | L4.1 (337)                                    | L4.3 (373)                      | L4.2 (339)                             | <a href="#">5.1</a>                                             |
| Full child; strict cube<br>hierarchy                                   | L4.2, T4.3 (367)                              | —                               | —                                      | <a href="#">5.3</a> (= PSG<br>cor:strict)                       |
| Binary coding length $\ell_2$                                          | D4.4 (393)                                    | —                               | —                                      | <a href="#">5.4</a>                                             |
| Real and binary codes                                                  | L4.5 (402)                                    | in P4.6                         | (5.1)–(5.3)                            | <a href="#">5.5</a>                                             |
| Cardinal identities;<br>cf $\kappa > \omega$                           | 02:425, R11.3                                 | in T9.3                         | L3.2 (288)                             | <a href="#">(4)</a> , <a href="#">12.2</a>                      |
| Strict real powers<br>(Kuhlmann)                                       | —                                             | —                               | T5.2 (477)                             | <a href="#">5.6</a>                                             |
| Eventual constancy                                                     | L5.1 (437)                                    | L5.1 (454)                      | L4.3 (355)                             | <a href="#">6.1</a>                                             |
| No $\nu^+$ -chain                                                      | T5.2 (451), R5.3                              | T5.2 (465), R5.3                | T4.4 (367)                             | <a href="#">6.2</a>                                             |
| Universality; ordinal<br>spectrum                                      | T5.4 (487)                                    | C5.4 (501), E4.5                | L4.2, C4.5 (394)                       | <a href="#">6.4</a>                                             |
| Universality in every slice<br>$\mathcal{W}_\beta(X)$                  | —                                             | T4.4 (389)                      | —                                      | <a href="#">15.1</a>                                            |
| Separating families of<br>initial segments                             | —                                             | 06:415–423                      | —                                      | <a href="#">15.2</a>                                            |
| $\mathcal{W}_\kappa \simeq_e \mathbb{B}_\kappa \simeq_e \mathcal{A}$   | T6.1 (516), C6.2                              | P4.6 (425)                      | (5.3), L11.1,<br>T11.2 (1129),<br>R1.2 | <a href="#">7.1</a>                                             |
| Finite tails $\mathcal{W}_{\kappa+n}$ ; capacity                       | T6.3, T6.4, E6.5                              | —                               | —                                      | <a href="#">7.3</a> , <a href="#">7.4</a>                       |
| Small cubes and real<br>powers below $\kappa^+$                        | L7.1 (665), C7.4<br>(732)                     | L4.1, P4.6                      | P5.1 (460)                             | <a href="#">8.1</a> , <a href="#">8.6</a>                       |
| Coding length<br>$\ell_2(\mathcal{W}) = \kappa^+$                      | T7.2 (682)                                    | —                               | —                                      | <a href="#">8.2</a>                                             |
| Representation ranks $\text{repr}_\mathbb{R}$                          | —                                             | —                               | T5.3 (505)                             | <a href="#">19.1</a>                                            |
| $\mathcal{W}$ into no fixed power<br>below $\kappa^+$                  | C7.3 (719)                                    | C9.5 (1053),<br>Q14.1           | C5.4 (525)                             | <a href="#">8.3</a> , <a href="#">8.4</a> , <a href="#">8.5</a> |
| All fixed strata                                                       | L8.1, T8.2 (782),<br>C8.3–C8.4, E8.5,<br>P8.6 | —                               | —                                      | <a href="#">9.2</a> – <a href="#">9.6</a>                       |
| Disjoint copies; power<br>absorption; closure                          | L9.1, T9.2 (946),<br>C9.3                     | —                               | L6.1, T6.2 (563),<br>C6.3              | <a href="#">10.2</a> , <a href="#">10.3</a>                     |
| Fixed-length powers                                                    | C9.4 (1005)                                   | —                               | P6.4 (613)                             | <a href="#">10.4</a>                                            |
| $\mathcal{W}_{\text{lex}}^\omega \simeq_e \mathcal{W}$ but not $\cong$ | —                                             | —                               | 07:628–633                             | <a href="#">19.2</a>                                            |
| Convex copies of $\mathcal{W}$                                         | P9.5 (1026)                                   | —                               | —                                      | <a href="#">10.5</a>                                            |
| Extremal enumerations                                                  | L10.1 (1044)                                  | L3.3 (263)                      | L8.1 (714)                             | <a href="#">11.1</a>                                            |
| Countable monotone real<br>sequences; countable ends                   | —                                             | L3.4, L3.5 (294),<br>C3.6       | L8.1 (714)                             | <a href="#">16.1</a> , <a href="#">16.2</a>                     |
| Adjacency criterion;<br>example                                        | T10.2 (1062),<br>E10.3                        | E6.1, T6.2 (548),<br>App. B     | T8.2 (747)                             | <a href="#">11.2</a>                                            |

| Result                                                                          | Source 02                  | Source 06                          | Source 07          | Here              |
|---------------------------------------------------------------------------------|----------------------------|------------------------------------|--------------------|-------------------|
| Finite blocks; isolated points                                                  | D10.4, T10.5 (1129)        | P6.3 (590), C6.4                   | T8.3 (770)         | 11.5              |
| Dense isolated points; perfect remainder                                        | —                          | L6.5, T6.6 (650), P6.7             | C8.4 (798)         | 16.3–16.5         |
| Global invariants of $\mathcal{W}$                                              | P11.1 (1166)               | C3.6, T6.6                         | C8.4               | 12.1              |
| Topology of $\mathcal{W}_\kappa$                                                | P11.2 (1204), R11.3        | L9.1, T9.2, T9.3, R9.4             | T7.1 (637), C7.2   | 12.2, 20.1–20.5   |
| Finite-tail strata topology                                                     | P11.4 (1264)               | —                                  | —                  | 12.4              |
| Topology of $\mathcal{A}$                                                       | L12.1, P12.2 (1330)        | —                                  | T11.3 (1163)       | 13.2, 23.1        |
| $(\omega, \omega)$ gaps of $\mathcal{W}$ , $\mathcal{W}_\kappa$ , $\mathcal{A}$ | P13.1 (1384), P13.2        | —                                  | E7.3 (703), P11.4  | 14.1, 14.2        |
| Small cuts; completion; complete cubes                                          | L13.3, T13.4, L13.5, R13.6 | C5.5, T7.6 (824)                   | C4.6               | 14.3–14.5         |
| Gap spectrum of $\mathcal{W}$                                                   | topic (1635)               | D7.1, L7.2, T7.3, L7.4, T7.5 (802) | L10.1, T10.2 (941) | 18.1–18.6         |
| Point-character spectrum                                                        | topic (1635)               | T8.1, P8.2, T8.3 (913)             | P9.1, T9.2 (861)   | 17.1–17.3         |
| Gaps of $\mathcal{W}_\kappa$ ; no convex copy                                   | —                          | P10.3 (1101)                       | T10.3 (996), C10.4 | 20.6, 20.7, 20.10 |
| Lexicographic-sum decomposition                                                 | —                          | D10.1, T10.2 (1078)                | —                  | 20.8, 20.9        |
| Dense quotient $\mathcal{W}/\sim_{\text{fin}}$                                  | question (1604)            | T11.1 (1133)                       | P10.5 (1072)       | 21.1              |
| Bounded surreal sign orders                                                     | —                          | T12.1 (1198)                       | —                  | 22.1              |
| Further gaps and fibres of $\mathcal{A}$                                        | —                          | —                                  | P11.4 (1218)       | 23.2, 23.3        |
| Well-ordering-based ultrafilter index                                           | —                          | —                                  | T12.1 (1265)       | 24.1              |
| Research questions                                                              | 10 topics (1573)           | Q14.1 answered; Q14.2–Q14.7        | Q14.1–Q14.7        | 26                |
| Choice; dependency maps; checks                                                 | §14, App. A, B             | §13                                | §13                | 25, 25.3, A, B    |
| Conclusions                                                                     | —                          | §15                                | Conclusion         | 27                |
| Summary tables                                                                  | Table 1                    | App. A                             | §2 table           | Tables 3, 4       |

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